
A Digital Twin Framework for Predictive Quality Assurance and Resilient Supply Chain Management in the Automotive Sector

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ARTICLE INFO

Keywords:

Digital Twin
Supply Chain Resilience
Predictive Quality Assurance
Automotive Industry
Disruption Risk Management
Machine Learning in Supply Chain
Supply Chain Optimization
Simulation and Optimization
IoT in Supply Chain
End-to-End Supply Chain Visibility
Decision-Support Systems
Supply Chain Digital Twin (DSCT)
Ripple Effect in Supply Chain
Digital Product Passport (DPP)
Battery Passport
Autonomous Vehicle Development
Supply Chain Risk Analytics
KPI-based Supply Chain Management
Resilient Supply Chain Design
Data Analytics in Supply Chain

ABSTRACT

The automotive industry faces increasing complexity in managing supply chain resilience, particularly in the face of disruptions and fluctuating market demands. Traditional quality assurance (QA) systems often fall short due to limited traceability and adaptability. This paper proposes a novel framework leveraging digital twin technology to enhance predictive quality assurance (PQA) within automotive supply chains, fostering greater resilience and operational excellence. The proposed model integrates real-time data from diverse sources, including IoT sensors and historical performance data, to create dynamic, virtual replicas of supply chain processes. By simulating potential disruptions and evaluating the impact on key performance indicators (KPIs), the digital twin enables proactive risk management and informed decision-making. This paper outlines an eight-phase, five-stage PQA framework, aligned with IATF standards, to enhance supply chain traceability, improve predictive capabilities, and optimize quality assurance processes. The results of a case study that will be present in future studies can confirm the framework's effectiveness in improving supply chain visibility, reducing operational risks, and ensuring product quality. Additionally, this research discusses the broader applicability of the digital twin model across industries and suggests future directions for integrating advanced technologies to further strengthen supply chain resilience.

1. Introduction

1.1. The Economic Importance of the Automotive Industry

The automotive industry has long been a cornerstone of economic growth, both in developed and developing countries. Since the end of World War II, the sector has been a major contributor to industrial development, providing employment and driving innovation. For instance, the UK's automotive industry generates an annual turnover of over £82 billion, contributing significantly to the economy, with £18.6 billion added to the GDP and more than 823,000 people employed in various roles (Vine, 2020, as cited in [Belhadi et al., 2021](#)). In the United States, the automotive sector has historically contributed between 3-3.5% of the national GDP, with over 1.7 million people employed directly in manufacturing, engineering, and service roles ([Hill et al., 2010](#)). The industry also fuels vast networks across other sectors, including raw materials, machinery, legal, healthcare, and electronics, which collectively enhance its economic footprint ([Hill et al., 2010](#)). The automotive industry alone spends between \$16 to \$18 billion annually on research and development,

a substantial proportion of which is self-funded by the sector ([Hill et al., 2010](#)).

As global supply chains grow increasingly interconnected and vulnerable to disruptions, the need for enhanced supply chain resilience (SCR) in the automotive industry has never been more critical. Disruptions—whether due to geopolitical tensions, pandemics, or natural disasters—pose significant risks to the timely production and delivery of automotive goods. In response, companies are turning to innovative technologies such as digital twin (DT) to improve visibility, traceability, and predictive capabilities within their supply chains ([Putha, 2022](#)).

A digital twin, a virtual replica of physical assets or systems, provides real-time monitoring, simulation, and predictive analytics, enabling automotive manufacturers to enhance operational efficiency, optimize supply chain management, and proactively mitigate risks ([Bakhshi et al., 2024](#)). By using IoT devices, machine learning, and simulation tools, digital twins offer a dynamic approach to quality assurance (QA), shifting from traditional static models to more agile and responsive systems ([Putha, 2022](#)). However, despite the promise of these technologies, challenges remain in integrating digital twin systems within existing supply chain infrastructures and ensuring scalability across global operations ([BLOMKVIST and Loenbom, 2020](#)).

This paper aims to explore the potential of Digital Twin technology in enhancing predictive quality assurance (PQA) and managing supply chain disruptions in the automotive industry. The proposed framework integrates data from various sources across the supply chain and uses simulation techniques to improve resilience, enhance traceability, and enable proactive decision-making. By aligning this framework with the international automotive task force (IATF) standards, the study aims to provide a practical and scalable model for improving SCR in the automotive sector. The goal is to develop a robust methodology that automotive companies can implement to better predict, respond to, and mitigate disruptions, ensuring continued operational excellence.

1.2. Supply Chain Complexity in the Automotive Industry

The automotive industry is characterized by an intricate and highly interconnected supply chain that spans across multiple tiers, regions, and industries. From raw material extraction to component manufacturing, assembly, and distribution, each stage involves a complex network of suppliers, manufacturers, logistics providers, and service partners. The complexity is further exacerbated by the global nature of the industry, where production facilities are often located in different countries, and components are sourced from various parts of the world. This extensive network leads to a high level of interdependence among suppliers, creating a delicate balance between cost, quality, and delivery times.

One of the primary challenges faced by the automotive industry is managing the flow of materials and components through a global supply chain that is often subject to disruptions. A minor delay at one point in the supply chain can ripple through the entire network, leading to significant production stoppages and financial losses. For instance, in 2020, the automotive sector experienced severe disruptions due to the COVID-19 pandemic, leading to supply shortages, production halts, and the need to reevaluate risk management strategies. The industry's reliance on just-in-time (JIT) inventory systems, while reducing costs, has made it vulnerable to such disruptions, as even small delays or unforeseen events can trigger a chain reaction throughout the entire production process.

Additionally, the automotive industry's supply chain is not only vast but also dynamic. It must constantly adapt to evolving consumer demands, technological innovations, and regulatory changes. For example, the push towards electric vehicles (EVs) and autonomous driving technologies has altered the demand for certain components, while stringent environmental regulations are driving changes in materials and manufacturing processes. At the same time, global trade dynamics, geopolitical tensions, and environmental concerns can impact sourcing decisions and production locations, adding further uncertainty and complexity to the supply chain (Irsyadillah et al., 2020).

In this highly dynamic environment, traditional supply chain management practices are increasingly inadequate. The automotive industry requires real-time visibility into every facet of its supply chain, from procurement to production to delivery, to ensure agility and responsiveness (Putha, 2022). As a result, companies are increasingly turning to advanced technologies like digital twins (DTs) to enhance their ability to simulate, monitor, and optimize supply chain operations (Ivanov et al., 2019). Digital twins allow for the creation of virtual replicas of physical assets and processes, enabling manufacturers to simulate potential disruptions, optimize processes, and identify inefficiencies before they become costly problems. By providing real-time insights into the entire supply chain, digital twins enable automotive companies to make data-driven decisions, reduce downtime, and enhance resilience.

The complexity of the automotive supply chain presents significant challenges in terms of risk management, cost control, and operational efficiency. As the industry faces increasing pressures from technological advancements, consumer expectations, and global uncertainties, the adoption of innovative solutions such as digital twins will be essential for maintaining competitiveness and ensuring long-term sustainability.

1.3. The Role of Quality Assurance in Resilience

Quality assurance (QA) plays a pivotal role in ensuring the resilience of supply chains, especially in industries like automotive, where the production processes are highly sensitive to disruptions. In the context of supply chain resilience, QA not only ensures that the products meet the required standards of quality but also contributes to building the capability of the supply chain to withstand, adapt to, and recover from various disruptions. Traditional quality assurance methods, which often rely on historical data, standardized procedures, and end-of-line inspections, have proven insufficient in today's volatile, uncertain, complex, and ambiguous (VUCA) business environment.

As supply chains become increasingly complex and interconnected, traditional QA processes are no longer capable of providing the timely insights required to prevent failures before they occur. The automotive industry has seen a shift towards more dynamic and proactive approaches to quality management, where continuous monitoring and real-time feedback are central to ensuring product quality and process integrity. As such, modern QA practices are moving away from a purely reactive stance, where issues are identified after they occur, to a more proactive model that integrates predictive capabilities, real-time monitoring, and adaptive problem-solving into the supply chain processes.

In this context, the integration of emerging technologies such as digital twins and artificial intelligence (AI) into quality assurance systems is becoming increasingly important. By creating virtual replicas of physical systems, digital twins enable real-time tracking and analysis of product quality and production processes across the entire supply chain. This integration allows for the simulation of various scenarios, including potential disruptions, and provides a framework for continuous monitoring of key performance indicators (KPIs) related to quality and resilience. The use of AI-powered analytics further enhances this capability by enabling the identification of patterns, predicting potential failures, and suggesting optimal corrective actions, even before a disruption occurs.

For instance, in the automotive supply chain, a digital twin can track the entire lifecycle of a vehicle, from raw material procurement to manufacturing and assembly, identifying any potential quality issues or bottlenecks that could disrupt the production flow. The predictive capabilities of digital twins, combined with AI algorithms, can analyze historical data, operational variables, and real-time inputs to anticipate quality issues, ensuring that corrective actions are taken promptly. This predictive approach minimizes the likelihood of defects reaching consumers and reduces the risk of costly recalls or production stoppages, thereby safeguarding the company's reputation and financial stability.

Moreover, a robust quality assurance system integrated with digital twins can provide greater visibility into the entire supply chain, fostering improved communication between suppliers, manufacturers, and distributors. This enhanced visibility allows all stakeholders to collaborate more effectively, enabling quick identification of quality issues and a coordinated response to potential disruptions. In the automotive sector, where supplier relationships are crucial, such transparency strengthens collaboration and enhances trust among partners, further reinforcing the overall resilience of the supply chain.

Ultimately, the role of quality assurance in building supply chain resilience is about more than just ensuring that products meet the required standards. It is about creating a culture of proactive quality management that can swiftly respond to challenges, optimize processes in real time, and enhance the agility of the supply chain. As the automotive industry continues to face increasing pressures from technological advancements, environmental regulations, and global disruptions, an integrated approach to quality assurance—one that leverages cutting-edge technologies like digital twins—will be essential for ensuring that automotive companies remain competitive and resilient in the face of uncertainty.

On the other hand, according to the ISO/TS 16949, in automotive supplying companies based on this standard, organizations shall have a defined process for problem-solving leading to root cause identification and elimination. It is up to the organization as to which method to use. However, if a customer-prescribed problem-solving format exists, the organization shall use the prescribed format (ISO/TS 16949, 2009).

There are several problem-solving methods derived from the PDCA cycle with some modifications, such as:

- PDSA: Plan, Do, Study, Act
- OPDCA: Observe, Plan, Do, Check, Act
- SDCA: Standardize, Do, Check, Act
- LAMDA: Look, Ask, Model, Discuss, Act
- DMAIC: Define, Measure, Analyze, Improve, Control

There are also several other methods that have been invented and applied by large companies, such as:

- OODA: Observe, Orient, Decide, Act
- FACTUAL: Focus, Approach, Converge, Test, Understand, Apply, Leverage
- 8D: Eight Disciplines Problem Solving

The 8 disciplines (8Ds) method is a tool of ISO/TS 16949:2009 and has been widely applied in automotive industries for service or products problem solving, including the problems with supplier qualification confirmation, manufacturing process deviations, defects, maintenance, customer complaints, returned purchases, etc. (Chen and Cheng, 2010). The 8D method is a method or model developed at Ford Motor Company used to approach and to resolve problems. Since Ford Motor is one of the largest automotive companies, 8D was chosen as the method of choice in our research to best align with the research objective. An analytical comparison will help to clarify our reasons for using 8D instead of the other methods mentioned highlighting their relevance and limitations concerning predictive quality assurance and supply chain resilience. A PDCA Cycle is a Six Sigma quality improvement methodology. PDCA aims to improve processes by identifying potential problems and implementing solutions continuously. PDCA can be a day-to-day method taught and used by front-line staff level to do fundamental problem-solving, but PDCA appears to be less complicated and more direct (Six Sigma Online, 2025).

LAMDA is actually specialized on product development. It also aims to be a PDCA replacement in this field, but it has some shortcomings too. Additionally, different sources define it differently. Luckily, it is rarely used. LAMDA never considers what we want to achieve, or which problem you want to solve. Due to the inconsistency of the Act part, it either never asks if it actually worked, or it never actually generates a finished product (Christoph, 2016).

DMAIC is a Six Sigma quality improvement. The DMAIC process aims to improve process quality by identifying and eliminating defects. The DMAIC philosophy is more in-depth. It tends to be useful for tackling data-driven complex problems on a large scale. DMAIC users must have the necessary technical expertise to lead-problem solving initiatives while employing this method (Six Sigma Online, 2025). This method emphasizes measurement and analysis, often involving the use of statistical tools, to improve process capability and address complex issues where large amounts of data are available. While the DMAIC itself does have controls to maintain the improved process, it does not have a specific prevention step. First and foremost, DMAIC is primarily used to enhance an existing process. It focuses on making an existing system more efficient and reliable. (Six Sigma Online, 2025).

Originally, the OODA framework was used for combat operations processes to make faster and more effective decisions in fluid, high-stakes environments and is now utilized in various settings, including business and crisis management, to aid in rapid decision-making and adaptive strategies (Croft, 2025).

FACTUAL is part of the Shainin toolbox developed by Dorian Shainin. The Shainin approach is using quite a bit of statistics to determine the relation between cause and effect, and to determine the most relevant cause, also known as the Red X. Hence, it is no surprise

that FACTUAL is structured around this statistic. We don't always have the luxury of a deep statistical understanding of the problem, which means we cannot always use FACTUAL. Even so, the focus is on the statistics of understanding the problem, and very little on developing a solution, let alone checking if it worked (Christoph, 2016).

Table 1 shows the distinct differences between the methods mentioned earlier. Also, PDSA, OPDCA, and SDCA methods are considered similar to the PDCA cycle due to their significant proximity to it. According Table 1, some methods are not data-driven and not useful for implementing digital twins. In addition, some methods such as LAMDA and FACTUAL have several shortcomings for use. Also, the DMAIC approach is used for existing systems without preventive steps, which conflicts with our goal of predicting quality assurance in the future. Moreover, the DMAIC methodology require a steering committee, tollgates, a champion, and a project sponsor (Six Sigma Online, 2025). In addition, PDCA is almost used for less complicated, more direct, and daily problems, which conflicts with complex supply chains such as the automotive industry. In contrast to the methods mentioned, 8D is particularly suitable for organizations that require a systematic and collaborative approach to addressing intricate issues, which can generate huge improvements. The methodology has been adopted widely in industries such as automotive, where the focus is on solving critical problems, implementing corrective actions, and preventing recurrence. It is employed to address an existing problem or issue and create a resolution. In DMAIC, the team collaboratively works on enhancing the process to make it more robust and efficient. The goal is continuous improvement in the process. In contrast, the 8D methodology engages the team to discover a permanent solution to a particular problem. The emphasis is on addressing and eliminating a specific issue that is inhibiting operational effectiveness. (Six Sigma Online, 2025).

Table 1

Analytical Comparison Between Problem-Solving Methods

Method	Scope	Purpose	Limitation	Data-Driven
PDCA	Various Supply Chains	Quality Improvement Projects	Less Complicated, More Direct, and Daily Problems	✗
LAMDA	On Site	Product Development	Inconsistency	✗
DMAIC	Various Supply Chains	Quality Improvement and Project Management	Complex Problems, Technical Expertise, Statistical Tools, without Prevention Step, and just for Existing Process	✓
OODA	Fast-Paced Environments and Military Strategy	Quick Decisions and High-Stakes, Complex Situations	Small Scale, Lack Depth, Resource Intensive, Context-Specific, Risk of Hasty Decisions, and Real-Time Data	✓
FACTUAL	Various Supply Chains	To Determine the Relation Between Cause and Effect	Deep Statistics	✗
8D	Various Supply Chains	Solving Critical Quality Problems, Implementing Corrective	Team Working	✓

The 8D methodology involves teams working together in order to solve quality problems, using a structured 8 phases as shown in Table 2 (Misztal et al., 2016). On the other hand, given the IATF standard's emphasis on organizational roles, responsibilities, and authorities in clause 5: Leadership that considers team collaboration, there is another reason why we see the 8D method as more appropriate than other methodologies .

Table 2
Phases of the 8D Methodology (Misztal et al., 2016)

No.	Phase Name	Description
1	Problem Description	Defining the internal or external problem with accuracy and concision. The root cause of the problem is found by using the .5 Whys WWWW (Who?, What?, Where?, When?, How?) methods. A problem well-defined is a problem half-solved."
2	Reference to Similar Products	This phase is performed with the purpose to detect other products with the same nonconformity risk. The following aspects must be taken into account: the list of the detected problems, the place of their detection, the technological process in question, what other products are realized in the same process, whether here is also a nonconformity risk in what they are concerned, where these products are used. Identifying the causes of the failure to detect the problem: -in what place should the nonconformity be detected?
3	The First Analysis	-why wasn't the nonconformity detected? Both the manufacturing process and the control process are taken into account. The point where the problem has escaped the control system (escape point) must be sought for and detected, as well as the moment when it could have been detected.
4	Immediate Action Plans	Defining and implementing setting immediate action plans to protect the internal/external customers from the problem until permanent corrective actions can be implemented. The work tools recommended for the performance of this stage are datasheets, control ranges, audit, planning.
5	Final Analysis	This phase is carried out with the purpose of obtaining a real and complete of the situation in order to find the root causes and to decide the optimum actions required for the treatment of the

6 Final Action Plans

Establishing an implementation plan for permanent corrective actions. Permanent actions are analyzed and applied in order to prevent the recurrence of the problem permanently. The final elimination action must be focused on the real main root causes (phase 5) and on the escape point (phase 3). Thus, the aim is to solve both the root causes of the problem, as well as the root causes of the failure to detect the problem. In order to prevent the recurrence of the problem, a number of actions are applied: updating procedures, training the personnel.

7 Confirmation of the Action Plans

Phase 7 of the 8D method is very important due to the fact that it allows for action plans the closing of the action plans. The effectiveness of the final action plans is checked. It is a key stage meant to avoid the recurrence of the quality problem. The Supplier Quality Service will require evidence of the effectiveness of the action plans and, in order to be able to validate the SD, may also perform an audit if it finds that the effectiveness of such actions is not trustworthy. So long as the effectiveness of the actions is not proven, the supplier should make sure that it supplies 100% good parts by carrying out checks. The Supplier Quality Service will be notified in what the results of such checks are concerned.

8 Preventing a Recurrence of the Problem

Identifying possibilities for improvement and establishing a process of initiatives for improvement: modifying specifications, update training, review work flow, and improve practices and procedures in the scope required to prevent recurrence of this and all similar problems in the future.

In Fig. 1, we show the proposed structure based on IATF standards with eight phases in the 8D method. This research considers eight phases to implement the predictive quality assurance framework and applies five stages to the design. It is worth noting that due to the extensive structure in clauses 7 and 8 of the IATF standard, both are displayed concisely. It should also be noted that clauses 1 to 3 provide a background to the standard, and clauses 4 to 10 provide the automotive quality management system requirements. These seven clauses are mandatory (TQP, 2022).

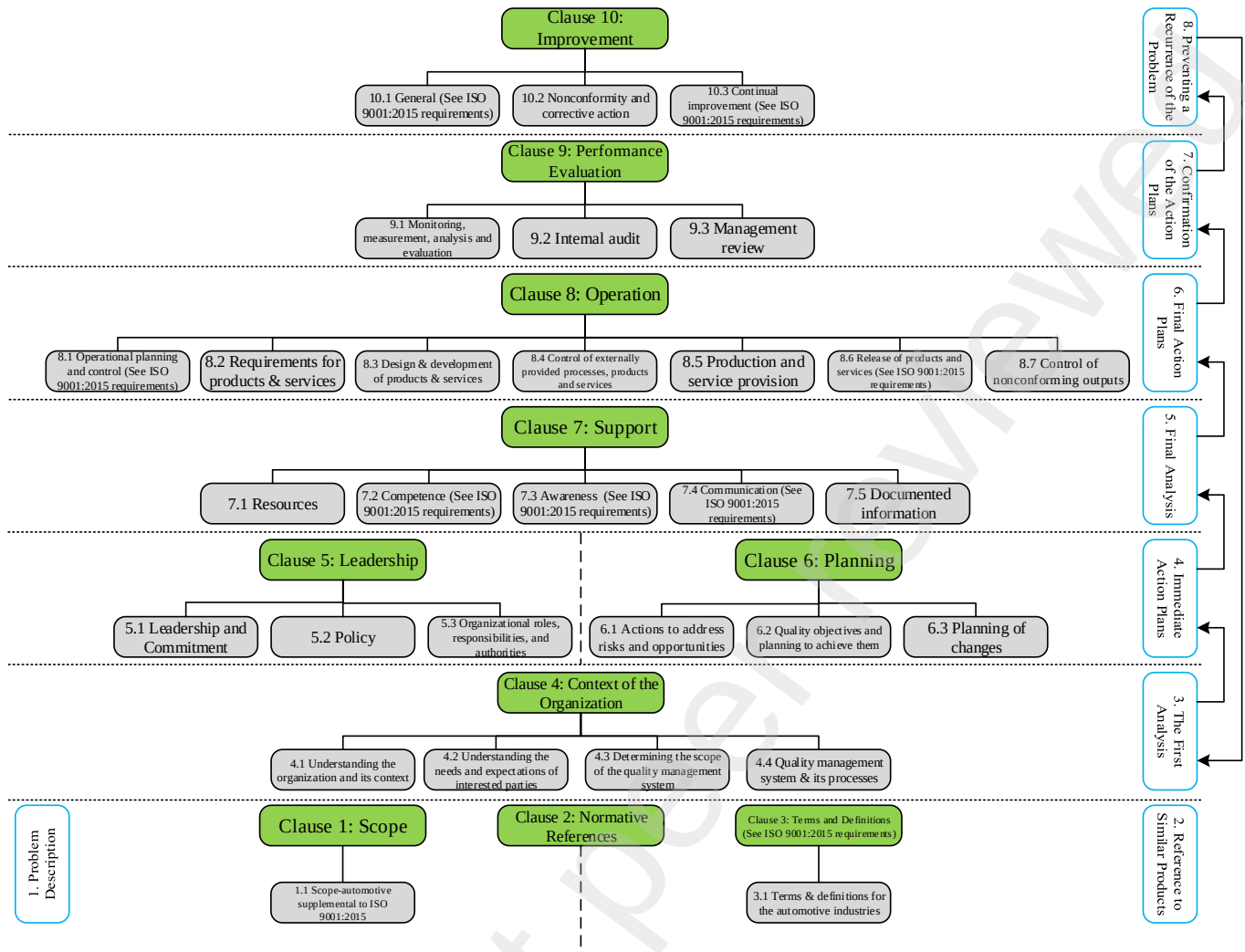


Fig. 1. IATF 19949 Clause and Requirements Arranged Based on 8D Method

According to the IATF model arranged by the 8D method, clauses 1 to 3 or problem description and use of references regarding the 8D method are applied to prepare an automotive company to use our proposed framework (Sections 1 of the model). Also, regarding the conceptual model shown in Fig. 15, clauses 4 to 6 or initial analysis and immediate actions phases in the 8D method are considered for business analysis and recognition (Sections 2 of the model). Moreover, clauses 7 and 8, or the final analysis and actions phases in the 8D method will help to implement the necessary IT infrastructure (Section 3 of the model). Finally, Therefore, clauses 9 and 10, or plans confirmation and preventing a recurrence of the problem phases in the 8D method are used to evaluate and predict quality (Sections 4 and 5 of the model) .

1.4. Data Quality and Predictive Quality Assurance in Automotive Industry

As competition intensifies, the automotive industry is faced with increasingly complex products and reduced product development lead times. Additionally, the growing trend of global alliances necessitates cross-company collaborations, often involving the sharing of common components and parts. These collaborations help companies navigate product variety and maintain competitiveness. Historically, original equipment manufacturers (OEMs) were responsible for creating and managing engineering data. However, this responsibility has increasingly shifted to suppliers. Reports indicate that major OEMs now outsource 60–80% of their physical parts and corresponding digital product data. To manage this complexity, OEMs are adopting integrated enterprise systems, such as C3PE (CAD, CAM, CAE, PDM, and ERP) (Crabb, 1998; Rezayat, 2000, as cited in Yang et al., 2006). As a result, ensuring the quality of data exchanged between OEMs and suppliers is critical for maintaining product integrity and meeting quality standards (Yang et al.,

2006).

The importance of high-quality product data is underscored by the substantial costs associated with poor interoperability. For instance, one U.S. OEM estimates that 453,000 exchanges of product data occur annually across its divisions and between the company and its suppliers. Imperfect interoperability is estimated to cost the automotive supply chain at least \$1 billion annually (Tassey, 1999, as cited in Yang et al., 2006). In Japan, the automotive industry spends approximately \$67 million per year addressing product data issues across 250,000 cases, with each case requiring an average of 1.5 days to resolve (JAMA, 2003, as cited in Yang et al., 2006). DaimlerChrysler reports that 200,000 exchanges of product data occur annually, and 10,000 of these require corrective action due to data quality issues, incurring an additional €5 million in downstream costs. Hyundai Motors estimates that poor interoperability due to product data quality (PDQ) issues costs \$6.8 million annually, excluding delays in product development (Yang et al., 2006).

The costs of engineering changes are particularly impactful. For instance, the cost of an engineering change during the design phase is \$3,500, but if made after the product data have been released to manufacturing, the cost increases by a factor of 10, and 100 times higher if the change occurs after production has started (Yang et al., 2006). Additionally, a significant portion of manufacturing costs is incurred during setup and maintenance, with some automotive programs, such as an engine program, reaching costs exceeding \$1 billion (Harley, 2008, as cited in Neal et al., 2021).

To address these challenges, emerging technologies such as smart manufacturing systems, "Intelligent Products," and the industrial internet of things (IIoT) are being utilized to reduce costs and enhance flexibility (Lee et al., 2013; García et al., 2016; Chaâri et al., 2016, as cited in Neal et al., 2021). These systems enable predictive, self-aware services, including self-

organization, real-time control, and predictive maintenance, contributing to enhanced efficiency and robustness (Monostori et al., 2016, as cited in [Neal et al., 2021](#)). Global initiatives such as Industry 4.0 in Germany and Smart Manufacturing in the United States aim to create next-generation manufacturing systems by integrating networks of distributed components with tightly coupled digital representations, or "twins," to enhance market competitiveness (Davis et al., 2012, as cited in [Neal et al., 2021](#)).

The use of cyber-physical systems (CPSs) and their derivative, cyber-physical production systems (CPPSs), offers potential improvements in production yield and waste reduction, aided by advanced defect detection mechanisms. These systems improve production efficiency, quality, and adaptability (Leitão et al., 2016; Lee, 2015; Barbosa, 2016, as cited in [Neal et al., 2021](#)). In addition, CPSs enable enhanced simulations, real-time manufacturing scheduling, and agile labor resource management, thereby improving responsiveness to demand fluctuations (Schuh et al., 2013; Li et al., 2017; Spath et al., 2013, as cited in [Neal et al., 2021](#)).

The automotive industry is increasingly adopting principles of Industry 4.0 to integrate digital technologies and optimize production processes ([Markulik et al., 2019](#)). Central to this evolution is the application of digital quality management, advanced process and statistical process control (SPC), which are essential for maintaining high-quality production standards. As quality assurance becomes a critical function across the supply chain, it must not only focus on ensuring product quality during manufacturing but also across the entire supply chain network ([Sohail et al., 2022](#)).

predictive quality assurance (PQA) plays a pivotal role in this context, moving away from reactive quality control to a proactive approach that predicts potential issues before they manifest. By leveraging data analytics and machine learning (ML), PQA aims to identify and mitigate product defects early in the production cycle, reducing waste and preventing flawed products from reaching customers ([Nalbach et al., 2018](#)). The integration of digital twin technology within PQA frameworks further enhances the ability to predict disruptions and maintain resilience throughout the supply chain, ultimately contributing to higher product quality and more efficient manufacturing operations.

1.5. The Role of Key Performance Indicators (KPIs) in Predictive Analytics

In automotive manufacturing, productivity management systems play a crucial role in assessing the performance of various processes and activities within the production network. These systems rely on key performance indicators (KPIs) to monitor and track performance

([Thakur et al., 2020](#)). KPIs are vital for understanding and improving manufacturing efficiency by minimizing waste from a lean perspective and aligning operations with the company's strategic objectives. They focus on different dimensions of organizational performance that are critical for long-term success and competitiveness. KPIs are typically measured at different intervals—ranging from real-time monitoring to daily, weekly, or monthly assessments—depending on the specific needs of the process ([ISO 22400, 2014](#)).

The large volumes of historical and real-time KPI data generated in automotive manufacturing create both opportunities and challenges. Advanced analytics techniques, such as machine learning (ML), can help identify previously undetected patterns and trends within this data, enabling manufacturers to make more informed decisions. By predicting KPIs, companies can seize future opportunities and mitigate risks, thereby enhancing business success. The prediction of KPIs is often performed using predictive analytics techniques, including probabilistic models, machine learning algorithms, and statistical methods, which help uncover patterns and forecast future performance based on historical data ([Soltanpoor and Sellis, 2016](#)).

Predictive analytics methods can be broadly classified into two categories: model-based and data-based approaches. Model-based methods rely on theoretical statistical knowledge to define relationships between variables and make predictions, whereas data-based methods leverage machine learning algorithms without relying on predefined models or assumptions. The selection of machine learning algorithms for predictive modeling depends on project-specific requirements, the type of data, the structure of the dataset, the presence of outliers, and the available computational resources ([Kotu and Deshpande, 2014](#)).

In the context of KPIs, predictive analytics can be applied in two main ways: KPI selection and KPI prediction. KPI selection involves identifying critical KPIs that are most closely linked to business objectives, allowing organizations to focus their efforts on improving those metrics. KPI prediction, on the other hand, focuses on forecasting the desired future values of KPIs based on historical data trends ([Thakur et al., 2020](#)). This research adopts the initial KPIs identified in various parts of the automotive supply chain management (SCM), based on the reference model shown in [Fig. 2](#) and aligned with clauses 2 and 3 in the IATF standard as well as step 2 in the 8d method. These initial KPIs will be refined and enhanced with the results derived from the predictive quality assurance framework, particularly after the design and implementation of the digital twin infrastructure.

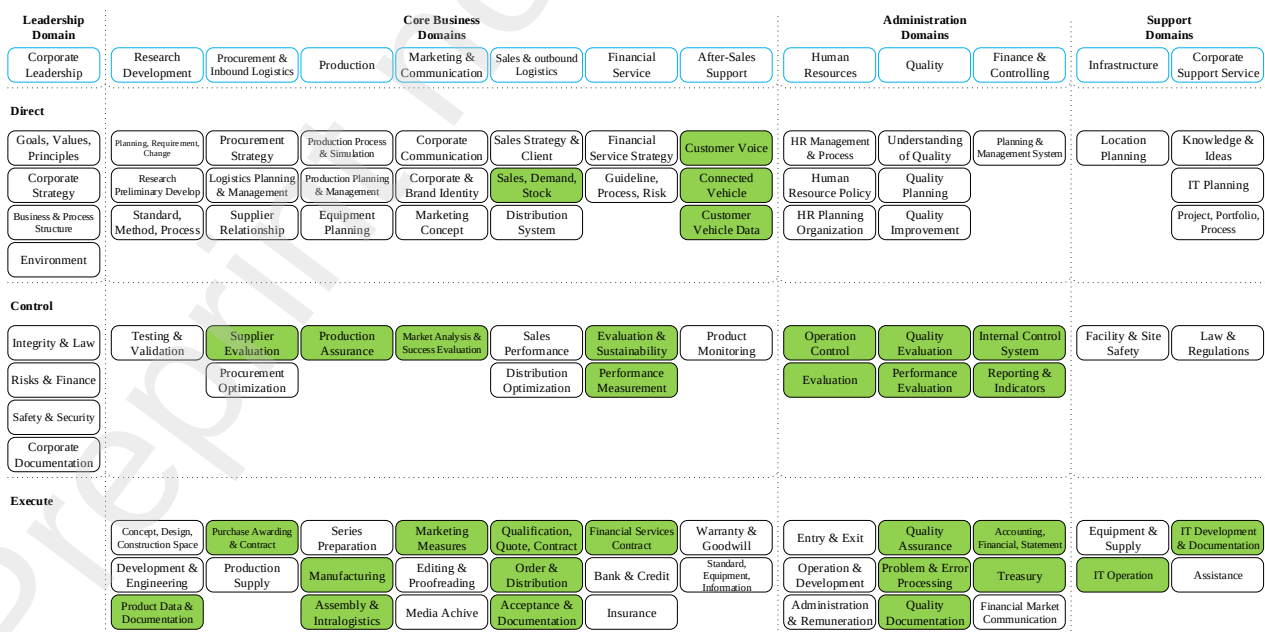


Fig. 2. Reference model for business architecture for today's Automobile industry. It represents an overview of some business competences of a fictitious automotive company which manufactures passenger vehicles ([Wedniwski, 2015](#))

Despite the increasing relevance of KPI prediction in industry, there remains limited research in this area, with fewer than ten studies

focusing specifically on KPI prediction within the industrial sector using predictive analytics. One possible explanation for the scarcity of research is the reluctance of companies to share confidential internal KPI data (Thakur et al., 2020).

Effective SCM is a critical factor for automotive companies to remain competitive, especially in an industry characterized by rapid technological advancements and fluctuating market demands. Traditional SCM approaches, which rely heavily on historical data and static models, are increasingly inadequate for managing the dynamic nature of modern automotive supply chains. The demand for real-time insights, adaptive strategies, and predictive capabilities has never been more pressing. Consequently, the integration of advanced technologies to optimize supply chain operations has become a strategic necessity for automotive firms aiming to achieve operational excellence and meet evolving market demands (Putha, 2022).

Examples of predictive analytics applications within the supply chain include real-time logistics and supply chain control, inventory management using sensing data, dynamic resource allocation in Industry 4.0 environments, and improved forecasting models using big data and machine learning techniques (Ivanov et al., 2019). These methods help improve supply chain visibility, risk control, and decision-making by integrating optimization algorithms with machine learning. According to Ivanov et al. (2019), supply chain competition will increasingly depend on the ability to apply analytics algorithms, optimization methods, and simulation modeling effectively.

However, before applying any analytical methods or simulation models, it is essential to first understand the business processes involved. A thorough analysis of processes helps identify bottlenecks, which can then be addressed by defining relevant KPIs for each specific bottleneck or department. Once the KPIs are established, advanced analytics and optimization algorithms can be applied to diagnose issues, predict outcomes, and recommend solutions, following a structured approach of description, diagnosis, prediction, and prescription.

1.6. Digital Twin and Technological Evolution

The pervasive adoption of analytics and its integration with operations research (OR) have demonstrated that simulation and optimization are not only key for modeling physical supply chain (SC) systems but also crucial for simulating cyber supply chain systems and deriving insights from them. A decision-support system that combines simulation, optimization, and data analytics can provide proactive solutions for SC risk management and disruption prediction. This approach, illustrated in Fig. 3, highlights how such a system can help anticipate disruptions and adapt the SC structure in real time (Ivanov et al., 2019a).

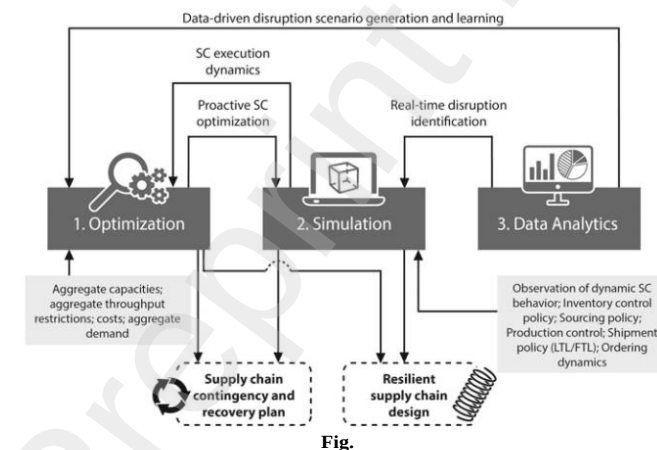


Fig.

3. Concept of a Decision-Support System for Supply Chain Risk Analytics (Ivanov et al., 2019a)

The decision-support system for SC risk analytics is designed to optimize SC resilience by proactively simulating potential disruptions and providing real-time decision-making support. It integrates simulation and optimization

models to simulate SC behavior under disruption scenarios, enabling predictive simulations of disruption impacts and recovery strategies. These simulations are further optimized using analytical models. The data analytics component, powered by real-time feedback (from sensors, RFID, etc.), is critical for identifying disruptions as they occur and feeding the data into the simulation models for more accurate recovery policy predictions. Additionally, data analytics supports the proactive stage by generating potential disruption scenarios to help design resilient supply chains (Ivanov et al., 2019).

This combination of simulation, optimization, and data analytics forms the foundation of a digital twin, a powerful, data-driven tool that helps manage SC disruption risks (Khan, 2024). As depicted in Fig. 4, a digital SC twin is an advanced model that mirrors the real-time state of a supply chain. By integrating data from multiple sources—sensors, IoT devices, and historical records—this virtual representation helps identify vulnerabilities and anticipate disruptions. This proactive approach allows companies to test various contingency plans and make informed adjustments to improve resilience (Ivanov et al., 2019; Khan, 2024).

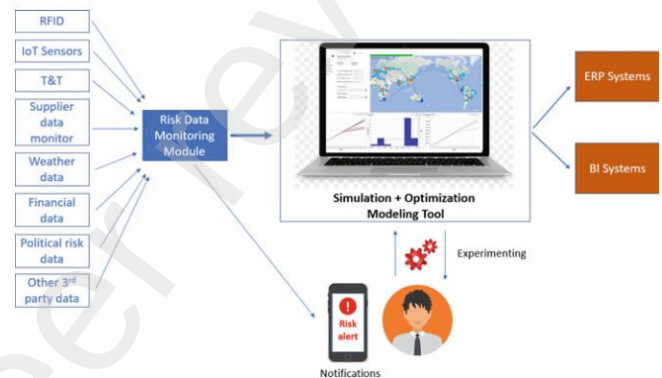


Fig. 4. Supply Chain Digital Twin (Ivanov, 2018c, as cited in Ivanov et al., 2019)

A digital twin is a virtual replica that offers continuous, real-time visibility into the supply chain, enabling businesses to monitor every step of the supply process—from suppliers to manufacturers to distributors and customers. This end-to-end visibility helps detect potential issues, optimize operations, and improve overall supply chain performance. Digital twins are becoming increasingly popular in SC management because they enhance transparency and enable real-time, data-driven decision-making (Rauniyar et al., 2023; Singh et al., 2023).

Belhadi et al. (2021) describe a digital twin as a tool for real-time monitoring, analysis, and optimization of physical products, processes, or systems. For instance, in manufacturing, a digital twin can track inventory levels, monitor production efficiency, and detect bottlenecks, all while supporting the identification and resolution of supply chain inefficiencies (Ivanov, 2024). Furthermore, digital transformation enables improved collaboration among supply chain partners, enhancing communication, reducing errors, and ensuring goods are delivered on time and to the correct locations (Rana and Daultani, 2023).

Digital twins leverage various technologies, such as IoT, sensors, and big data analytics, to create a virtual representation of the physical supply chain. This virtual model not only aids in real-time control but also supports long-term strategic decisions. By continuously comparing the digital twin's simulation with real-world performance data, companies can optimize their operations, predict and manage risks, and improve overall supply chain resilience (Kamle et al., 2022; Ivanov et al., 2019).

For example, the digital product passport (DPP) has emerged as a critical concept in industries like automotive manufacturing, with applications for tracking a product's lifecycle and ensuring compliance with standards. A notable example is the battery passport required by the European Union for electric vehicle batteries, which integrates information about the battery's composition, performance, and environmental impact. Digital twins play a key role in developing and managing such digital passports, offering a comprehensive, standardized record for product performance (GBA, 2024).

Additionally, the development of autonomous vehicles relies heavily on digital twin technology. Using IoT sensors, radar, and machine learning algorithms, manufacturers can simulate vehicle performance under various

conditions throughout its lifecycle. These simulations, which include environmental interactions, system responses, and user behavior, contribute to refining vehicle designs and improving performance before the vehicles are physically produced (IBM, 2021).

While digital twin technology is still an emerging area for supply chains, its implementation is showing promise, especially in addressing disruptions and optimizing supply chain operations (Khan, 2024). However, challenges remain. One of the significant hurdles is the ripple effect—the cascading impact of disruptions that spreads across the supply chain, often amplifying the severity of problems in downstream processes. This phenomenon, which can cause structural changes in the supply chain, requires effective modeling to understand its dynamics and mitigate its impact (Dolgui et al., 2018).

To effectively manage these challenges, the proposed research will focus on developing a framework for predictive quality assurance (PQA) utilizing digital twins in the automotive industry. The objective is to create a predictive model that addresses the specific performance areas of the supply chain and develops KPIs suitable for each area, integrating them into the digital twin for real-time risk management and decision-making.

This research aims to bridge existing gaps in digital twin applications by pursuing the following objectives:

Objective 1: Identify the most critical factors for implementing digital twins in the automotive industry's resilient supply chain, based on the IATF standards and reference models.

Objective 2: Design a conceptual framework for predictive quality assurance through digital supply chain twins, utilizing the proposed eight-phase, five-stage framework.

The structure of this paper includes a detailed discussion of digital twin applications across industries, the challenges in implementing resilience strategies in automotive supply chains, and the methodologies used in predictive quality assurance. Sections 3 and 4 outline the conceptual model of the framework, while Section 5 explores its integration with existing automotive supply chain processes. Section 6 examines the framework's impact on resilience, and Section 7 offers conclusions, practical implications, and future research directions.

2. Literature review

2.1. Digital Twins in Supply Chains

An initial observation during our data collection shows that scholars' works for Physical Internet (PI) and digital twin developments in SC and operations management have surged extraordinarily in recent years. However, despite the exponential growth, the extant literature remains largely unstructured and scattered across various disciplines, thus limiting the knowledge transfer and innovation of the subject area. For this reason, it is necessary to have systematic literature reviews which can synergize current research findings, detect research forefronts and draw on future directions (Nguyen et al., 2022).

This section provides detailed descriptions of several studies and papers related to the use of digital twin technology in various supply chain contexts. According to examined studies that their results have been summarized in Table 3, the most focus is on its application in improving resiliency, efficiency, and sustainability in supply chain operations. Regarding the research conducted, it is clear that most of the studies are from recent years and some studies have examined a specific industry case study. Moreover, the quality of the supply chain has been the missing link in previous research. Given the importance of the automotive industry supply chain as a useful indicator for measuring the current wealth of the international economy and the use of digital twin to make it resilient to the disruptions described in the previous section, it is necessary to study the history of this convergence in detail. As mentioned, we summarized key points from the studies and outlined specific use cases to show the potential of digital twin technologies in modern supply chains.

2.1.1. Practical Applications and Case Studies

Many of the studies review real-world applications of digital twins in industries ranging from manufacturing to logistics, including the food supply chain, medical supplies, and even the wind turbine industry. These studies focus on how digital twins optimize processes such as inventory management,

logistics, and even environmental factors like temperature control in the cold chain. The studies cover a variety of industries and use cases, such as semiconductor, pharmaceuticals, manufacturing and most of all about food, especially agricultural food (Agri-Food) like, coffee, fruit and vegetable. Chart 1 shows the frequency of the case studies presented in Table 3.

- Semiconductor Supply Chain: Using DTs to manage disruptions and enhance resiliency by conducting experiments and scenario testing via the Anylogistix software platform (KEK et al., 2023).
- Pharmaceutical Supply Chain: Demonstrating how DTs can reduce the ripple effect in the pharmaceutical supply chain, enhancing agility and response to disruptions (Chen, 2024).
- Sustainability in Manufacturing: DT applications to enhance sustainability in manufacturing supply chains, integrating IoT, cloud computing, and blockchain technologies (Kamble et al., 2022).
- Coffee Value Chain: Integrating DTs for real-time monitoring and optimization, ensuring quality control, and considering sustainability in the coffee supply chain (Kamble et al., 2022).
- Industrial Symbiosis: Exploring the application of DTs in industrial symbiosis networks, with a focus on user needs and the design of digital twins to support collaboration (Liu et al., 2023).
- Food Supply Chains: Using DTs for real-time planning, monitoring, and optimization in food supply chains, considering procurement, production, and distribution strategies. Also, some studies have taken advantage of leveraging DTs to enhance resilience and sustainability in food supply chains through a grey causal modelling approach (Singh et al., 2023).

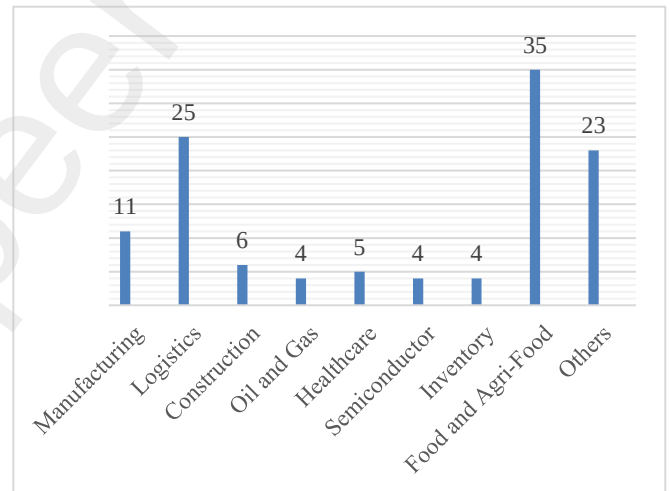


Chart 1. Frequency of the Case Studies in Integration Digital Twin and Supply Chain

Each study presents different methods, frameworks, and technologies that can be used to improve various aspects of supply chain management, from prediction and optimization to ensuring sustainability and resilience in the face of disruptions. These innovations are crucial for creating more efficient and agile supply chains in diverse industries.

2.1.2. Digital Twin Frameworks and Methodologies

Several studies present conceptual frameworks that guide the adoption and application of digital twins in supply chains, often structured in multiple layers (e.g., SCDT interaction and cognition layers). These frameworks aim to align technology, scope, and organizational factors to improve performance across supply chain processes. In the following, we present several examples to clarify our proposed model.

According to the principles described by Ivanov and Dolgui (2021b), the following framework for SC resilience analysis and stress-testing can be generalized (Fig. 5) (Ivanov, 2023).

Fig. 5 shows that a digital twin integrates different data sources from external systems, SC models, and performance analysis for SC stress-testing. Moreover, it illustrates a dynamic deployment of the modelling in the SC stress-testing. External disruption data and decision-maker knowledge build scenarios for stress-testing. At the proactive stage, the digital twin uses statistical data in optimization and simulation models for resilience analysis and stress-testing. For example, an alternative SC design and back-up logistics can be analyzed.

With the help of discrete-event simulation, disruptive events build dynamic scenarios to assess the impact of disruptions and the ripple effect on SC performance. The post-processing of results is implemented using some external performance visualization software (Ivanov, 2023).

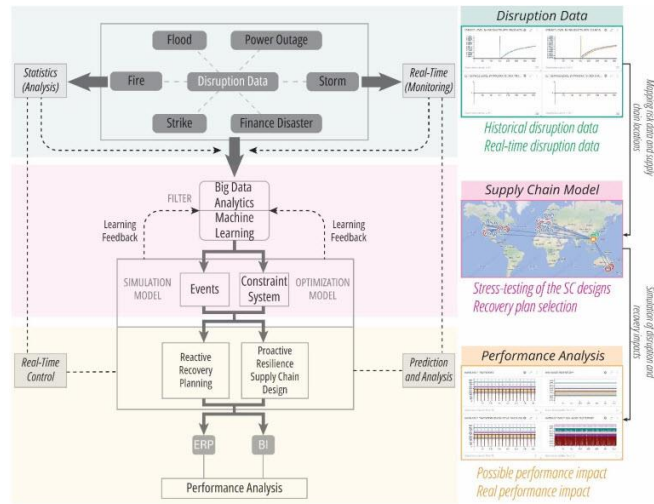


Fig. 5. Supply Chain Stress-Test Framework Using a Digital Twin (Based on Ivanov and Dolgui (2021b)) (as cited in Ivanov, 2023)

As Fig. 6 shows, in the model presented, Ivanov (2024) has considered the seven major elements of a digital twin in SC and operations management are defined as technology, people, management, organization, scope, task, and modelling.

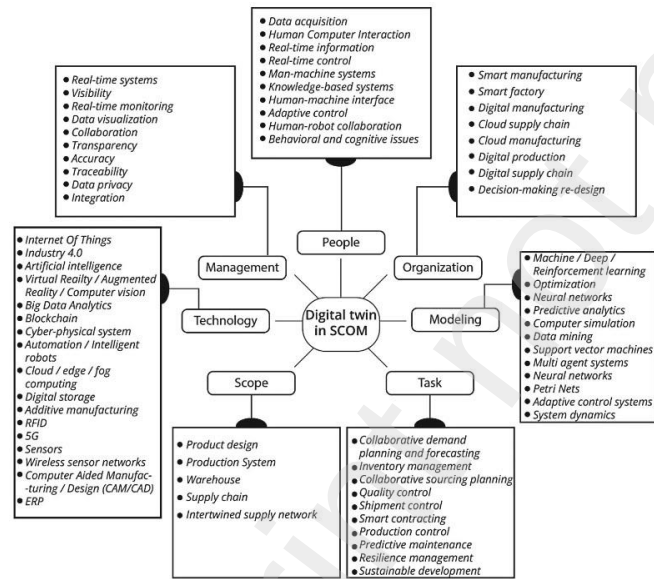


Fig. 6. Seven-element SCOM digital twin framework (Ivanov, 2024)

It also distinguishes five major types of digital twins in Supply Chain and Operations Management (SCOM): product, process, organization, supply chain and network-of-networks. Illustration of a SCOM digital twin is provided using an Anylogistix example. It concludes that digital twins in SCOM are not merely a simulation-based replica of a real object but a complex socio-technical phenomenon involved in continuous human-artificial intelligence interactions. This leads to an understanding of the role of digital twins through the lens of Industry 5.0, reconfigurable and viable supply chains (Ivanov, 2024).

In another example, according Fig. 7, Wang et al. (2022) highlights a motivating case from JD.COM, China's largest retailer by revenue, in applying the DTSC platform to address supply chain network reconfiguration challenges during the COVID-19 pandemic (Wang et al., 2022).

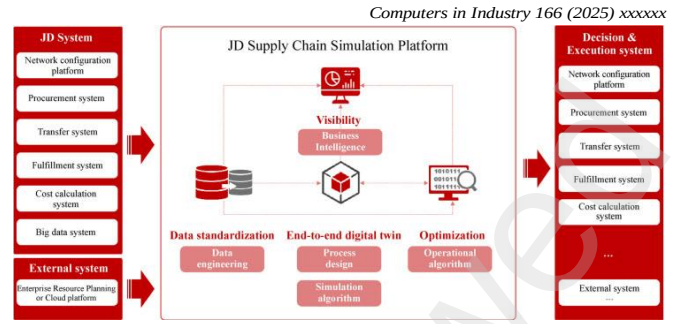


Fig. 7. Framework of JD.COM's DTSC platform (Wang et al., 2022)

As shown in Fig. 7, the data sources used to synchronize the physical supply chain and the digital models include network configuration platform, procurement system, transfer system, and fulfillment system. Optimization algorithms and simulation algorithms work together to support integrated supply chain design and intelligent operations. By directly connecting with decision and execution systems, these optimized insights can be quickly implemented to respond to changes. Meanwhile, insights are visualized for better understanding. The DTSC platform is expected to support a wide range of business decision-makings in JD.COM, from long-term strategies, to mid-term plans and short-term operations. The next subsection takes the supply chain network reconfiguration as an example (Wang et al., 2022).

According to Cimino et al. (2024) research, the framework proposed in this paper arises from the need to support global and local supply chains in achieving sustainability and resilience goals. The main objective is to increase their level of readiness in the face of unexpected and disruptive events (e.g., COVID-19 pandemic, wars, interruptions in the supply chain flow such as the case of the ship stranded in the Suez Canal in 2021, etc.). It is aimed at supply chain managers and has a holistic and cyclic nature. The framework is also visually represented in Fig. 8. Basically, it is characterized by some fundamental steps, listed and described below (Cimino et al., 2024):

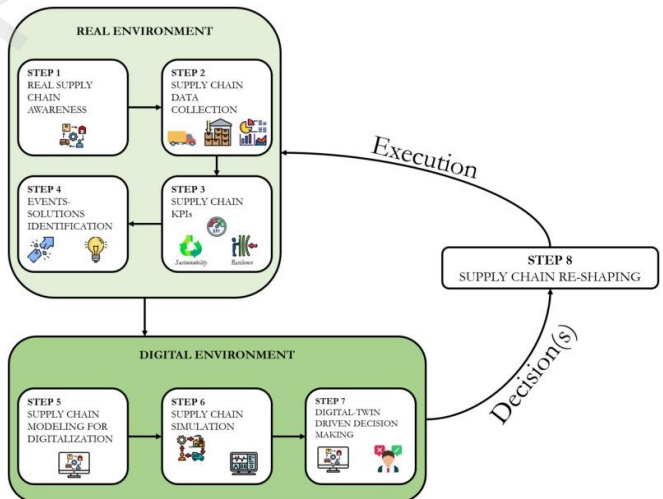


Fig. 8. Framework of JD.COM's DTSC platform (Wang et al., 2022)

The first 4 steps of the framework refer only to the existing physical supply chain. The next steps describe instead address the digital twin of the real supply chain. As can be seen, the first 4 steps refer to the real environment and represent the foundation for the next 3 phases, which are carried out within the digital environment. Finally, there is a phase of supply chain re-shaping which jointly represents the output of the digital environment and the input of the real environment. Basically, based on the analyzes conducted through the use of simulation, it is possible to make decisions that have an impact on the real environment (47).

Fig. 9 shows a framework of digital twin-driven smart lean supply chain management developed on the basis of the four-layered digital twin

reference architecture (Guo et al. 2023, as cited in Guo and Mantravadi, 2024) and the seven-element digital twin framework (Ivanov, 2024, as cited in Guo and Mantravadi, 2024).

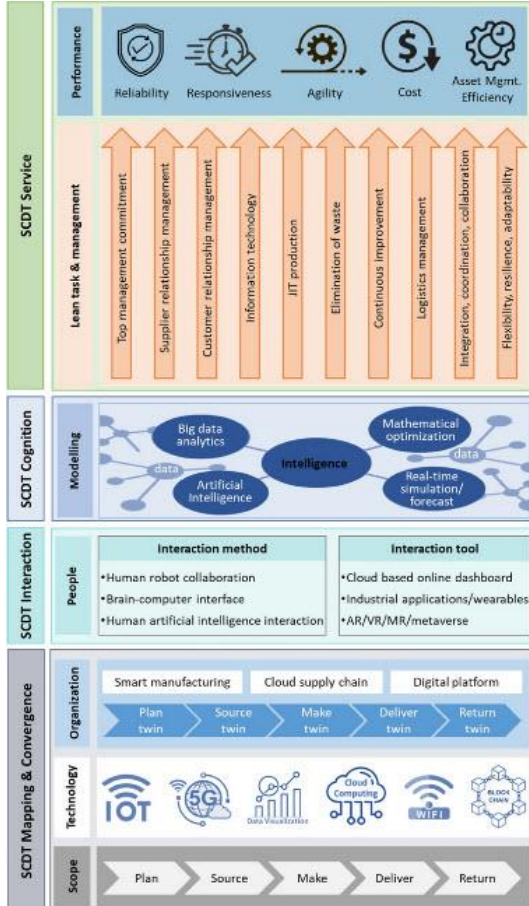


Fig. 9. A Developed Framework of Digital Twin-Driven Smart Lean Supply Chain Management (Guo and Mantravadi, 2024)

The proposed framework incorporates seven major elements of scope, technology, organization, people, modelling, management, and task into four layers: the supply chain digital twins (SCDT) mapping and convergence layer, SCDD interaction layer, SCDD cognition layer, and SCDD service layer. Based on the four-layered digital twin reference architecture, the SCDD mapping and convergence layer acts as the technical infrastructure of the framework, focusing on real-time perception and mapping of a physical supply chain into a digital twin while maintaining convergence between them. The SCDD interaction layer supports real-time interaction between the SCDD and its physical counterpart. The SCDD cognition layer acts as the brain of the framework by offering digital twin-driven insights from prescriptive and predictive analytics. The SCDD service layer augments the capabilities of lean task and management for enhancing supply chain performance metrics such as reliability, responsiveness, agility, cost, and asset management efficiency (Guo and Mantravadi, 2024).

One of the few frameworks proposed in the automotive industry is the model proposed by Khot and Thiagarajan (2024). According to Fig. 10, it presents developing a resilient and sustainable non-linear closed-loop supply chain management framework for the automotive sector industry using a Gaussian fuzzy optimization based non-linear model predictive control approach.

Also, this research proposes a novel non-linear closed-loop supply chain management framework tailored to the tire industry supply chain from the automotive sector to address the issue of exploring interrelationships. Framework employs trapezoidal linguistic cubic fuzzy Z-score technique for order of preference by similarity to the ideal solution ranking approach to prioritize resilience strategies to maintain sustainability performance during sudden disturbances. Furthermore,

Gaussian fuzzy optimization-based non-linear model predictive control acts as a feedback controller to integrate sustainability and resilience by providing a stable output based on the objective function related to sustainability dimensions. An experimental study assesses the impact of resilience strategies on total supply chain costs, highlighting significant cost savings. Adopting strategies like multiple sourcing, information sharing, and improved design quality of the supply chain keeps total expected costs optimal for various sustainability levels.

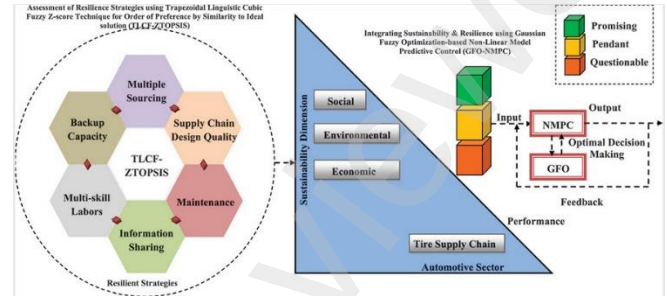


Fig. 10. Non-linear closed-loop supply chain management framework tailored to the tire industry supply chain (Khot and Thiagarajan, 2024)

2.1.3. Challenges and Research Directions

A number of the studies also highlight the challenges that come with implementing digital twins, such as information asymmetry, the complexity of integration, and the need for substantial infrastructure. These works propose future research directions, aiming to improve digital twin applications and resolve current gaps. We will discuss implementation challenges in section 5, but some challenges are :

- Data Privacy and Ethical Concerns: Addressing the challenges related to data privacy, security, and ethical considerations in the use of digital twins, especially when integrating sensitive business and customer data (Zhen and Yao, 2024).
- Costs: Companies would like to keep inventories at a level that ensures smooth order fulfilment while minimizing their own costs (Mielczarek et al., 2023).
- Data Quality: Most State Departments of Transportation (DOTs) use fixed acceptance thresholds to evaluate data quality at the network level and project level (Tao et al., 2023).

2.1.4. Emerging Trends

The intersection of digital twins with other technologies, such as blockchain, AI, and the metaverse, is also discussed in the feedback. These integrated technologies are seen as critical to improving supply chain resilience, transparency, and sustainability, especially in the context of disruptive events like pandemics.

- Blockchain and DTs: Analyzing the integration of blockchain with DTs to improve data integrity, traceability, and security in supply chain management (Roumeliotis et al., 2024).
- AI-Driven DTs: Investigating how AI-powered digital twins can analyze real-time data to predict events, optimize decision-making, and enhance operational efficiency (Mandala et al., 2020).

2.1.5. Commonly Used Approaches

A comparative analysis of different methodologies (e.g., simulation vs. machine learning approaches) across the studies could provide insights into the strengths and weaknesses of each approach in specific supply chain contexts. This would aid practitioners in choosing the most suitable methods for their particular challenges. Several approaches are shown in Table 3 that researchers have used for various purposes .

- Graph-Enabled Digital Twins for PLM: A multi-dimensional approach to product lifecycle management (PLM) with a focus on design, manufacturing, and supply chain transformation (Lim, 2023).
- Quantum Approach in DTs: Applying quantum-classical hybrid approaches to manage uncertainty within supply chain DTs and financial risk assessment (Abdullah et al., 2024).
- Business Process Simulation and Optimization: Presenting a framework for simulating and optimizing business transactional

processes in supply chains using DTs (Perez et al., 2021). Gerlach (2023) according to Fig. 11 has used an exemplary comparable process model the Cross Industry Standard Process for Data Mining (CRISP-DM) presented by Herden (2020) (as cited in Gerlach, 2023), which is being used widely in the industry as a guideline for conducting analytics initiatives. Combining and adjusting existing process models leads to a process model for conducting DSCT initiatives.

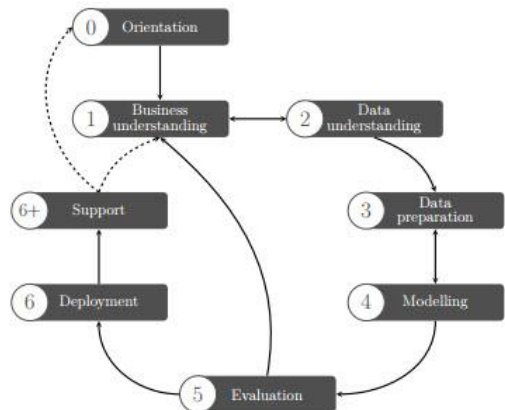


Fig. 11. sCRISP-A Approach Presented by Herden (2020)(as cited in Gerlach, 2023)

The sCRISP-A features eight phases instead of six, extending the CRISP-DM by the orientation phase (phase 0) as well as the support phase (phase 6+). Additionally, specific tasks and actions are presented for every phase to guide users over the course of the Supply Chain Analytics (SCA) initiative (Gerlach, 2023).

Table 3
Examining Integration Between Supply Chain and Digital Twin

Study	Application Field	Purpose	Concern	Approach	Procedure	+ DT
(Singh et al., 2023)	Manufacturing	Measuring Sustainability	Output and Efficiency	Identifying the Mediating role of SCR and SCP upon SC Sustainability	Partial Least Square Structural Equation Modeling (PLS-SEM) and Collecting Data from Respondents Using Closed-Ended Questionnaires	
(Khan , 2024)		Improving Resilience	Global Diversification Risk, Continuity and Competitiveness	Thesis, Conceptual Framework, Designing a Phase to Integration and Modeling	Machine Learning (ML) and Simulation Models, and Validating the Model by Case Study	
(Ivano v et al., 2019)		Improving Resilience and Test Contingency by Integrating Lean Thinking	Completeness, Fullness, Validity, Consistency, Timely Availability and Ripple Effect	Low-Certainty-Need (LCN) Framework for Risk Analytics by Data-Driven Optimization, Simulation, and End-to-End SC Visibility		Blockchain
(Nguy en et al., 2022)	Logistics	Bibliometric Knowledge Mapping	Business Ecosystem, Sustainability, Efficiency, Complexity, Downstream Management, Cognitive Thinking in Industry 5.0, Citizen Twin in Digital Society and Resilience	Systematic Literature Review by Clustering 10 Key Research Streams on PI/DT Applications in: Job Shop Scheduling, Smart Manufacturing Design, PI-Based SCM, Manufacturing Virtualization, Information Management, Sustainability Development, Data Analytics, Manufacturing Operations Management, Simulation and Optimization, and Assembly Process Planning		PI
(Abid een et al., 2021)		Building Prescriptive Decision Support Platforms by Reinforced ML	Flexible, Smart, Agile and Lean Production, and Lead Time	Data-Driven Framework and Systematic Review	Strict Keyword Selection, Search Strategy, and Exclusion Criteria in Scopus Database	
(Bhan dal et al., 2022)		Taking Stock of the Emerging Research Stream Identifying Trends, and Capturing the Value Potential for Operations and SC Management	Business Processes and SC Capabilities	Bibliometric Literature Review by Identification of Four Value Clusters and one Enabler Cluster	Bibliographic Coupling and Keyword Co-Occurrence Network Analysis	
(Gai et al., 2022)		Reengineering SCM	Energy and Time Savings	Designing Intelligent Switch-Based Algorithms	Comparing Proposed Algorithm with Three Baseline Algorithms, Including Effective Iterative Greedy (EIG), Two Dimensional Genetic (TDG), and High-level Task Scheduling Dynamic Programming (HTSDP)	Blockchain
(Ivano v, 2023)		SC Stress-Testing, Resilience, and Viability by Intelligent DT (iDT)	Ripple Effect	Generalized Decision-Making Framework by Building Dynamic Scenarios	Analytics Methods, Mimics Human Decision-Making Rules, and New Decision-Making Algorithms	
(Bakh shi et al., 2024)		Overcoming Construction SC (CSC) Challenges	DT Applications	Extensive Literature Review	Investigating Examples of Other Industries, Utilizing Scopus CSC-Related data, and Employing VOSviewer to Extract and Visualize CSC Hotspots	
(Ivano v, 2024)		Conceptualization of A 7-Element DT Framework in SCOM	Seven Major Elements: Technology, People, Management, Organization, Scope, Task, and Modelling. Five Major Types: Product, Process, Organization, SC, and Network-of-Networks	Framework and Conceptualization	Lens of Industry 5.0, Reconfigurable, Viable SC, and Anylogistix	
(Ogun soto et al., 2024)		SC Recovery and Resilience to Anticipate Floods and Predict Recovery Indicators	Natural Catastrophe and Hydrological Disruptions	Three-Phase Conceptual Framework	ML Algorithms, Including Logistic Regression and Long Short-Term Memory (LSTM), Flexsim for model service-level failures, Multilayer Neural Perceptron Network (MLPNN) for Predicting Production Network Recovery Post-Disruptions, and Mean Squared Error (MSE) for validation	Integration of Simulation Software, ERP, ML, and IoT
(Peron , 2024)		Transition from Traditional Spare Parts SCs (TSPSCs) to Digital Spare Parts SCs (DSPSCs) for Availability	Overcoming Lack of Expertise in Three Main Areas: How to Design, Operate, and Control and Monitor			Additive Manufacturing (AM)
(Wan g et al., 2022)	Retailer	Creating DT-Driven Smart SC	COVID-19	Discussing Three Research Opportunities: SC Modelling, Real-Time SC Optimization, and Data Usage in SC Collaboration	JD.COM Platform for SC Network Reconfiguration Challenges	
(Rah manza deh et al., 2023)	Clothes Manufacturing Industry	Conceptualization of Open SC Management (OSCM)		Open Innovation	Fuzzy Tactical Planning Model for Epistemic Uncertainty and Validating by Investigating a Subset of OSCM	

Study	Application Field	Purpose	Concern	Approach	Procedure	+ DT
(Perez et al., 2022)		Online Optimization of Business Processes	Orders Flowing	Framework with Bringing Together Contributions from Digital SC, Discrete Event Simulation, Business Process Management Notation, Queueing Systems, and Online Optimization	Stochastic Simulation and Feedback Loop	
(Park et al., 2021)	Logistics	Resilient SC Control	Make-To-Order (MTO) Environment and Bullwhip and Ripple Effects	Multi-Level Architectural Framework of a Cyber Physical Logistics System (CPLS)	Dividing Simulation into Type and Instance Stages, and suggesting operation procedures of the DT application and technical functionality of the DT engine	
(Abou zid and Saidi, 2023)	Manufacturing Factory	Modelling and Simulation of Business Processes for Technology and System Evaluation at the Process Study Stage, and Demonstrating Stylized Business Case Application in Real-World Scenarios	Order Fulfilment	Conceptual Implementation Method	(BPMN) to Simulate "As-Is" and "To-Be" Processes	
(Bada khshan and Ball, 2024)		SC Master Planning Maximizing SC Service Level and Minimizing Total Cost under Disruptions	Demand and Lead Time Disruptions	Framework for Integrating Simulation, ML, and Optimization to Identify the Production, Storage, and Distribution Values	Hybrid Modelling	
(Galle go-Garcia et al., 2023)	Agri-Food and Farmers	Increasing Resilience of Smallholders Through the Improvement and Optimization of Agri-Food SC, and Related Environment of any Smallholder Organization Considering all Related Stakeholders, and Securing Viability in the Long Term	Internal or External Risks in Three Levels: Farmer's Organization, SC, and the Related Environment Based on the Corporate Social Sustainability, Stock Levels, and Feedback Loops and Information Flows, and COVID-19	Framework Derived from Five Pillars: Organization, Human Factor, Process Management, New Technologies, Product, Process Improvement, and Their Modelling with Simulation Capabilities	System Dynamics Modelling to Test Model Constructs and Potential Scenarios, Vensim as Simulation Software	
(Heza m et al., 2024)	Gas Company in Egypt	SC Sustainability Risk Assessment and Management in Supplier Selection	Risks: Supply, Environmental, Financial, Regularity, Political, Ethical, and Technology, and Their Sub-Criteria	Fuzzy Framework	Spherical Fuzzy Numbers (SFNs) to Evaluate the Criteria and Suppliers And Multi-Criteria Decision-Making (MCDM) Under a Spherical Fuzzy Set (SFS) Environment to Overcome Uncertainty and Incomplete Data in the Assessment Process, and SFS-Combinative Distance-Based Assessment (CODAS) Method to Rank the Supplier, and Comparative Analysis with Other MCDM Methods	
(Yevu et al., 2023)	Construction, Building Projects	Prefabrication SC for Smart Construction and Carbon Emissions Evaluation	Information Requirements for Materials/Components, Machinery, and Processes	Two-Steps Mixed-Method Review: Scientometric and Qualitative Analysis	Integrating Emissions And Energy Control, Artificial Intelligence-Based Decision-Making and Blockchain for Prefabrication, and RFID, GPS, Laser Scanners and Sensors at The Production, Transportation, And On-Site Assembly Stages of PSC for Buildings	Blockchain
(Zhen and Yao, 2024)		Impact of DT Technology (DTT) and Blockchain Technology (BCT) on Achieving Sustainable and Efficient SC Operations	Innovation and Sustainability, Resource Optimization, Waste Reduction, and Barriers: Data Privacy, Scalability, and Interoperability	Offering Insights into the Mediating Role of Sustainable SC Management (SSCM) Practices and Dynamic Capabilities in the Relationship Between Industry 5.0 Technologies and Operational Resource Utilization (ORU) Performance	Examining Real-World Case Studies Across Various Industries	Blockchain and Industry 5.0
(Tebaldi et al., 2021)	Agri-Food	Assessing the State-of-Art		Literature Review by Descriptive Metrics: Year of Publication, Research Type, Geographical Origin and Keywords Analysis, and Qualitative Aspects: Subdivision According to the SC Phase Involved and Data-Driven Versus Physic-Based Modelling of the DT Solutions Presented in the Documents		
(Oehlschlager et al., 2024)		SC Optimization	Customer Demands	Framework Model with Three Hierarchical DT Levels with Different Degrees of User Integration and Examines Determinants for Their Respective Acceptance	Quantifying the Impact of Radio-Frequency Identification-Enabled DT from the Business Benefits Perspective, and Assessment Identifying Key Contribution Areas and the Dynamics Explaining How the Benefits Are Expected to Land on a Business Level	
(Cimino et al., 2024)	Oil and Gas	Enhancing Internal SC (ISCs) Management	Flexibility and Scalability	Modular Architecture for Designing, Develop Test and Validate a Multi-Plant Simulation-Based DT Production Planning Platform for ISCs Management	Object-Oriented Structure and Enables What-If Analyses, Involving Several Scheduling Rules and ISC Configurations, Make-To-Order Production Strategy and Comparison with a Baseline Scenario and Testing and Validating Through a Case Study	
(Sani et al.,		Improving SC Visibility		Framework	Genetic Algorithm (GA)	

Study 2024)	Application Field	Purpose	Concern	Approach	Procedure	+ DT
(Cimino et al., 2024)	Agri-Food	More Resilient and Sustainable Future with Increasing Readiness for Unexpected and Disruptive Events		Eight-Stages Framework with Cyclic and Holistic Methodology	Simulation, SC KPIs, and Testing and Validating Through a Case Study using Anylogistix Software	
(Zhang et al., 2023)	Healthcare, Vaccine Logistics SC (VLSC)	Improving Environmental, Social, and Governance (ESG) Evaluation		Analyzing Systematically the Influence of DT Service (DTS) On ESG Evaluation and Analytically Investigating the Long-Term Behavior of Sustainability-Concerned Stakeholders in the VLSC	Evolutionary Game Analysis in a Two-Tiered VLSC Including Cold Chain Logistics Service Providers (CCLSPs), the Public, and Government Regulators	
(Almazmi et al., 2024)	Borouge Petrochemical Company	Optimizing Inventory Holding Point (IHP) Selection Between Al Ruwais and the Destination Port	Minimizing Network Cost Leading to Better, Visibility, Capability, Autonomy, and Cost Savings	SC Network Optimization and Simulation Model	12-Month SC Baseline Information, Optimized Network, Transportation Modeling, Logistics Cost Information, Constraints, Scenarios, Frequency, and Processes: Ingestion of Data, Cleansing and Transformation of Data, Model Building, Scenarios Building and Transforming Model Output to User Consumable Reports	
(Maheshwari et al., 2022)		Measuring Optimal Inventory Policy (OIP)	Cost and Customer Satisfaction	Agent-Based Simulation Method to Create Administrative Asset Shells for Small and Medium-Scale Factories	OIP Optimization Method and Essential Measures for Developing the SCDT Distribution Strategy	
(Valero et al., 2022)	Food, Circular Meat SC	Reducing Food Waste	Zero-Waste	Conceptual Framework for Integrating Key Technological Enablers Across Different Areas		
(Nguyen et al., 2023)	Semiconductor	Reinforcing SC Resilience	Shortages, Ripple Effect, and Operational and Disruption Risks	Balanced Approach	Simulation of Semiconductor Shortages with Combining Case Study and Simulation	
(Li et al., 2024)		Minimizing the Impact on the Surrounding Environment and Maximizing the Use of Production Raw Materials	Processes and Services, Specified Time, and Green Financial System		Big Data Analysis of Enterprise SC Under the Green Financial System, and Proving Usability Through Experiments	Big Data
(Yadav and Majumdar, 2024)	Agri-Food	Analysis of Barriers and Strategy Development for Mitigation	Technology Infrastructure and Immaturity, and High Capital Investment	Framework with a Triple Helix Approach Through Literature Review and Experts	Weighted Influence Non-Linear Gauge System (WINGS)" Method	
(Voipio et al., 2023)	Manufacturing Firm	Evaluating Business Implications		Framework Model for Identifying Key Contribution Areas and Dynamics Landing Benefits on a Business Level	Quantifying the Impact of RFID-Enabled DT in a Single Case Study	
(Agarwal et al., 2024)		Prioritizing Challenges		Strategic Framework to Make Effective Decisions Around Resource Allocation	Best-Worst Method (BWM) And Decision-Making Trial and Evaluation Laboratory (DEMATEL)	
(Bo et al., 2024)	Renewable-Based Community Based Self-Sufficient Utility	Sustainable Development with Minimizing Operational Expenses Using Storage Batteries	Environmental Stewardship, Smart Island (SI) System and Summer Season	Analytical Review of Social and Environmental Impacts and Trade-Offs	Teaching Learning-Based Optimization (TLBO) Algorithm to Balance Migration Rates and Habitat Suitability Indices, Fortify SI Grid Stability, Gated Recurrent Unit (GRU) Deep Learning Model for Accurate Predictions	
(Agrawal, 2024)	IOTA' SC in Footwear and Apparel	Supporting Direct-to-Consumer Growth with Improving Forecast Coverage and Smooth Supply	Customers' Expectations in Terms of Service, Cost-Effectiveness, and Sustainability	Thesis, Planned Establishment of a Cross-Dock Facility, and Digital Infrastructure Through the Development of a Decision Engine for Routing Products Within the New Decentralized SC Network and Decision Postponement in the Inventory Flow From Asia to the US	Heuristic-Based Algorithm and Synthetic Datasets	
(Serrano et al., 2021)		Optimizing and Prescribing a SC's Master Production Schedule (MPS)	Flexibility, Resilience, Sustainability, And Quality Criteria	Conceptual Framework in a Zero-Defect Environment	Intelligent Optimization Algorithms	
(Longo et al., 2023)	Agri-Food	Enhancing Resilience and Sustainability	COVID-Like Crises	Simulation-Based Decision-Making Framework	Validating with a Case Study	
(Zhang et al., 2024)	Semiconductor	Sustainability with Improving ESG Ratings and Mitigating Risks	Fairness among Stakeholders, Information Disclosure, Investment, and Tax Rates	Analytical Analysis of the Dynamic Strategic Interplay among Semiconductor Manufacturers, Consumers, and Government Regulatory Agencies (GRAs)	Evolutionary Game Model	

Study	Application Field	Purpose	Concern	Approach	Procedure	+ DT
(Mielczarek et al., 2023)	Inventory Control	Keeping Inventories at a Level and Minimizing Costs	Delivery on Time, Customer Needs, and Optimal Parameters	Simulation-Based Optimization Model to Improve The Quality Of Inventory Management Decisions		
(Liu et al., 2025)	Shop Floors, Warehouses, and Logistics	Exploring Applications across SC Processes from Smart Manufacturing to Intelligent Maintenance	Security, Traceability, Transparency, and Efficiency	Chapter, Literature Review for Benefits of Fusion in Blockchain-Driven DTs		Blockchain
(Mingawa et al., 2023)	Management Accounting	Satisfying Target Customers' Needs at Customers' Perceived Value-Based Selling Price	Sustainability Management, Social Responsibilities, and Sustainability Development Goals	Chapter, New Product Development (NPD), Method Based on Target Pricing and Value-Based Pricing (VBP)		
(Vishwakarma et al., 2023)				Chapter, Analysis of Opportunities and Challenges with Identifying 15 Benefits of a DT to Build an Efficient SC		
(Raja et al., 2024)	Logistics	Optimizing the Transportation and Routing Network	Demand, Credit Transactions, And raw materials	Book, Project for Real-Time Tracking of the Transport Vehicles		Cloud System, Microsoft Azure, and IoT Sensors
(Melesse et al., 2024)	Agri-Food	Improving the Management of Perishable Food Items by Enhancing Visibility	Waste of Fresh Produce	Detailed Analysis for Enhancing SC Interoperability		
(Samayametri and Vaddy, 2024)		Sustainable Development with Optimizing SC Processes	Environmental Impact, Resource Efficiency, and Responsible Practices	Predictive Analytics, Adaptive Decision-Making, and Fostering Efficiency and Resilience		
(Serrano-Ruiz et al., 2023)		Simulating the Behavior of the Zero-Defect Planning System		Modelling Domain up to the Second-Tier Supplier in SC at the Tactical and Operational Decision Levels	New Models and Optimization Algorithms for Zero-Defect Manufacturing (ZDM) Planning Problem	
(Yenew Melese, 2020)	Agri-Food, Fruit	Reducing the Waste of Fresh Produce	Waste and Visibility	Thesis, General Framework with an Integrated Solution for Monitoring Fruit Quality Status, Inventory Planning, and Delivery Optimization	Italian Food Bank, ML to Track Fruit Quality Changes Throughout Storage, and Prediction Accuracy, and Historical Time-Series Data	
(Ashraf et al., 2024)		Enhancing Resilience		Cognitive Framework	Hybrid Deep Learning, Deep Autoencoder Neural Network Combined with a One-Class Support Vector Machine for Detection, and Long-Short-Term Memory Neural Network to Predict Time-to-Recovery	
(Meng and Li, 2024)		Change Control	Interruptions and Environmental Influences	Dual-Space Four-Layer Framework and Dynamic Model Based on SC Structural Characteristics	Hysteresis Switching Law	
(Coching et al., 2023)	Logistics	License Plate Recognition (LPR) System	Traffic Flow and Violations	Automated no-Contact Apprehension Policy (NCAP)	MATLAB Simulation, AI-Based System to Detect Traffic Violations and Link to the License Plate Information, and Neural Network-Based LPR Architecture For Traffic Control And Management	
(Tyapukhin et al., 2024)		Sustainable Prototype For Digitalization	Structure of Value, Demand	Methodological Prerequisites with the Determination of Qualitative Attributes and Dichotomies of the Subject and Objects of Chain Management, and Study of Binary Matrices to Clarify the Structure of DT Prototype	AI in the Management of Non-Physical Objects of Chain Management	
(Juned et al., 2022)	Healthcare, Vaccine	Improving Visibility And Connectivity	Shortage and Wastage	Framework to Control Vaccine Shortage and Wastage		5G Technologies
(Mridha, 2023)	Ready-Made Garments	Revitalization	Data Transparency	Cloud-Based Web System Manages Hardware Data and Tracks the Order, Production, Logistics, and Delivery process	Robotic Process Automation (RPA) Architecture	IoT
(Erel-Ozçevik et al., 2023)	Warehouse	Optimal Location Assignment and Maximum Space Utilization	Energy, Locations of Products (Blocks)	Block Location Problem (BLP) Using Drones as a Block Transportation Strategy	Formalizing the BLP in a 3D Grid Model and Genetic Algorithm	

Study	Application Field	Purpose	Concern	Approach	Procedure	+ DT
(Perez et al., 2023)						
(Perez et al., 2021)		Optimizing Business Processes	Delays, Bottlenecks, and Fulfillment Dates	Three-module Framework with Process Definition, Simulation, and Optimization	Heuristics and Mixed-Integer Linear Programming (MILP) to Optimize, Julia Programming Language, APIs, Multi-Stage Transactional Network, and Order-to-Cash Case Study and Sensitivity Study	
(Belal, 2024)		Enhancing SC Resilience (SCR) with Strategic Fit in Focus		Conceptual Framework with Dynamic Capability View (DCV) as the Theoretical Foundation	Survey and Partial Least Square (PLS)	
(Guo and Mantravadi, 2024)		Enhancing Coordination and Bolstering Resilience in Lean SC Processes and Performances	Efficiency, Costs, Responsiveness, and COVID-19	Systematic Literature Review and Supply-Chain Operations Reference (SCOR) Framework		
(Chen Z and Huang, 2021)		Information-Sharing in Remanufacturing SC	Information Asymmetries	Systematic Literature Review		
(Zhang et al., 2022)		Improving Resilience	Cloud-Based Systems, Digital Platforms	Chapter		
(Le and Fan, 2023)	Logistics	Deploying a Transparent, Trustworthy, and Resilient DT	Effective, Efficient, Transparent, and Timely Decision-Making Process	Conceptual Framework with Literature Review, Research Potential, and Practical Challenges		
(Huang et al., 2024)	Food	Meeting the Key Challenges in Food Supply Chains (FSCs)	Monitoring, Real-Time Simulation, and Scenario Analysis	Conceptual Framework with Systematic Literature Review (SLR), Theory and the Technology-Organization-Environment (TOE) Framework, and Five-Step Three-Stage Implementation Process	Typology-Driven Thematic Analysis	
(Orozco-Romero et al., 2020)		The Use of Agent-Based Models Boosted	Customer Satisfaction, Operating Costs, Financial Standing, and Competitiveness	Literature Review	Agent-Based Models	
(Shoji et al., 2022)	Agri-Food, Fresh Fruit	Optimizing the Postharvest SC of Imported Produce	Quality, Air Temperature Data and Cold Chain	Comparing and Quantifying the Impact of the Different Measures in Preserving the Quality		
(Klar et al., 2023)	Ports	Energy Savings	Environmental Footprint and Smart City	Extensive Cross-Domain Literature Review		
(Aggarwal et al., 2024)	Metaverse	Empowering Blockchain	3D Evolution, Security and Privacy Vulnerabilities, and Data Immutability	Conceptual Design with Use-Case Scenario	Validating with Various Computational Metrics	Blockchain
(Shrivastava et al., 2022)	Food, Refrigerated Fruit	Quantification Of the Trade-Offs in Maintaining Citrus	Quality and Marketability		Sensor, Actionable Metrics, and Principal Component Analysis (PCA)	
(Bargavi and Mathivanan, 2023)		Discussing the Various Definitions of DT, Types, and Potential Benefits	Visibility, Accuracy, and Efficiency	Chapter, Ten-Step Process of Implementing		
(Singh et al., 2024)	Manufacturing	Developing Resilient Sustainable Manufacturing SC	Influence Score, Visibility, and Coordination	Identifying the Seventeen Resilient Sustainable Factors Through Extant Literature And Experts	Grey Influence Analysis (GINA) for Causal Analysis	
(Frees and Ludwig, 2024)		Development and Evaluation with Addressing Lack of Application Framework	Transparency and Responsiveness	Conceptual Framework with Multiple Layers, Dimensions, and Stakeholder-Technology Dependencies for Early-Phase Adoption, and literature and expert interviews	Validating with Expert Evaluations, and Key Insights and Guidelines	
(Frees and Ludwig, 2021)		Creating More Robust and Flexible SC	Transparency, Production Interruptions and Delivery Bottlenecks	literature review with Six Dimensions: Scope, Actor, Asset, Flow Reference Object, Performance Measurement, and SC Process	Sensors	
(Patil et al., 2023)	Indian	DT Readiness to Enable SC	Visibility And Operational	Theoretical Model	Questionnaire, Confirmatory Factor Analysis	

Study	Application Field	Purpose	Concern	Approach	Procedure	+ DT
(2024)	Manufacturers	Transparency	Inefficiencies		(CFA) and Structural Equation Modelling (SEM)	
(Dy et al., 2022)	Food, Aerospace and Automotive, Construction, Pharmaceutical, and General Manufacturing	Reviewing the Implementations for Resilience and Agility	Case Studies and Actual Applications	Literature Review	Survey	
(Sujendi and Siagian, 2024)	Oil and Gas in Pertamina Company	Analyzing the Effect of SC Capabilities, Resilience, and Performance on Sustainability	Maintenance, Production Reliability, and Information Integration	Structural Equation Modeling with Analysis Moment of Structure Program	Questionnaires	
(Aguilar-Ramirez et al., 2021)		Empowering SC		Identifying the Main Characteristics for Working together DT and Blockchain		Blockchain
(Alguirat et al., 2024)		Presenting the Concept of DT in Various Fields	Humans and Machines			
(Botta et al., 2023)	Food	Strengthening the SC Management of Physical Goods	Traceability	Model with Various Scenarios	Formalism of Secure Computation	Blockchain
(Zarnitz et al., 2022)	Logistics, Automotive	Sustainable Network Planning	Predictive Planning and Control	Literature Review	Interviews and Case Study	
(Buratini et al., 2024)	Oncologic Pharmaceutical Company	Engineering a System to Monitor Effectiveness and Efficiency	Therapy Personalization, on-Demand Drugs, Timing, Delivery, and Responsiveness	Identifying the Main Challenges with a Public Cancer Care and Research Institute		
(Yu et al., 2024)	Logistics	Enhancing Asset and Operations Management	Resource Utilization, Stakeholder Coordination, and End-to-End Services	Systematic Literature Review, and Three-Axis Framework for Key Aspects of Effective DT Application with Technology Readiness Level (TRL)		
(Jeyalakshmi et al., 2024)	Logistics	More Efficient, Intelligent, and Flexible Manufacturing Infrastructure	Waiting Times, Agility, and Responsiveness	Organizing and Systematizing the Existing Research	Case Studies	
(Muñoz-Díaz et al., 2024)	Wind Turbine Tower	Maximizing Performance	Global Warming, Fossil Fuels, and Layouts	Three-Level Conceptual Framework: Physical, Logical Processes, and Plant Logic	Lean Manufacturing Techniques with new Planning and Sequencing Methodologies and Algorithms	
(Corradini et al., 2024)	Food	Developing and Applying Virtual Food System	Quality, Safety Data	Chapter, Comprehensive Framework	Spectroscopic-Based Techniques for Mapping and Tracking, and Mathematical Modeling	
(Luo and Ball, 2023)		Sustainability and Resilience in Manufacturing Systems		Review, and Three-Dimensional Framework: DT Integration Levels, SC Structural Hierarchies and Processes		
(Lv et al., 2022)	Healthcare, N95 Medical Masks	Resilience with Reducing Operating Costs and Increasing Profits	COVID-19, and Cost and Interruption of Public Health Emergencies	Emergency Strategies		
(Auer, 2024)		Enhancing Resilience in Agent-Based SC		Three-Phase Framework with Articles Analysis	Large Language Processing (LLP)	
(Khan and Seger, 2024)		Enhancing Resilience	Visibility, Agility, and Collaboration	Thesis, Systematic Literature Review and Supplier Management Digital Twin (SMDT) Framework	KPIs for Visibility, Case Study, and Semi-Structured Interviews	
(BLOMKVIST, 2020)	Logistics in Scania Company	Improving Logistical SC Visibility		Thesis		
(Arishi and Krishnan, 2024)	Manufacturing	Increasing Visibility and Resilience	Medical Supplies, Society Survivability, and Demands	Analyzing the SC Behavior and Quantifying the Impact of COVID-19 with Disruptions Scenarios	System Dynamics (SD) Simulation and Artificial Neural Networks (ANN)	

Study	Application Field	Purpose	Concern	Approach	Procedure	+ DT
(Mar molej o- Sauce do et al., 2019)		Current Context of DTs	Business Processes	Brief Literature Review		
(Wan g et al., 2020)	Aircraft Landing Gear (ALG)	Planning	Demand Forecast, Aggregate and Inventory Planning	Elaborating DT with Existing Approaches	Case Study	
(Santo s et al., 2020)	Pharmaceutical Company	Estimating Monthly Capacity			Graphical User Interface and Case Study	
(Wan g et al., 2020)	Food, Industrial Hemp SC (IHSC)	Improving Transparency and Quality Control	Complexity, Variability, Data and Information Tracking	Two-Layer Blockchain for Authority Smart Contract in Supporting the Process Learning and Risk Management		Blockchain
(Ivano v and Dolgu i (2021 b))		Mapping Supply Networks, and Ensuring Resilience and End-to-End Visibility	Post-Pandemic Recoveries	Uncovering the Interrelations of Risk Data, Disruption Modeling, and Performance Assessment		
(Daei Niaki et al., 2021)			Complexity, Dynamism and Agility		Review	
(Lind quist and Haas, 2021)		Enhancing Resilience		Book, Four Scenarios Planning: Covid-19 Lockdown, Brexit without Deal, Conflagration at a Dairy and Political Regulations on Transport	SC Guru Software and AS-IS Model	
(Gerla ch, 2021)	Logistics	Research Conceptual Clarification	Application Areas and Use Cases	Systematic Literature Review		
(Lee and Lee, 2021)	Logistics and Construction	Predicting Logistics Risks and Delivery Time	Project Scheduling and Costs	Framework with Building Information Modeling (BIM)	Geographic Information System (GIS) and Case Study	IoT
(Burg os and Ivano v, 2021)	Food Retail	Improving Resilience and End-to-End visibility	COVID-19, Governmental Measures, Inventory- Ordering Dynamics, and Customer Behaviours	Several Directions and Practical Implementation Guidelines	Anylogistix	
(Moaz zeni, 2021)	Toilet Tissues in Luxxeen Company	Ensuring Resiliency with Multiple Sourcing, Changing Inventory Control Policy and Buffering	COVID-19, SC Survival, Demand Satisfaction, and Delivery Time	Thesis, Framework with SCOR Reference Model, and Disruption Scenarios	Discrete Event Simulation with Arena Software, Process Analyzer, Optquest, and Case Study	
(Binsf eld and Gerlac h, 2022)	Food	Quantifying DSCT Benefits		Three Volatility Scenarios for Multi- Echelon Inventory Management	Simulation Experiments	
(Marti n and Oger, 2022)	Healthcare, Drug	Supporting Supply and Delivery Decisions	Volatility, Uncertainty, Complexity and Ambiguity	Framework	Reinforcement Learning	
(Thiya garaja n et al., 2022)		Enhancing Resilience and Visibility	Connectivity, Data Generation and Value Creation, Planning and Monitoring	Examining to Develop a Roadmap	Qualitative Interviews and Qualitative Survey	
(Sipol a et al., 2023)	Food	Building Cybersecurity Exercises		Realistic Global Cyber Environment (RGCE) Platform	Node.js Services and Case Study	
(Lam et al., 2023)	Logistics		Key Clusters of Research and COVID-19	Bibliometric Analysis		

Study	Application Field	Purpose	Concern	Approach	Procedure	+ DT
(Gerlach, 2023)	Logistics	Enhancing Resilience with Examining the Application of DSCTs in Road Freight	Data, Complexity and Model Building Time	Thesis, CROSSROAD Framework with Qualitative Content Analysis (QCA)	Eight-Phase CRISP-DSCT Process Model	
(KEK et al., 2023)	Semiconductor	Managing Resilience	Loss and COVID-19	Real Disruption Scenarios with Prediction, Metrics for Evaluation, Strategies for Recovery, Diagnostics, Projections, and Visibility	Anylogistix	
(Ghasemi et al., 2023)		Operations for Supply Chain Management (OSCM)	Efficiency, Downtime, Customer Satisfaction and Data Infrastructure (DI)	High-Level Introduction to Monitor, Analyze, and Optimize	Sensors	IoT and Cloud Computing
(Le et al., 2024)	Food, Coffee	Improving Monitoring, Quality Control and Management, and Operational Efficiency	Complexity, Fluctuations and Sustainability	Conceptual Framework into a Smart Agriculture platform for Value Chains and SC	Microsoft Azure	
(Espinoza-Jaramillo et al., 2024)		Integration of AI with DTs by Underscoring the Transformative Potential	Operations Efficiency, Costs, Responsiveness Processes, Data Privacy, Complexity, and Standardized Protocols	Architecture of AI-Driven by Highlighting Components, Data Flow, and Interaction Mechanisms		
(Liu et al., 2023)	Norwegian Wood	Collaboration in Industrial Symbiosis Networks	Achieving Sustainable Operations	Analytical Framework, Intellectual Support	Interviews	
(Lim, 2023)		Intelligent Product Lifecycle Management (PLM) with Resilience Evaluation, SC Re-planning, and Shop Floor Rescheduling	Efficiency, Responsiveness, Sustainability, and Customizability	Thesis, Four-Tier Technology Stack with knowledge graphs (KGs), product family (PF), Smart Product-Service Systems (Smart PSS) and Circular Economy, and Maintenance Operations Using Cognitive Digital Twins (CDTs)	Benchmark, Case Study and Two Use Cases: Additive Manufacturing, and Gearbox Assembly Line	
(Abdullah et al., 2024)		Attaining Resiliency and Financial Risk Assessment	Complexity and Dynamism	Hybrid Framework for Uncertainty Quantification (UQ)	Quantum-Classical Hybrid ML	
(Chen, 2024)	Pharmaceutical Company, Aspirin Tablets in London	Reducing Ripple Effects and Improving Agility	ERP and Patient Safety	Quantitative Analysis	Anylogistix, t-Test ML and KPIs: Inventory Level, Order Fulfillment Rate, and Delivery Time	
(Roumeliotis et al., 2024)		Enhancing Reliability and Scalability	Data Dependability, Integrity, Traceability, Security, and Efficiency	Comprehensive Analysis with Qualitative Research Methodology	Desk Research, Case Studies, and Interviews	Blockchain
(Hirata et al., 2024)		Determining SC Management Priorities	Macro Perspectives	Identifying Core Technologies AI/ Reinforcement Learning, Picking Robots, Cybersecurity, 5G Networks, The Physical Internet, Additive Manufacturing, and Cloud Manufacturing	ML	
(Bribian, 2019)	Semiconductor in Infineon Technologies AG Company	Understanding to Build Simulation Models Automatically	Reusability	Thesis, Framework with Ontologies	KGs and Use Case	Semantic Web
(Kamble et al., 2022)		Sustainable Manufacturing	Quality, Time, and Customized Products	Systematic Literature Review		IoT, Cloud Computing, and Blockchain
(Maheshwari et al., 2023)	Food	Optimizing the Make-Span and Lead Time	productivity	Procurement, Production, and Distribution Strategies (PPDs) by improving Make-Span Time, Data Redundancy (DR), Optimal Scheduling Plan (OSP), Overall Operations Effectiveness (OOE), Overall Equipment Effectiveness (OEE), and Capacity Utilization	Mixed-Integer Linear Programming (MILP) and Agent-Based Simulation (ABS)	
(Singh et al., 2023)	Food	Enhancing Resilience and Sustainability by Identifying Fifteen Factors by Experts	Causes, Effects, Objectives, Outcomes and Waste	Grey Causal Modelling (GCM), and Procurement, Production, and Distribution Strategies (PPDs)		
(Perez et al., 2021)		Mitigating Operational Bottlenecks	Business Transactional Processes	Framework for Test Optimization Strategies, Forecast Potential Delays, Accurate Fulfillment Dates, and Identifying Bottlenecks	Three Mathematical Programming Models and Greedy Heuristic Algorithm, and Case Study	
(Mar molejo-Saucedo et al., 2024)	Packing	Solving Large-Scale Optimization Problems	Bin-Packing and Vehicle Routing Problems	Framework with Commercial SC Platform	Heuristic Optimization Algorithms and Verifying with Hypothetical Case Solution	

Study 2022)	Application Field	Purpose	Concern	Approach	Procedure	+ DT
(Guidani et al., 2024)	Agri-Food	Sustainability by Virtualizing, and Enhancing Practitioners' and Consumers' Knowledge of Entities and Operations	Safety, Quality, Affordability, Transparency, Visibility and Complexity	Strategic Bottom-Up Pressure and What-If Simulation Analysis for Preserving Natural and Anthropogenic Ecosystems	Four Virtual Scenarios and Multi-Dimensional Dashboard of Indicators	
(Barat et al., 2024)	Food, Apple	Optimizing Climacteric Perishable Fruit SCs	Harvesting and Stakeholders	Three-Fold Challenge for Ensuring Quality, Improving Profit Margins, and Reducing Waste	Modeling Abstractions and System Dynamics Model	
(Valerio et al., 2023)	Food	Optimizing Distribution in Perishable Food SC	Costs, CO2 Transportation, Storage and Waste	Dynamically Optimized Distribution (DOD) Methods	Anylogic and Case Study	
(Bijlma et al., 2018)	Coil Warehouse in Steel Industry and Marine Shipments in Oil Industry	Optimizing manufacturing	Save Money, Customer Satisfaction and Risk	Simio Simulation Model	Scenario Navigator User Interface, Case Studies and Arena	
(Jayalakshmi et al., 2024)	Fashion and Textile	Greater Efficiency and Sustainability	Visibility, Quality Control, Agility, Customer, Inventory, Data Security, Implementation Costs, Workforce, and Interoperability	Chapter		
(Pan et al., 2023)		Enhancing Resilience and Human Rights	Post-COVID-19, Dynamic Capabilities, Green and Ripple Effects	Knowledge-Based Framework Road Mapping Taxonomy to Enrich Sustainability-Practice	Web of Science Core Collection	
(Valk et al., 2022)			Visualization and Optimization Potentials	Review for Answering Use Cases, Purposes, and Technological Readiness		
(Defraeye et al., 2021)	Agri-Food, Fruits and Vegetables	Quantifying Quality Loss of Fresh Horticultural Produce	Postharvest, Temperature, Gas Atmosphere and Airflow	Preventive Shipment-Tailored Measures to Reduce Retail and Household Food Losses		
(Marmolejo-Saucedo et al., 2020)	Pharmaceutical Company	Connecting Simulators, Solvers and Data Analytic Tools in an Integral Interface	Business Processes and Effectiveness		Case Study	
(Zaidi et al., 2024)	Manufacturing and Logistics	Improving Performance	Real-Time Synchronization Autonomous, Decision-Making and Complexity	Three-Layered Conceptual Framework and Systematic Literature Review		IoT, RFID, Cloud Computing, CPSs, Cybersecurity
(Buss et al., 2021)		Optimizing Multi- and Intermodal SCs	Optimization Potential	Initial Framework and High-Level Architecture		
(Jesus et al., 2024)	Battery	Recycling	Visibility, Traceability and Dashboard Screen	Literature Review and Synthesis Systems Architecture	Case Study	Blockchain
(Ivanov and Dolguin, 2019)		Risk Analytics for Ripple Effect	Redundancy, Flexibility, and Resilience	Control Framework with Mitigation Strategies and Resilience		
(Defraeye et al., 2019)	Agri-Food, Mango Fruit	Minimizing Environmental Impact of the Cold Chain and Optimizing Logistics	Quality, Waste, Energy, Temperatures, Airflow, Refrigerated Transport and Storage, and Export		Mechanistic Modeling for Simulating Thermal Behavior	
(Bada khshan and Ball, 2023)	Inventory and Cash Management	Identify Inventory and Cash Replenishment	Demand Capacity and Credit Purchase	Framework to Examine Physical and Financial Flows	ML And Simulation	
(Kala boukas et al., 2023)		Improving Sustainability	Three Views: Business and Sustainability, Data and Cognition AI Models Governance	Holistic Governance with by Circular Strategies	Cognitive	
(Pernici et al., 2023)		Reconfiguration with Reactivity and Resilience in Internal	Customers, Suppliers, Efficiency and	Conceptual Framework with AgileChains Paradigm, and Combining Service Oriented Architecture (SOA) With Cyber-Physical	Data Management Mechanism with Governance	

Study	Application Field	Purpose	Concern	Approach	Procedure	+ DT
(Luga resi et al., 2023)		Processes	Effectiveness	Systems (CPS)	Tools	
			Complexity, Planning, Control and Value Chains	Overview for Current Level of Development and the Future Research Challenges		
(Band ara et al., 2024)	Logistics	Sustainable Inbound and Outbound Logistics by Identifying Key Enablers	Technological Infrastructure, Processes, Improvement, Knowledge Workers and Managerial Support	Comprehensive Review with Person- Environment-Occupation (PEO) Model and Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) Framework		IoT, IIoT and Augmented Reality/ Virtual Reality (AR/VR)
(Trian tafallo u et al., 2024) ^	Automotive Industry	Enhancing Lifecycle Management and Optimizing Processes	Maintenance Disposal and Interoperability	Integration with Other Disruptive Technologies and Developing Robust Data Management Strategies		IoT and Big Data
(Habt emich ael et al., 2024)		Tracing Value Added Creation	Consumers, Globalization, Sourcing, Production, Sale Traceability, Visibility, Raw Materials and Complexity	Secure Data Flow among Stakeholders for Transferring Origin of Products Information	Non-Fungible Token (NFT) and Coupled with Smart Contracts	Blockchain
(Chatt erjee et al., 2024)	Electronics Industry and Metaverse	Improving Security and Efficiency	Processes, Waste, Customer Satisfaction and Smart Contracts	Creating a Transparent, Secure, and User- Friendly System to Verify Product Details and Access Warranties and Payment Settlements		Blockchain
(Kapil et al., 2024)		Information and Risk Management	Visibility, Traceability And Resilience	Systematic literature review (SLR) and Bibliometric Analysis		
(Gong and Zhang , 2024)	Hybrid Production Systems		Response Time, Operating Costs, Competitiveness and Efficiency	Framework for Creating DT Spaces, Reconstructing Old Models, and Algorithm Testing by Data Assurance Strategies and Hybrid Modeling	Data Collection and Fault Feature Diagnosis Algorithms	

2.2. Supply Chain Resilience in Automotive Industry

The review of relevant research summarized in Table 4 shows that not only is the use of KPIs for resilience low (Raaymann and Spinler, 2024, and (Kenanga and Ardi, 2022) in the automotive industry, but its application in other fields is also rare .

Table 4
Examining Resilience in Supply Chain Automotive Industry

Study	Case Study	Purpose	Concern	Approach	Procedure
(Raay mann and Spinle r, 2024)		Measuring SCR with KPIs	COVID-19, Lead Time, OTIF (On time in full), Suppliers Flexibility and Stock Level of High-Risk Parts	Comparative Research by Investigating Literature's Perspective	Text Mining, Analytical Hierarchy Process (AHP), KPIs and Interviews
(Khot and Thiag arajan, 2019)	India	Sustainability and Resilience by Management Commitment to Government Rules	Responsiveness, Vulnerabilities, Disturbances and Managerial Activity	Literature Review for Evaluating the Role of SC Management and Survey	DEMATEL for Inter-Relationship
(Agar wal and Seth, 2024)	India	Enhancing Resilience for Better Adoption and Resource Allocation of Enablers	Financial Performance, Business Continuity, Customer Satisfaction Profitability and Market Share Growth	Identifying Eight Enablers and Developing Structural Model for Enhancing Flexibility, Agility, Preparedness and Responsiveness	Delphi Method, Fuzzy ISM and Fuzzy MICMAC
(Kavi ani et al., 2020)	Iran	Evaluating Two Resilience Key Elements of Vulnerability and Capability	Resisting Five Elements of Vulnerability: External Pressures, Sensitivity, Connectivity, Supplier/Customer Disruptions, and Resource Limits, and Embracing Nine Elements of Capability, Flexibility in Order Fulfillment, Capacity, Efficiency, Visibility, Adaptability, Recovery, Dispersion, Organization, Market Position, and Security	Benchmarking Model	Fuzzy Approach for Statistical Hypothesis Testing, and Questionnaire
(Soare s et al., 2021)	Brazil	Resilience by Evaluation of Scenario Post COVID-19	COVID-19 and Post-Pandemic Scenario	Reference Model For Advanced Product Quality Planning and Control Plan (APQP/CP)	
(Ali and	Sweden,	Better Resilience	COVID-19, War, Complexity, flexibility and Multi-	Thesis, Systematic Literature Review by Investigating the Trajectory of SC	Single-Case Study and Semi-Structured

Study	Case Study	Purpose	Concern	Approach	Procedure
(Aslani, 2023)	Volvo		National Corporation (MNC)	Disruptions Compare with the Theoretical Suggestions	Interviews
(Fabbre-Costes et al., 2021)	France, Renault	Improving Robustness and Resilience	COVID-19, Complexity,	Lessons by Forty-Three Key Informants	Single-Case Study
(Jum'a et al., 2024)		Moderating Effect of Reactive and Proactive Resilience Strategies	Transportation, Continuity, Maintaining Production, Delivery Schedules, and Environmental Impact	Theoretical Framework by Identifying Four Major Risks, Seven Proactive, and Twelve Reactive Strategies	Stakeholder Theory Perspective and Quantitative Method
(Mandala et al., 2020)		Resilience by Predictive Maintenance	Auto Parts Manufacturers	Survey-Based Study by AI-Driven Strategies	
(Khot and Thiagarajan, 2024)		Prioritizing Resilience Strategies to Maintain Sustainability Performance	Interrelationships, Multiple Sourcing, Information Sharing, and Design Quality	Non-Linear Closed-Loop Framework by Gaussian Fuzzy Optimization-Based Non-Linear Model Predictive Control as a Feedback Controller	Trapezoidal Linguistic Cubic Fuzzy Z-Score Technique for Order of Preference
(Sadra badi et al., 2024)	Iran, Battery	Designing a Green-Resilient Closed-Loop SC	Interrelated Disruption, Pricing Decisions, Environmental Impact Recycling Cycle, Continuity and Mathematical Complexity	Multi-Objective Optimization Model	Fuzzy C-Means Clustering Method, Augmented E-Constraint and Case Study
(Li et al., 2024)	China	Forging Resilience Through SC Collaboration	COVID-19, Information Sharing, Resource Sharing and Adjustments	Resource Dependency Theory (RDT)	Case Studies and Interviews
(Kengana and Ardi, 2022)	Indonesia	Developing KPIs for Resilience	COVID-19 and Demand	Content Validity Index (CVI) by Eleven Indicators and Forty Six Sub-Indicators	KPIs and Questionnaires
(Kasselia and Somi, 2023)	Electric Vehicles in OEM X	Resilience by relationship between the purchasing department and new product development projects	Technologically Uncertain Environment, Resource, and Stakeholders	Thesis	Interviews and Single Case Study
(Ciavarella, 2021)		Greater Resilience Development		Thesis, Survey	

2.3. Predictive Quality Assurance

According to the studies conducted in the field of predictive quality assurance summarized in Table 5, it is clear that predictive quality assurance in all research is considered as a case for data analysis, not as a model or framework .

Table 5
Examine Prediction for Quality Assurance

Study	Application Field	Purpose	Concern	Approach	Procedure
(Burggraf et al., 2021)	Assembly Processes	Predicting Product Quality	Complexity, and Expensive, Time-Consuming, or Incomplete Tasks	Lessons	Machine-Learning for Classification, Machine Sensor and Case Study
(Nalbach et al., 2018)		Assuring Product Quality in a Preventive Way	Failures, Costs, Public Image and Design Process		Machine Learning and Case Study
(Awujoola et al., 2024)	Wine	Optimizing Production Processes and Maintaining High-Quality Standards	Smart Manufacturing and Predicting Product Failures	Chapter	Scikit-learn ML Library for Accurate Categorization of Attributes, Random Forest and Decision Tree Models and Comprehensive Case Study
(Puyati et al., 2020)		Proactive Preventative Maintenance by MPC Tests	Feasibility, Optimal Linac Performance and Downtime	Machine Performance Check (MPC)	Statistical Process Control and Autoregressive Integrated Moving Average (ARIMA) Forecast Modeling
(Maglupaz and Lacatan, 2023)	Education	Development of Predictive Models	Services, Local Higher Institutions and Standards	Basic Framework for Ten Areas	Data Mining Models and Statistical Methods

Study	Application Field	Purpose	Concern	Approach	Procedure
(Smith, 2024)	Healthcare, Radiotherapy	Proactive Software	Defects, Higher Customer Satisfaction, Reduced Time to Market and Reliability	Testing Process, Enabling Continuous Integration and Delivery	Machine Learning and Agile Test Automation
(Li et al., 2024)		Warning Target Failure	Integrity of X-Ray Target, Patient, Wedge-Directed Beam Profiles and Linear Accelerator (Linac)		SPC, ARIMA, Data Collecting by Open Fields and Enhanced Dynamic Wedge (EDW), and Non-Uniform Magnification Factors
(Bejar et al., 2020)		Modeling Rater Behavior	Feasibility, Operational Conditions and Essay Scoring	Evaluating the Relationship Between Rater Predictability and Inter-Rater Agreement and Performance on Pre-Scored Essays	Linguistic Attributes and Linear Regression
(Olive s and Pagan o, 2013)	Healthcare	Monitoring Health Programs	Producer and Consumer Risks, and Positive Predictive Values (PPVs)	Bayesian-Lot Quality Assurance Sampling (B-LQAS)	
(Tao et al., 2023)		Acceptance Thresholds for Screening Automated Pavement Data	Distress Data and State Departments of Transportation (Dots)	Conventional and Bootstrapping Prediction Interval Methods	
(Luo, 2007)	Software in SAP Inc	Increasing Customer Satisfaction	Cost Penalty, Defects, Lost Productivity and Post-Release Maintenance, Products, and Efficiency	Thesis, Software Reliability Growth Modeling by Geometric Insight	Anisotropic Laplace Test Statistic, Operating Characteristic Curves and Case Studies by SAP Projects
(Kumar and Rath, 2015)	Software in Eclipse JDT Core	Assessing the Effectiveness of Fault Prediction Techniques	Software Development	Cost Evaluation Model For Object-Oriented Software	Chidamber and Kemerer (CK) Metrics Suite Using LR and Hybrid Approach of Neural Network, Particle Swarm Optimization (Neuro-PSO and Modified Neuro-PSO), and Case Study
(Xu and Khoshgoftaar, 2001)	Software	Timely Predictions of Quality Indicators on a Module-by-Module Basis	Cost Function and Reliability	High-Assurance Network Telecommunications System	Fuzzy Nonlinear Regression (FNR) by Integrating Fuzzy Logic and Neural Networks Techniques, Pascal-Like Proprietary Language, Multiple Linear Regression (MLR), and Case Study
(Fu and Ma, 2018)		Better Design to Facilitate Early Ejection	Deformation	Computer-Aided Engineering	Integrating Moldflow™ and Ansys™, and Case Study
(Sasankar and Sakar kar, 2023)	Software	Cross-Project Defect Prediction to Enhance Software Quality and Maintenance	Data, Project-Specific Characteristics, and Software Repositories	Knowledge Transfer Strategies: Instance-Based, Feature-Based and Model-Based, and Analyzing Key Factors	
(Harris et al., 2019)		Complex Geological Modeling Using Unstructured Grids	Fault/Horizon Input Data, Depositional/Geological Grid Geometries and Spatial Measures	Reservoir Modeling and Simulation Workflows Based on Pillar Gridding or Stair-Step Grid Technologies, and Volume-Based Structural	Case Studies
(Denninger et al., 2011)	Automotive Industry, Hemming Process	Optimizing Bendability Prediction	Failure Criterion	Finite Element Analysis (FEA) and Process Robustness Analysis	Gradient Based Strain-Criterion
(Pickard and Dieter , 2008)	Product Design	Reliability Prediction over the Complete Product Design Cycle	Possibility and Lifetime Distributions	Transfer of the Gained Information into a Failure Mode and Effects Analysis	FMEA for System Behavior Description
(Raeis , 2021)		Delay Distribution Prediction	Demand Patterns, Service Capacity, Scheduling Discipline and Customers	Thesis, Ensuring Quality of Service (QoS) by No Knowledge of The System Model or Parameters, and Traffic Signal Control Methods	Gaussian Mixture Model, Queue Length and Reinforcement Learning

3. Methodology

3.1. Framework Development:

The automotive industry is one of the largest and most dominant manufacturing sectors worldwide (Zailani et al., 2015, as cited in Ghadge et al., 2022). Fig. 2 summarizes the model, serving as our reference for business architecture in today’s automobile industry. With few exceptions in business competences, this reference model is also applicable to other types of vehicle manufacturers (Wedeniowski, 2015). It is crucial to understand the supply chain not merely as a relationship between suppliers and customers but as an interconnected network where the essentials—cost, quality, and risk—of a product are shaped

by the performance of the entire network network (Irsyadillah et al., 2020). A company’s desired goal is often reflected in its strategy. Similarly, a business model captures how different business sectors fit together at a particular point in time, shaping the company’s holistic orientation and future direction. Successful companies distinguish themselves not only by how they develop business models but also by how they implement them within their strategies. A business model aggregates all relevant business aspects at a high level of abstraction, illustrating how a company creates and delivers value while generating profit (Wedeniowski, 2015) . This approach facilitates clear communication of business activities in terms of value creation and performance areas with similar functions. To incorporate Clause 1 of the IATF standard in our research, and in line with the first step of the 8D method, it is essential to determine the model scope and problem description.

3.1.1. Business Domains

The primary purpose of business domains is to consolidate critical aspects of business into individual focus areas. These domains are purely functional, designed independently of processes, organizations, and applications. A business domain represents a functional summary of the competencies required to implement the business model. Each domain serves a specific business goal and has a logical role within the company. Like the business model, the business domain model is created for a particular point in time (Wedeniowski, 2015).

Business domains are categorized into four groups: leadership, core business, administration, and support. The leadership domain forms the framework for business model implementation, addressing profit generation, risk management, and cost control. Core business domains encompass primary business competencies necessary for value proposition delivery. Administration domains include operational competencies that indirectly support value creation by ensuring seamless process design and supervision. Support domains incur costs without directly creating tangible benefits but provide foundational support for business implementation (Wedeniowski, 2015).

Additionally, each domain's competencies are classified into three responsibility levels:

- **Direct:** These competencies define the strategic mission and guidelines for long-term business continuity and intrinsic value preservation. They promote cooperation between primary and secondary competencies, whether internal or external.
- **Control:** This middle layer monitors performance, implementation results, and compliance. It deals with exceptions, enforces consequences for non-compliance, protects assets, and validates information through relevant performance indicators.
- **Execute:** Execution-level competencies apply specific knowledge, skills, and professional behaviors to create value for the end customer (Wedeniowski, 2015).

The value chain concept revolves around monitoring organizational processes, viewing the company as a system composed of subsystems with inputs, transformation processes, and outputs (Porter, 1996, as cited in Roumeliotis et al., 2024). These processes encompass resource procurement and use—such as financial capital, human labor, materials, and infrastructure—directly influencing costs and profitability. (Roumeliotis et al., 2024). This study focuses on identifying inputs and outputs in automotive industry subsystems to determine suitable key performance indicators (KPIs).

Business processes drive a company's actual value creation through a series of individual activities, increasing economic value and utility for internal or external customers (Bach et al., 2012, as cited in Wedeniowski, 2015). Many industrial companies have embedded business processes deeply into their organizational structures, with numerous reference works providing established models (Wedeniowski, 2015). Our study focuses on existing reference models for business processes in the automotive industry, linking business processes to business components and services. Formal process modelling languages, such as Business Process Modelling Notation (BPMN) developed by IBM and later adopted by the Object Management Group (OMG), facilitate the translation of business processes into technical models. A useful starting point is the Process Classification Framework (PCF) for Automobiles (OEM), developed by IBM and the American Productivity and Quality Center (APQC), which provides a high-level portfolio of around 1000 processes. PCF is linked to Open Standards Benchmarking (OSB), offering comprehensive performance metrics for process evaluation (Wedeniowski, 2015).

The purpose of this research is to design a framework for predictive quality assurance by identifying KPIs within the automotive supply chain. Using the IATF standard and the eight phases of the 8D method, we begin with quality management and quality objectives in Clauses 1–3. Clause 5 emphasizes leadership characteristics and process analysis, aligning with the leadership domain in Fig. 2. Therefore, the business process must start with a focus on the quality domain.

3.1.2. Business Quality Domain

The quality domain ensures that vehicles remain defect-free from development through manufacturing and maintain long-term reliability post-sale. Quality encompasses not only product performance but also the entire spectrum of company services, embedding itself within organizational culture and guiding staff behavior (Wedeniowski, 2015). Quality assurance within supply chains is a vital subsystem of the automotive industry's value chain, involving planning, quality control, evaluation, reporting, and continuous improvement. It ensures product and service adherence to specific standards, fostering consumer confidence and competitive advantage (Flynn et al., 2010, as cited in Roumeliotis et al., 2024).

As illustrated in Fig. 12, the quality domain includes eight key business competencies.

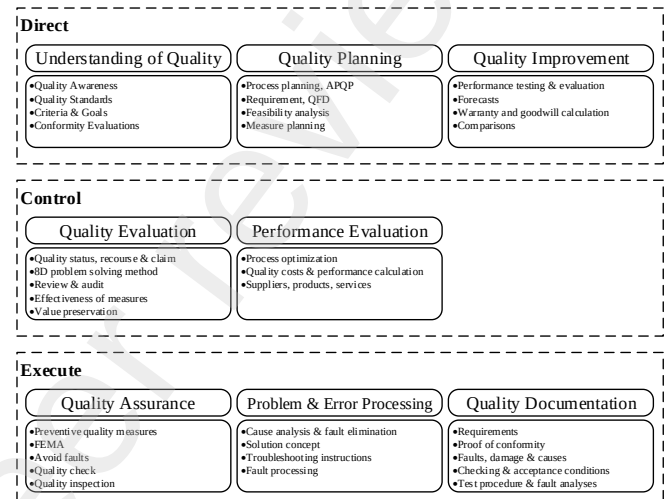


Fig. 12. The Business Competencies of the Domain "Quality" (Wedeniowski, 2015)

Quality assurance resides at the execution level, ensuring consistent product quality through systematic measures. Section 4 elaborates on these quality domain components.

To enhance predictive quality assurance, the integration of digital twin technology is essential. Digital twins align closely with IATF 16949 goals, particularly in continuous improvement, risk management, and precision monitoring (Advantive Co, 2024). Digital twins enable manufacturers to meet quality objectives with greater accuracy and control.

In designing a digital twin for business quality, specialized analytical tools are required for preemptive quality evaluation. Tools like InfinityQS, WinSPC, and PQ Systems support this requirement through Statistical Process Control (SPC) and compliance with IATF standards (Advantive Co, 2024). These tools offer capabilities such as:

- **Automated Data Collection:** Real-time data acquisition from diverse sources, including metrology tools.
- **Real-Time SPC Charts and Alarms:** Immediate, actionable intelligence through real-time control charts and automatic alerts for process deviations.
- **Built-In Quality Reports:** Accessible, professional quality reports for quick dissemination across the organization.
- **System-Wide Audit Trail:** Comprehensive, time-stamped records of data changes and process history.
- **Data Analysis:** In-depth trend investigation and historical data analysis, supporting both real-time and retrospective quality assessments (Advantive Co, 2024).
- **Executing tests using the digital copy of the product or vehicle and simulating crash tests, autonomous driving and other scenarios to enable a better understanding of the various aspects of the vehicle that could be improved (Damjanovic-Behrendt, 2018, as cited in SUNDARAM and KARTHIK, 2023).**

Fig. 13 describes the process of developing our framework that integrates digital twins with quality assurance.

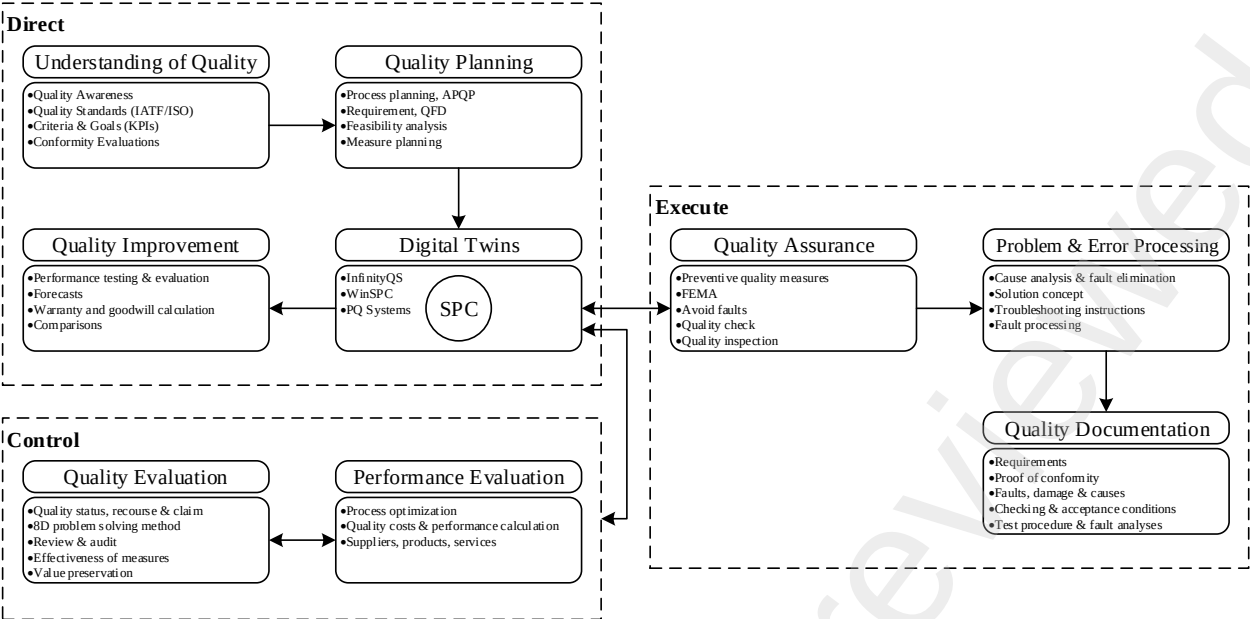


Fig. 13. The Framework That Integrates Digital twins with Predictive Quality Assurance (Based on Wedeniwski, 2015)

The digital twin is placed in the direction level of responsibility because it can ensure control of the business domains in the company context. It also makes the necessary data for quality assurance predictions available at the beginning of the framework design. As shown in Fig. 13, quality assurance and the digital twin are continuously related .

3.2. Data Collection

As explained in the previous sections, the quality of model-based decision-making support strongly depends on the data, its completeness, fullness, validity, consistency, and timely availability. These requirements on data are of a special importance in supply chain risk management for predicting disruptions and reacting to them (Ivanov, 2018b, as cited in Ivanov et al., 2019).

Alongside the product master data and product structure data, the Product Data Management (PDM) system forms the core of the internal, technical product documentation. As part of this, numerous different technical documents from different manufacturing systems, such as drawings, models and calculation results, are connected with the vehicle parts. Additionally, the increasing product complexity requires a higher degree of data networking due to the great diversity of models in the vehicle classes (Wedeniowski, 2015). The rather technical PDM system can be given the primary focus of the virtual product and of supporting product development. In contrast, the material master sets and bills of materials, which serve the purposes of production planning and management, are updated into an economic ERP system (Wedeniowski, 2015).

We can use digital twins to collect data across the supply chain in various forms, such as geometry data noises and vibrations. We will also consider the new possibilities offered by data processing and networking to change the production environment of logistics, engineering and manufacturing. For example, the following technologies for the calculation and simulation of product properties are nowadays used as standard as part of Computer-Aided Engineering (CAE) to test and validate developments, engineering and product lifespans in the automotive industry (Wedeniowski, 2015):

- The finite element method for the simulation of physical properties of materials. These include, for example, the deformation of a vehicle during a crash, mechanical stress, and the vibration behavior of parts and components.
- Computational fluid dynamics is an established method of fluid mechanics. Simulations of vehicle aerodynamics are common as an alternative to wind tunnel trials. Alongside conservation laws of physics for mass, impulse and energy, empirical approaches such as turbulence or heat transfer are also used. Fluid dynamics also makes it possible to observe flow processes in systems which are optically inaccessible, such as motoric combustion processes.
- Multiple-body simulation is used for the numerical calculation of the kinetics of systems which are made up of non-malleable

components, the movement of which is limited by joints. The ability to move is described by categories such as position, speed and acceleration. These can also be used to define certain bearing forces in accelerations.

- Measurement of the overall noise level by simulating audible noises and noticeable vibrations, summarized using the terms noise, vibration, harshness. Sources of disturbance-causing noises and vibrations as well as their effects on passengers and environments are sought, for example by calculating the extent of the disturbance-causing noise component which is caused by the running motor or by the wind
- A Digital Mock-Up (DMU) is used for simulating assembly and dismantling, in addition to testing for collisions and buildability. For this purpose, all geometry data of all vehicle parts is brought together in their planned installation positions in a digital model for tests. Specifically, in the case of assembly and dismantling simulations, production suitability is tested, and critical distances are measured (Wedeniowski, 2015).

In addition to the PDM system, the data and documents must also be validated, administrated in a structured way and made available in downstream phases in the Document Management System (DMS) and Content Management System (CMS). The product documentation is a very extensive part of the manufactured product, and must be delivered at a consistent level of quality. There are standards such as (German Institute for Standardization) DIN 6789, guidelines, such as the (Association of German Engineers) VDI guideline 4500, and various publications (Kothes, 2011, as cited in Wedeniowski, 2015) which structure the technical documentation and place it in a specialized context. This system can be used for the business domains and competencies concerning determining initial KPIs for the proposed model in Fig. 15. Table 6 provides the documents for identifying the initial KPIs created in the life cycle of a vehicle (Wedeniowski, 2015) . Also, according to the reference model used in this research, the information of each subsystem, such as corporate data, customer data, contracts and databases that will be susceptible to use for analysis is highlighted in green in Fig. 2 .

Table 6
Documents Used to Identify Initial KPIs (7 Wedeniowski, 2015)

Business Domain	Business Competence	Documents
Research and Development	Planning, Requirement, Change	Market Study, Target Catalogue Requirements List Target Prices
	Research and Preliminary Development	Research Results Patents
	Standard, Method, Process	Guidelines Certificates

Process Descriptions	
Testing and Validation	Test and Measurement Data Test/Audit Reports
Sketches, Drafts	Concept, Design, Construction Space CAD Data Concept Booklet
Development and Engineering	Product and Requirements Specifications Bills of Materials Circuit Diagrams
Procurement and Inbound Logistics	Calls for Tender Orders Contracts
Production	NC Programs Assembly Instructions
Marketing and Communication	Brochures, Photo Material Campaigns
Sales and Outbound Logistics	List Prices Sales Catalogue
Finance Services	Financing Contract Account Statement
After-Sales Support	Repair Orders Operation Instructions Maintenance Intervals Repair Shop Equipment Complaints Disposal Order
Quality	Quality Test Data Warranty and Goodwill Calculation

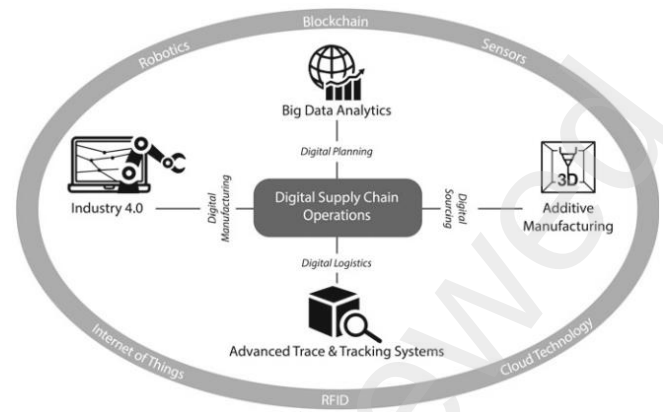


Fig. 14. Digitalization Framework of Supply Chain Risk Management (Ivanov et al., 2019)

Industry 4.0, Big Data Analytics (BDA), Additive Manufacturing, and advanced Tracking and Tracing (T&T) technologies can be considered as digital enablers of the four major SC processes in the Supply Chain Operations Reference (SCOR) model, i.e., plan, source, make and deliver, respectively. A digital version of the SCOR model would therefore consist of digital planning, digital manufacturing, digital sourcing, and digital logistics. IoT, cloud technology, robots, and sensors form the technical guts of a digital SC (21 Ivanov et al., 2019). This classification and division of the automotive supply chain into subdivisions will be further used in the paper to analyze subsystems or performance areas with similar functions according to related KPIs. Performance areas should be considered by a specific range that can be a part of a business system and not the whole system. The main benefit of dividing a system into several performance areas is the quick identification of bottlenecks and the compilation of more accurate and achievable goals to solve those problems.

For each of the areas, Fig. 14 suggests possible applications of digital technology with regards to SC disruption risks. For example, additive manufacturing can reduce supply risk by creating the opportunity to replace missing materials with the 3D printed components. BDA can be used at the planning stage to identify supplier risk exposure. T&T systems can help at the reactive stage to monitor and identify disruptions. At the same time, it needs to be noted that digital technologies may have multiple applications, which are not restricted to a particular SCOR process (Ivanov et al., 2019).

Network design, with regard to the geographical location of sites, is the core of most SC simulation models. GIS maps are used in simulation models to locate the sites, and calculate distances, routes, and travel times along real roads. In addition to geospatial calculations, they provide visualization and transparency in a model. (Ivanov et al., 2019)

From a methodological perspective, some of the machine learning techniques that are applied in supervised learning are K-NN, logistic regression, Support-Vector Machine, and Decision Tree to improve resilience supplier selection (Khan, 2024). Although some studies have developed a hybrid technique of machine learning and simulation to improve resilience using unsupervised learning, they have had limitations in using a limited number of parameters, such as delivery time and quantity, as well as small datasets and often without the use of relevant KPIs.

A simulation model that considers all the aforementioned factors can be the basis for building a successful digital twin of a physical SC that can be used for complex analysis of SC failures. This study proposes a predictive quality framework for the automotive SC industry and shows a perspective on how digital twins can help integrate resilience and quality into one framework.

3.4. Validation Approach

Digital validation is based on the data of CAD and CAE systems that makes it possible to use virtualization to create interactive models in the context of the whole vehicle. Such virtualizations can begin at a very early stage of development, for example with the exterior design as part of Computer-Aided Styling (CAS), but also in detailed designs of exteriors and interiors (Braess and Seiffert, 2007, as cited in Wedeniwski, 2015). Despite the technological possibilities and advances made in terms of digital models, physical validations, such as those provided through test series, remain necessary and indispensable.

3.3. Technology Integration

The intelligent networking of machines and processes with the help of digital technologies is creating autonomous, Internet-linked, self-regulating production systems, popularly termed as Industry 4.0. Industry 4.0 seeks to predict and visualize the performance of processes, plants, SCs, and product properties on the basis of information available in real time (Ivanov et al., 2019b, as cited in Ivanov et al., 2019). For this purpose, smart sensors are applied to capture and communicate information and requirements comprehensively to any recipient in real time. Intra and external connectivity are provided by IoT via digital communication devices. (Ivanov et al., 2019) Industrial robots are a part of Industry 4.0, found mainly in series production and warehousing applications. Robots perform high precision tasks independently as also support employees in their work, by handing over tools, for example. Their use accelerates, facilitates, and simplifies production activities. Unlike conventional industrial robots, which require time-consuming training, flexible robots learn quickly from people. They communicate with one another via the cloud, and support optimal production planning. In practice, however, questions still remain on several issues such as the ownership of cloud data among SC partners, or machine intercommunication protocols (Andelfinger and Hänisch, 2017, as cited in Ivanov et al., 2019)

In practice, new cloud-based analytics platforms such as SupplyOn Industry 4.0 Sensor Clouds make it possible to control the SC in real time, and plan and adjust processes using up-to-date information. By simply clicking on a container type, the graphs indicate whether there has been a violation of the defined temperature or humidity limits along the time axis. The data analysis in this chart allows a quick identification of all orders where the lead time was exceeded, allowing for a quick identification of questionable transports. Summarizing, the following SC digitalization framework can be presented (Fig. 14) (Ivanov et al., 2019).

Even if this is only about confirmation readings, these are still necessary for proofs of feasibility and validating product liability. In addition to the measures for minimizing the consequences of accidents, to limit the harm which can be expected to come to passengers (passive safety elements such as the seatbelt, airbag or passenger space), it is increasingly frequently the case that measures for the prevention of accidents are gaining significance thanks to assistance systems (active safety systems such as the anti-lock braking system ABS, traction control system TCS or electronic stability program ESP) (Braess and Seiffert, 2013, as cited in Wedeniwski, 2015). Due to safety systems in particular, in practice, a combination is needed of digital computations and physical tests in order to validate the vehicle to be developed in a targeted way (Kramer, 2009, as cited in Wedeniwski, 2015). The final integration and product validation will occur after the sales of the vehicle and not anymore before as it is today (Wedeniwski, 2015).

Moreover, we use a multi-layer perceptron neural network (MLPNN) presented by (Ogunsoto et al., 2024) and (Ogunsoto, 2023) that is trained using simulated case scenario data features to forecast supply chain recovery after disruptions. The MLPNN considers multiple disruptive inputs and monitors mean squared errors (MSE) during training and validation (Ogunsoto, 2023).

4. Framework for Predictive Quality Assurance

4.1. Conceptual Model

While many supply chain researchers have so far focused on investigating

risk and resilience strategies (Bakshi and Kleindorfer, 2009, as cited in Raaymann and Spinler, 2024), only some have investigated how to assess and measure the performance of a supply chain towards resilience (Hohenstein et al., 2015, as cited in Raaymann and Spinler, 2024).

For companies to know the resilience of their supply chain, an easy and quantitative way to measure supply chain resilience (SCR) must be found. Such measures must be implemented into the operations of a company's management so that SCR becomes an integral part of management processes. So far, companies along the automotive value chain measure the performance of their supply chain and, in some cases, pursue project-based risk assessments. This approach should move towards constant resilience monitoring through measurable KPIs. This study derives which KPIs to include in SCR monitoring for the automotive value chain. (Raaymann and Spinler, 2024)

We aim to use a set of KPIs that can be used in various divisions to manage the automotive industry for quality assurance and does not require complex data collection methods to measure SCR. This approach allows automotive companies to implement the SCR measurement method and KPIs defined in this study, and thus ensure SCR. Also, by comparing with the IATF standard, it is possible to compare the current and desired status of an automotive company and use the necessary predictions to identify system disruptions.

Considering the problems faced by traditional quality assurance technology, such as low traceability, and unpredictability, a digital twin approach towards traceability and dynamic assurance for supply chain quality is proposed in this paper. Thus, we designed a conceptual framework, as shown in Fig. 15, to clarify our research avenue.

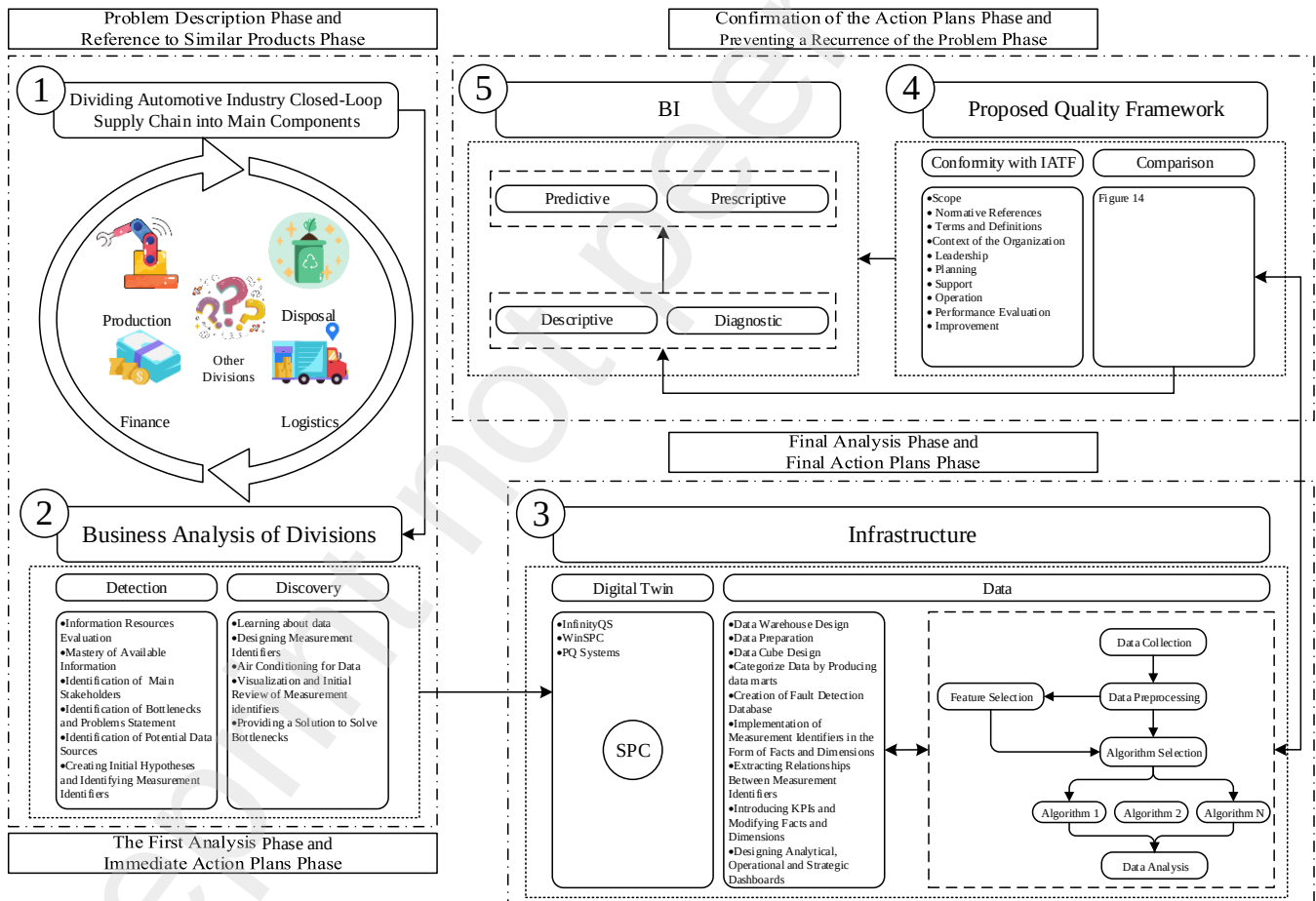


Fig. 15. Conceptual Framework for Automotive Industry Supply Chain Resilience through Predictive Quality Assurance

4.2. Components of the Framework:

As mentioned in the previous sections, we have designed our proposed framework based on the clauses and requirements of the IATF standard arranged according to the 8D method. In the following, we explain each section concerning the conceptual model.

4.2.1. Dividing Automotive Industry Closed-Loop Supply Chain into Main Components

In Fig. 2 which is our reference model for business architecture, we showed the main elements of the supply chain for today's automobile industry. This section is implemented according to clause 1 to 3 of the IATF standard as well as steps 1 and 2 based on the 8D method.

4.2.2. Business Analysis of Divisions

To implement this section, we consider clause 4 of the IATF standard for initial analysis, which corresponds to step 3 in the 8D method. Before analyzing a system, like automotive industry supply chain, knowledge must be obtained from it, and the system must be dissected to understand. In other words, the system must be divided, and its subsystems (performance areas with similar functions) determined. Performance areas should be considered by a specific range that can be a part of a business system and not the whole system. The main benefit of dividing a system into several performance areas is the quick identification of bottlenecks and the compilation of more accurate and achievable goals to solve those problems. For this division, in addition to knowing the totality of the system, identifying and determining the following items is of interest:

- Boundaries of the system
- Elements and components of the system
- Types of mutual relationships between the system components
- Behavior and performance of the components alone and concerning each other
- Characteristics and specifications of each component
- Hierarchy of system constituent elements
- External elements affecting the system
- Positive and negative feedback flows
- System resources
- Factors which cause disturbance

Furthermore, determining specific steps to recognize the business in the relevant sectors and the support of the stakeholders is essential for their realization. In addition, based on clauses 5 to 6 in the IATF standard and also, step 4 for the 8D method, we need to analyze and act immediate actions with the help of the roles and responsibilities regarding teamwork as well as business analysis, determining initial KPIs and objectives for different divisions .

– System Engineering

Various techniques can be used to obtain information. The important step is that the key stakeholders of the processes are identified, and it becomes clear what information and inputs are needed to do each task and determine the sources of each input. It is also necessary to specify the outputs of each activity, as well as its recipients and why they receive it. With the help of comparison and the use of specialized tools, the activities will be analyzed. Then the performance of each activity will be evaluated to see whether or not they are consistent with the current goals. The results obtained are compared with those of the current conditions. In these comparisons, specific contradictions and activities that have unnecessary outputs will be identified, and processes will be designed, which will cover the discovered needs of the business in question .

– Determining Goals

In planning to develop a business, one of the important steps is to adopt appropriate goals for the units and processes of the relevant organization to solve existing problems and dilemmas. For this purpose, the definition of goals is a basic prerequisite. After identifying the bottlenecks of a business, it is possible to encounter a large number of them during the compilation of goals, and only those that are critical should be selected based on importance and prioritization. Goals must be specific, measurable, achievable, relevant, and time-bound (SMART). To determine the goals, the success factors of the organization or the department under study should be determined. Success factors are several key activities that a company, organization or even an individual should focus on to achieve success. Success factors are specified conditions that measure the extent to which a business's goals are achieved over periods. A success factor starts with an action and then specifies the things that must be considered. Success factors always combine the elements of a measurable activity and a specific period .

– Determining Measurement Indicators

Measurement indicators are to emphasize and focus on the dimensions that play a vital role in the present and future success of the organization. A measurement indicator is a measurable value that shows how efficient the organization is in achieving its goals. For the measurement indicators to be meaningful, they should be determined according to the desired goals and the success factors of each goal. A key indicator for measuring performance is

meaningless without a clear goal because it does not help in determining the current position of the organization and its relationship with the desired path and position. Also, measurement indicators are calculations of metrics, and metrics are calculations of criteria and are always expressed as rates, averages, ratios, or percentages.

It should be noted that there are many criteria and metrics to measure. However, only a few of these metrics provide useful information about performance in a given subject. It is important to mention that all measurement indicators are metrics, but not all metrics are measurement indicators. After the measurement indicators have been determined, the constructive criteria of the measurement indicators should be determined. Therefore, these criteria and necessary calculations must be determined to specify them .

– Designing the birth certificate of measurement indicators

The birth certificate of the indicators will help to resolve the ambiguities, speculations and contradictions that may occur during the collection of required data or reporting of the indicators. If key performance indicators (KPIs) become the basis of strategic decisions, everyone in the organization will know what each indicator means, why they are measured, how reliable each indicator is, where the data comes from, and how they are prepared. The issues raised in the birth certificate include the following :

- Goal name
- Indicator name
- Responsible for indicator and access limits
- The question that the indicator answers and how to use the data
- Data collection method
- The purpose and performance limits of the indicator
- Formula
- Scale
- Indicator evaluation limits
- Data source
- Frequency of data collection and frequency of reporting
- Measuring person
- Reviewing person
- Date of review and use
- The degree of reliability of the indicator
- Ease and practicality of using the indicator

– Determination of data related to measurement indicators

To achieve any measurement indicators, related data that are determined by the desired criteria are needed. In this situation, regardless of the method of storing, retrieving and maintaining data, the only thing that is considered important is whether the data that is needed is available or not. It is expected that the data will be complete, without defects, and regular to obtain a correct evaluation of the current and future state of the business system. In the absence of necessary data, it is urgent to provide them, otherwise, it will lead to disregarding the relevant measurement indicators and accordingly the desired goal. Data can be quantitative or qualitative. It should be noted that qualitative data are suitable for a deeper understanding of business topics, but quantitative data are specifically used to calculate metrics. Therefore, it is necessary to quantify qualitative data according to the need and to be careful in preserving its nature. It is also important to check whether the required data exists explicitly .

4.2.3. Infrastructure

According to clauses 7 and 8 of the IATF standard and steps 5 and 6 of the 8D method, supporting the implementation of IT infrastructure, proper use of resources, communication, and documentation is essential. Ultimately, improved KPIs will be achieved using AI and operational data mining. That is why delivering digital twins can help us achieve our goals. Before using digital twins, the following points should be considered :

– Determining how to collect and record data

Apart from using digital twins in this research, according to each business and the importance and need of related data, system users are obliged to collect and record the desired data. Also, after identifying the criteria, it is necessary to consider a mechanism to record the required data that has not been in the system so far and to ensure the accuracy and correctness of their registration. If those responsible for data collection and recording do not act based on the established

mechanism, the next steps will not guarantee the creation of a healthy system for decision-making in any way .

– Examining suitable tools for data collection and analysis

To use the appropriate tools, special attention should be paid to the existing software and hardware platforms of each business. Paying attention to factors such as economy, required abilities, receiving new versions of software tools, etc., will result in getting the most productivity related to that tool at the lowest cost. It is possible that despite all the economic and security considerations, it is necessary to use tools with special features that will require decision-making at the top levels of each organization .

4.2.4. Proposed Quality Framework

Since our goal in this research is a predictive quality assurance framework to prevent possible defects in the future, we need to check the results obtained from the previous section before applying them, to modify them if needed. So, according to Fig. 14, We have considered a quality framework for their comparison. On the other hand, the results should be based on the IATF standard. Therefore, this section is considered according to clause 9 of the IATF standard as well as step 7 for the 8D method. The following are the components of the proposed framework:

– Understanding of Quality

The understanding of quality represents what a company can and wants to do. Every automotive company needs a framework for the quality awareness it would like to represent in its orientation. Alongside the foundations of the normalized quality standards such as ISO 9000:2015 (ISO, 2005, as cited in Wedeniwski, 2015), companies must also think about the criteria and goals they need in order to achieve far-reaching quality on the markets (Hoyle, 2009, as cited in Wedeniwski, 2015). Limitations on the company's capabilities come about for the company as a result of the skills of staff members, and processes. This is how the company's understanding of quality is derived, and it recognizes the areas it needs to take action in. For products which are subject to basic legal safety requirements, as is the case with vehicles, the conformity evaluation has crucial significance in terms of being able to prove compliance with legal requirements internationally, and achieving approval in different target markets (Wedeniwski, 2015).

– Quality Planning

Prevention is the basic principle of quality planning, the implementation of which is an important element of quality in the automotive industry in order to be able to turn requirements into specific quality characteristics. The quality required is guaranteed by test procedures and methods on the basis of process planning. For this purpose, American vehicle manufacturers have created the reference handbook "Advanced Product Quality Planning and Control Plan (APQP/CP)". A similar approach for German vehicle manufacturers can be found in the German Automotive Industry Association (VDA) (VDA Verband, 2011, as cited in Wedeniwski, 2015). In collaboration with the supplier, a feasibility analysis of the manufacturability of all requirements from different sources must be carried out and documented. This includes, for example, the suitability of the design and material, in addition to the possibilities provided by machines and processes in order to manufacture products to comply with the required specifications. As part of quality assurance-focused measure planning, one concentrates on the influencing factors of flexible elements throughout the entire value creation chain, deviations from which are recognized early so that measures can be introduced to counteract them (Wedeniwski, 2015).

– Quality Improvement

Quality improvement should constantly ensure that the quality of the product is increased, with the help of suitable measures within the company. An important element in this is the building up of corresponding structures which promote the involvement of employees. Alongside preparation and planning, the criteria for the performance testing and evaluation of results are significant in terms of goal-oriented, long-term quality development. There are numerous models which can be used to achieve a higher degree of precision in forecasts. The following questions usually play a role in predicting the behavior of the system in the event of quality problems which can be expected to occur:

- How many and which faults can customers expect to encounter after

the vehicle has been delivered?

- Under what circumstances and in which system parts can faults occur?
- Which risks are hidden behind faulty system parts?

While vehicles fulfil high quality requirements, they are not free from faults. This is why a warranty and goodwill calculation is indispensable in order to evaluate risks. Often, the evaluation of warranty cases to be expected is calculated on the basis of experience values and payment flows from over the past years. On the one hand, comparisons on the market are important in order to market what makes a product stand out, on the other, they are also vital for identifying measures which are necessary to improve quality on the market. Internal benchmarking is also important in order to bring about successful learning from past mistakes. Things become difficult when we come to the connected vehicle, which constantly features new functions which nobody has experience of (Wedeniwski, 2015).

– Quality Evaluation

The quality status is monitored using quality indicators from target and actual statuses. The figures collected show, for example, how high the percentage of fault free performance units is in comparison to the overall number of products made or sold. Using these evaluations, a desired quality level can be planned, managed and controlled. The systematic complaints evaluation of recourse and claims pursues the goals of safeguarding endangered customer relationships, while also improving the quality of products, services and processes. This can be the case, for example, using the Eight Disciplines (8D) problem solving method (Brückner, 2011, as cited in Wedeniwski, 2015), the report of which is standardized by the VDA. Specifically, planned reviews and audits check the effectiveness of measures in quality planning, and all business competences in execution. If a high enough degree of effectiveness cannot be proven, then the measure planning needs to be looked at again. Long-term quality includes vehicles' durability and reliability. From this follows an optimal value preservation of used vehicles – the checking of which demands comprehensive quality evaluation (Wedeniwski, 2015).

– Performance Evaluation

The performance evaluation is oriented towards operative and tactical control, and has the aim of maximizing profit in the short to medium term. It therefore orients itself primarily towards the evaluation of in-house process optimizations and their economic viability, as well as efforts which need to be made to warranty customer satisfaction. High quality should be guaranteed while keeping costs competitive by using a quality costs and performance calculation. This is why continual measurement and evaluation is needed at the in-house interfaces of suppliers, products and services (Wedeniwski, 2015).

– Quality Assurance

Quality assurance refers to all measures which warranty a constant level of product quality. These include "preventative quality measures" in continual execution, with the help of the "FMEA" (Werdich, 2012, as cited in Wedeniwski, 2015), a structured, systematic work technique for the recognition of functional contexts in order to "avoid faults" early on. As mentioned in the introduction section of the paper, FMEA is one of the five pillars considered by automotive companies based on the QS 9000 system and the basis of the IATF standard. Moreover, FMEA is standardized as part of the standard DIN EN 60812 for analysis techniques for system reliability. The "quality check" serves to determine the extent to which the test characteristics and requirements of the quality plan are being fulfilled. The "quality inspection" takes place over several phases in order to prevent faulty parts or assemblies from being passed on or processed (Wedeniwski, 2015). This paper effectively considers the combination of FMEA and digital twinning to provide an online vehicle diagnosis, thereby realizing predictive quality assurance .

– Problem and error Processing

During problem and error processing in terms of content, the problems identified during quality assurance, individualized, unrelated faults, or far-reaching structural problems, must be formally prepared for the cause analysis and fault elimination. Then, after analysis and classification, they must be put on an effective path to a solution concept which can lead to troubleshooting instructions both within and outside the company. In customer service, dealing with an enquiry rapidly and respectfully during fault processing can affect

4.2.5. Business Intelligence (BI)

4.3. Digital Twin Architecture

A conceptual framework that supports an integrated product-process DT has been proposed where it is envisioned that the process DT can be used to develop the product DT, and the product DT can be used to provide process configuration specific to the product to the process DT (Onaji et al., 2022, as cited in Visser et al., 2024). To better integrate the physical and virtual spaces in a workshop, three types of DTs have been proposed (Bao et al., 2019, as cited in Visser et al., 2024): a product DT, a process DT, and an operation DT. These DTs are then later integrated to achieve collaborative control of ‘people-machine’, ‘machine-machine’, and ‘machine-object’ in the workshop. Integration between DTs is achieved through a data bus that enables the DTs to synchronize and exchange data (Bao et al., 2019, as cited in Visser et al., 2024). A similar approach, based on the product lifecycle management, proposed a digital twin of the manufacturing process that comprises of all the related DTs within that process, e.g. the DTs of the manufactured products, machine resources, and human resources (Schützer et al., 2019, as cited in Visser et al., 2024).

A more general architecture shown in Fig. 16 was developed to represent a complex physical system (Human et al., 2023, as cited in Visser et al., 2024) and is one of the most broadly applicable architectures. Also, according to this research's concerns, all of the data related to the thorough supply chain have been considered.

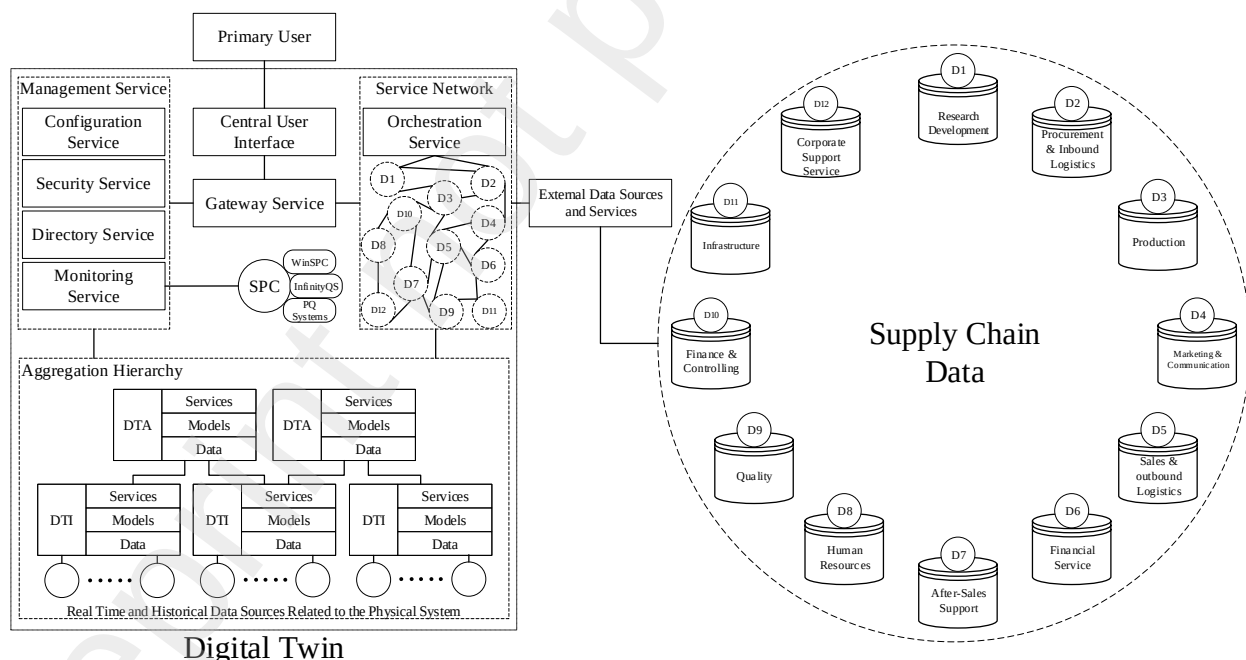


Fig. 16. Digital Twin Architecture (Adapted from Human et al., 2023, as cited in Visser et al., 2024)

Data integration refers to the practice of combining data from possibly

The data providers could be internal or external suppliers of product related

data, enterprise resource planning systems, production systems, etc. The DT system serves as the single point of truth for product and production related data for the data consumers. The context of an automotive manufacturing industry and discussions with an industry partner led to the functional requirements (FRs) for a DT system that integrates a product and productions DTs in Table 7 (Visser et al., 2024).

Table 7

Functional Requirements (Visser et al., 2024)

ID	Description
FR1	Ingest selected product and production related data from multiple, heterogeneous, external data sources with varying levels trust related to the longevity and integrity of the source data
FR2	Before loading data into the DTs, transform the newly ingested data to a data model common within the DT system and, if required, each DT's specific data model
FR3	Integrate the related product and production data generated during the production phase within the DTs
FR4	Validate the ingested data through predefined rule-based data consistency and data completeness checks
FR5	Provide decision support through: tracking the progress of the ingestion of production data and identifying products based on user-defined production conditions (e.g. if a product recall is required)
FR6	Generate data exports and real-time notifications to data consumers, e.g. internal DTs, external systems (such as DPP generators), or user interfaces

In addition to the FRs, the DT system should meet selected Non-Functional Requirements (NFRs). ISO 25010 (ISO, 2023, as cited in Visser et al., 2024) is a widely accepted standard for software quality attributes. For DT applications, extensibility, scalability (not expressly included in an earlier version of ISO 25010), and re-configurability should be considered in addition to the software quality attributes from the ISO 25010 (Ferko et al., 2022, as cite in Visser et al., 2024). For this paper, the most significant NFRs are performance efficiency, (re)configurability, and scalability, while the other NFRs are more relevant to implementation decisions (Visser et al., 2024).

4.4. Predictive Quality Assurance Model

We consider the presented method by (Zhang and Ouyang, 2024) according to Fig. 17 that proposes a quality performance evaluation method based on graph convolutional neural networks to analyze the quality performance represented by knowledge graphs. This method not only integrates and analyzes complex multi-source heterogeneous data but also synthesizes information on key indicators of quality performance through graph neural networks to achieve accurate evaluation and optimization. The core innovation of this paper lies in embedding the knowledge graph as input to the Graph Convolutional Network (GCN), aiming to enhance the accuracy of performance evaluation through the rich data structure and semantic relationships of the knowledge graph. In terms of multisource heterogeneous data fusion processing, this paper employs a representation learning method based on knowledge graphs, using GCN for embedding learning of multi-source heterogeneous data, which involves dividing the knowledge graph into hierarchical levels to construct both a global knowledge graph and domain-specific knowledge graphs. The global knowledge graph represents key entities and relationships in the entire supply chain (Zhang and Ouyang, 2024), such as product design, production steps, quality inspection, etc. The domain-specific knowledge graph provides a more refined description for specific functional modules and data differences. This process uses multiple layers of GCN to extract and learn features from each node in the graph (such as equipment, quality issues, quality indicators, etc.), aggregating information from adjacent nodes layer by layer. This allows each node's representation to include not only its own information but also information from nodes it is directly or indirectly connected to, thereby capturing global performance evaluation indicators (Zhang and Ouyang, 2024).

In the training process of the performance evaluation model, this paper optimizes the GCN model using supervised learning. The actual performance from historical data is used as a supervisory signal, and the model is optimized by minimizing the error between predicted results and true performance indicators. Additionally, the paper employs a regularized loss function design, integrating knowledge graph embedding and local performance data to create a hybrid loss function that constrains performance indicators at different levels

with embedded representations. This allows the model to focus not only on the fitting effect of local data but also to fully utilize the information from the global knowledge graph, thereby enhancing the overall accuracy of performance evaluation. To further enhance the model's predictive capabilities, the paper also introduces a multi-task learning mechanism, which allows the model to learn the potential relationships with other related indicators while evaluating a certain type of performance indicator, forming a multilevel evaluation system. In this way, GCN can not only make accurate predictions for a single indicator but also identify the coupling relationships between different quality indicators, providing a more comprehensive performance evaluation.

To enhance the GCN model's generalization ability in different scenarios, this paper will also explore how to strengthen the model's performance through the expansion and optimization of the knowledge graph. This involves dynamically updating entities and relationships in the knowledge graph to ensure its timeliness and accuracy. Furthermore, the paper will introduce knowledge distillation techniques, using the pre-trained language model RoFormer to optimize embedded representations. By combining self-attention mechanisms with positional encoding, the paper can further enhance the GCN model's perceptual capabilities, enabling it to more accurately evaluate performance in complex quality performance evaluation scenarios (Zhang and Ouyang, 2024).

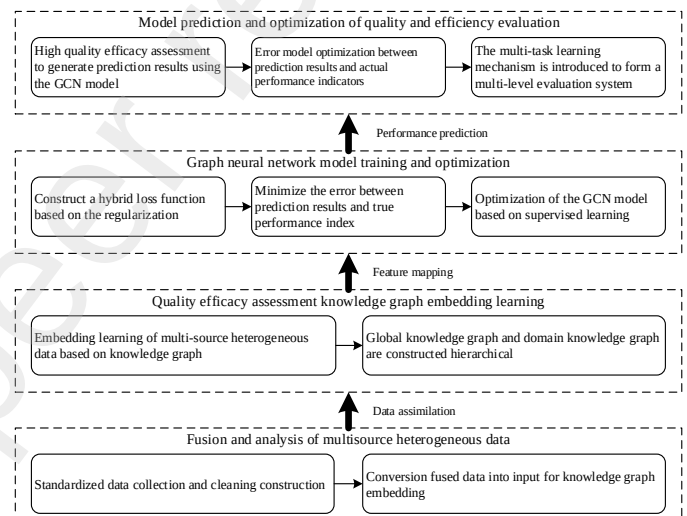


Fig. 17. Quality and Efficiency Assessment Technology Based on Graph Convolutional Neural Network (Zhang and Ouyang, 2024)

4.5. Resilience Metrics

According to done research by (Kenanga and Ardi, 2022) presented in Fig. 18, From the 11 indicators and 46 sub-indicators submitted through a questionnaire, all indicators and 27 sub-indicators were declared valid. At least 5 of the 6 experts agreed with the proposed sub-indicators. The total value of Scale-Content Validity Index (CVI) is 0.81, according to the minimum limit determined by Polit et al. (2006) (as cited in Kenanga and Ardi, 2022) for the universal agreement approach. It indicates that the consensus from the expert was good. The results of indicators that were declared valid are shown in Table 8.

Table 8

List of valid indicators and sub indicators by expert (Kenanga and Ardi, 2022) (Visser et al., 2024)

Indicators	Sub Indicators	Number of expert agreeing	I-CVI
Security	Cyber Security	5	0.83
	Redundancy in IT System	5	0.83
Knowledge Management	Training	6	1.00
	Team Work	6	1.00
Visibility	Information Technology	6	1.00
	Knowledge of Operating Asset	5	0.83
	Supply Chain Visibility	6	1.00
Risk Management	Monitoring	5	0.83

Collaboration	Contingency Planning	6	1.00
	Risk Sharing	5	0.83
	Production Planning	5	0.83
	Supplier Delivery Efficiency	6	1.00
Agility	Quick Response	6	1.00
	Stock Level	6	1.00
	Capacity Utilization	6	1.00
	Order Lead Time	6	1.00
Flexibility	On Time Delivery of Goods	5	0.83
	Product Variation	6	1.00
	Supplier Contract	6	1.00
Efficiency	Product Variability Reduction	5	0.83
	Preventive Maintenance and Repair	6	1.00
Redundancy	Change in Production Plan	6	1.00
	Recovery to Shutdown	6	1.00
Financial Strength	Insurance	5	0.83
	Price Margin	6	1.00
Market Position	Product Differentiation	6	1.00
	Customer Relationship	6	1.00

4.5.1. Security

Security is an essential element in the supply chain and must be designed in such a way as to continuously secure the company assets. Developing operational security means protecting companies from various types of harm related to manmade disruptions, cyber-attacks, or physical (Chowdhury and Quaddus 2017; Karl et al. 2018; Lam and Bai 2016, as cited in [Kenanga and Ardi, 2022](#)). Cyber security is one of the examples that companies must pay attention to because it protects computers and networks from malicious cyber-attacks, prevents security breaches, and avoids unauthorized access to sensitive information. Moreover, redundancy in the IT system is vital as backup resources to support the system if there is a disturbance and ensure the company's stored data security ([Kenanga and Ardi, 2022](#)).

4.5.2. Knowledge Management

Knowledge management is a way to understand the organization's environment (Karl et al. 2018, as cited in [Kenanga and Ardi, 2022](#)). Understanding supply chain operations, needs, threats, and human resources are the first step to creating SCR indicators. Under certain conditions, resilience must be prepared in the pre-disruption phase through practices like education and training from past experiences (Christopher and Peck 2004; Pereira et al. 2014, as cited in [Kenanga and Ardi, 2022](#)). With knowledge management, the company can focus on documenting knowledge and experiences from lessons that have been learned. It will help managers cope with the diverse types of interruptions. Training will accumulate knowledge and skills to finish each task to stay consistent or even better than the set target under any circumstances. It will form good cooperation between individuals and teams to support the company in overcoming any disruption ([Kenanga and Ardi, 2022](#)).

4.5.3. Visibility

Visibility is an important indicator to achieve resilience because it can visualize the inventory situation, demand, and supply from upstream to downstream of the supply chain (Christopher and Peck 2004). It functions as a warning strategy that gives companies critical time to align their capabilities in reducing the impact of a disruption. Thus, this indicator can be achieved and improved by using KPIs to monitor organizational assets, both tangible and intangible (Karl et al. 2018). Experts consider several sub-indicators essential and relevant, including information technology, knowledge of operating assets, and supply chain visibility. Information technology can help companies track every operational activity. Of course, this can ease companies in making decisions whenever a disturbance happens. Potential disturbances can come from within or outside the company. Hence, knowledge of operating assets is vital for operators working in production as a preventive measure. It is because all assets in the company must be used in a good and right way so that the assets can last for a long time and minimize disturbances from within the company. In addition, supply chain visibility allows companies to analyze and visualize customer expectations and follow global market business developments to

constantly improve over time (Sahu et al. 2017, as cited in [200 Kenanga and Ardi, 2022](#)).

4.5.4. Risk Management

Risk management is considered part of supply chain resilience by Christopher and Peck (2004) (as cited in [Kenanga and Ardi, 2022](#)) because many risks cannot be predicted or avoided. It can help lessen vulnerabilities through forecasting, monitoring, and risk mitigation (Karl et al. 2018, as cited in [Kenanga and Ardi, 2022](#)). Thus, with the risk management culture, the company must understand and share risks among partners. In this indicator, monitoring all the operations is considered essential and must be paid attention to together with contingency planning. It can increase preparedness and create mutual commitments between suppliers ([Kenanga and Ardi, 2022](#)).

4.5.5. Collaboration

Supply chain collaboration refers to working effectively with other partners for mutual benefit in forecasting, postponement, and risk sharing (Pettit et al. 2013, as cited in [Kenanga and Ardi, 2022](#)). Collaboration can also involve exchanging information, lessening uncertainty, increasing transparency, and facilitating the invention and sharing of knowledge, such as topics discussed supply chain risks and uncertainties (Christopher and Peck 2004, as cited in [Kenanga and Ardi, 2022](#)). Risk sharing, production planning, and supplier delivery efficiency are sub-indicators of collaboration that experts consider most relevant to the current situation. Risk-sharing between supply chain partners is conducted to share risk in an unexpected disruption. As we know, many companies invest in facilities and equipment at supplier's plants, so they are prepared to share risks with both suppliers and customers. Because each partner is related, production planning is also a problem when disruption happens. It refers to estimating the required resources, such as preparing detailed plans to achieve production goals economically, efficiently, and within the scheduled timeframe (Sahu et al. 2017, as cited in [Kenanga and Ardi, 2022](#)). Delivery of goods from suppliers is also an important thing that continues to be monitored and measured, especially during the disruption time when the rate of this indicator typically drops (Karl et al. 2018, as cited in [Kenanga and Ardi, 2022](#)). The greater the level of supplier delivery efficiency, the more effective cooperation between suppliers and company ([Kenanga and Ardi, 2022](#)).

4.5.6. Agility

In today's conditions, companies are required to be agile to survive. Christopher and Peck (2004) ([Kenanga and Ardi, 2022](#)) describe supply chain agility as a quick response to unexpected changes in demand and supply. It can be achieved by changing the company's systems and business processes (Karl et al. 2018, as cited in [Kenanga and Ardi, 2022](#)). Several sub-indicators stated as relevant in this indicator are stock level, capacity utilization, quick response, and delivery lead time. When a disturbance suddenly occurs in one of the supplier partners, the stock in the manufacturer's warehouse may not meet production demand due to a lack of supply. Hence, monitoring stock levels from suppliers and customers is an important KPI to avoid or mitigate the bullwhip effect and help managers mitigate risk and reduce inventory costs (Karl et al. 2018, as cited in [Kenanga and Ardi, 2022](#)). Understanding the capacity utilization every month can also improve the schedule and utilize the resources to increase the efficiency of operations. In this indicator, delivery lead time is also necessary for customer service by serving as feedback to control the day-to-day operations. Companies must swiftly and quickly respond to customers to increase their satisfaction ([Kenanga and Ardi, 2022](#)).

4.5.7. Flexibility

Flexibility is an enterprise's ability to adapt to the changing requirements of its environment and stakeholders with minimum time and effort (Tukamuhabwa et al. 2015, as cited in [Kenanga and Ardi, 2022](#)). The current pandemic pushes companies to be flexible with many changes that affect conditions within. It affects the on-time delivery of goods, product variations, and supplier contracts. On-time delivery of goods will help managers accomplish flexible operations to meet customers' needs (Karl et al. 2018, as cited in [Kenanga and Ardi, 2022](#)). In addition, the pace in replacing or adding product variations according to customer orders will also affect the effectiveness of operations within the company. Therefore, supplier contracts need to be flexible to minimize shortages (Karl et al. 2018, as cited in [Kenanga and Ardi, 2022](#)).

4.5.8. Efficiency

Efficiency is the ability to produce output with minimum resources (Pettit et al. 2010, as cited in Kenanga and Ardi, 2022). The pandemic has pushed companies to run their operations as efficiently as possible continuously. That is why reducing product variability, preventive maintenance, and repair measures are essential to current conditions. Reduced product variability in operations can prevent the bullwhip effect. Meanwhile, preparing all preventive actions for product maintenance and repair is necessary as the company's anticipation if a disturbance happens (Kenanga and Ardi, 2022).

4.5.9. Redundancy

Creating redundancy is the strategic and selective use of spare capacity and inventory that can be used to cope with disruptions (Christopher and Peck 2004, as cited in Kenanga and Ardi, 2022). Even though redundancy can be seen as duplicating the capacity to continue operations during a disruption, it is considered flexible (Tukamuhabwa et al. 2015, as cited in Kenanga and Ardi, 2022). Redundancy is considered to increase flexibility because it facilitates response by deploying resources and assets that can be customized (Singh et al. 2019, as cited in Kenanga and Ardi, 2022). It helps to avoid delays in processing operations and therefore improves SCR. Change in production plan and recovery to shutdown are examples of this indicator. Production plans can change in unexpected events, so they can be moved to another facility to keep production operating. In addition, recovery to shutdown is important because the shorter the recovery period, the smaller the loss due to damage caused by the disturbance (Kenanga and Ardi, 2022).

4.5.10. Financial Strength

Financial strength is the capacity to absorb fluctuations in cash flow (Christopher and Peck 2004, as cited in Kenanga and Ardi, 2022). The overall cash flow of any business tells whether that business is generating what it needs to sustain and grow. Pettit et al. (2013) (Kenanga and Ardi, 2022) wrote that two sub-indicators included as the company's financial strength are insurance and high margin, where both are relevant to the current situation. Companies with significant insurance coverage for facilities, equipment, goods, and personnel selling their products at a relatively high margin can support this indicator (Kenanga and Ardi, 2022).

4.5.11. Market Position

One indicator that is considered essential for creating SCR is market position. It is a company's status or its products in a specific market (Pettit et al. 2013, as cited in Kenanga and Ardi, 2022). A strong market position is related to the increase in market share. It allows investment in SCR, which undoubtedly helps maintain the relationship with the customers after an unexpected disruption happens (Pereira et al. 2014, as cited in Kenanga and Ardi, 2022). As we know, each company already has specific target markets so that each product produced can sell well in the market following customer needs. Product differentiation is required in every company because it distinct the products and makes customers more interested (Pettit et al. 2010, as cited in Kenanga and Ardi, 2022). That is why the Customer Relationship sub-indicator as a company's effort towards its customers should be paid attention to maintaining good relationships and retaining their customers (Chowdhury and Quaddus 2017, as cited in Kenanga and Ardi, 2022).

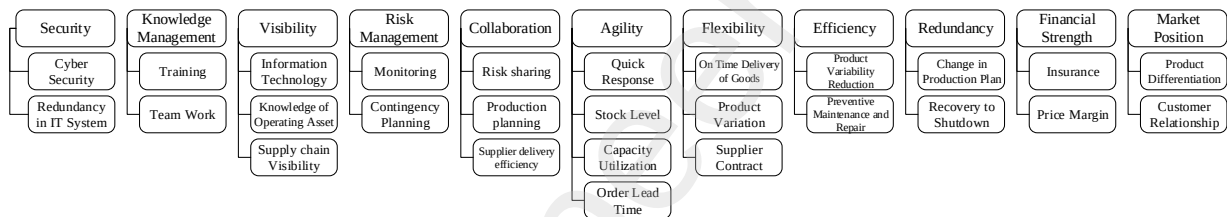


Fig. 18. Resilience Metrics in Supply Chain (Kenanga and Ardi, 2022)

4.6. Data Flow and Process Design

The data within the DT is stored within the warehoused data or federation log. The warehoused data is used to store any data that is stored long term within the DT, e.g. the bill of materials (BOM) of a product. The federation log maintains a record of all the federated requests made by the data federation service. Additionally, the federation log acts as a buffer for the data federation service in the case of data related to the same request arriving incrementally (Visser et al., 2024). The model for the data flow of the proposed DT system

is illustrated in Fig. 19. Its architecture is an application of the reference architecture for a system of DTs presented in Fig. 16. The architecture consists of the DT hierarchy, management services, services network, gateway service, source adapters, dashboards, external software system, available data in the supply chain, data warehousing, data analysis, identifying and improving KPIs, and an API service. Also, the relation between the supply chain elements has been determined based on the reference model presented in Fig. 2 (Visser et al., 2024).

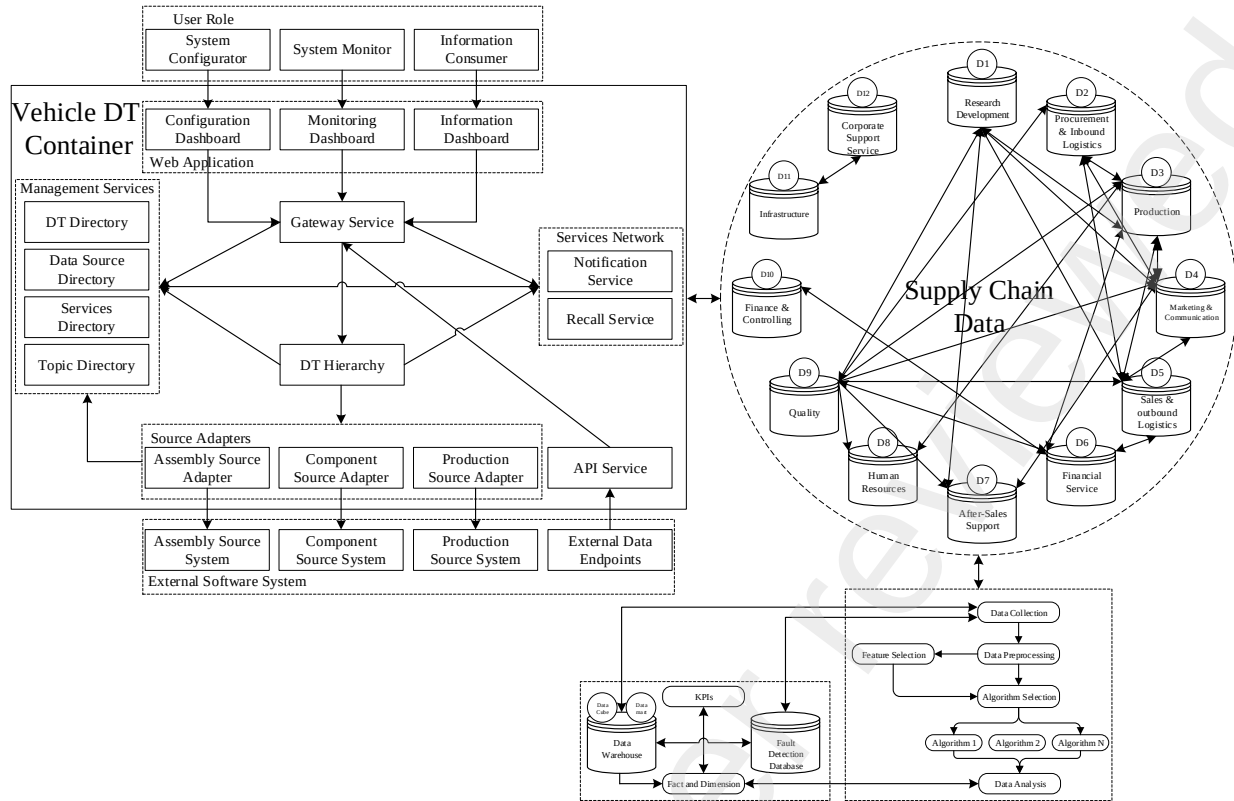


Fig. 19. Data Flow and Process Design (Visser et al., 2024)

4.6.1. Management Services

the management services support the components within the system. Here, four management services were identified: the DT directory, services directory, data source directory, and topic directory. The DT directory maintains all metadata related to the DTs within the system. This metadata could include the state of a DT (e.g. active or inactive), a description of the DT, how to interact with a DT (e.g. the communication protocol, what input parameters the DT is expecting, and the services offered by the DT), and the relationship the DT has with other DTs within the system. The topic directory contains all the metadata related to the publish/subscribe communication within the system. This could include the topics, the brokers used, a list of the publishers and subscribers for a given topic, and a description of the topic. The services directory contains all the metadata related to services within the services network, e.g. what input parameters the service is expecting and what the service can offer. The data source directory contains a complete view of all the interacting sources, what data they can provide, what method of data integration is required, how to interact with the sources, and their up-to-date states. Additionally, the source description, as described by Doan (2012) (as cited in Visser et al., 2024), is maintained within the data source directory which can be queried by the DTs in the scenario of data federation (Visser et al., 2024).

4.6.2. Source Adapters

In an automotive manufacturing enterprise, data sources typically function as standalone entities, with the interaction between the systems ensuring the overall needs of the enterprise. To isolate the peculiarities of each data source, source adapters are included in the architecture. These adapters are similar to the wrappers used in related research (Doubell, 2023, as cited in Visser et al., 2024): lightweight, custom developed software that provides a standardized interface between the DT hierarchy and the heterogeneous set of data source. The source adapters include additional functionally proposed by Doan (2012) (as cited in Visser et al., 2024) for a wrapper in the context of data virtualization or federation. In the context of an automotive manufacturing enterprise, types of source systems have been considered (Visser et al., 2024). There may be multiple source systems in each of these categories. It is assumed that the source systems are responsible either for interacting with the physical twins directly or for other databases related to physical twins indirectly (Visser et al., 2024), and,

therefore, all the DT data is ingested via the various source systems (Visser et al., 2024).

The source interface component serves as the entry point and is responsible for all communication with the source system. The source systems can publish their information using a variety of methods, e.g. Apache Kafka, APIs, etc. Furthermore, it is often not possible to force the source system to use a specific communication method. Therefore, the source interface can be configured to interact with the required communication method. Additionally, multiple source interfaces can be used within the same source adapter in the scenario where a source system uses different communication methods. The source interface transfers the incoming raw data into a raw data queue, with the queue acting as a temporary cache to ensure no data loss, even in the case of high traffic (Visser et al., 2024).

Once data enters the raw data queue, the data transformer component is triggered. The data transformer reads the raw data from the queue and determines what transformation is required to get the data to match the standardized data model used by the internal system. These transformations could include converting data from XML or CSV to JSON, mapping data fields to match the internal data schema, data aggregation, etc. After transformation, the data transformer loads the transformed data into the transformed data queue and removes the data from the raw data queue. When data enters the transformed data queue, the topic classifier is triggered and it reads the transformed data from the queue, identifies the appropriate topic(s) for the data, and publishes the data to the identified topic(s). Therefore, the topic classifier maintains the knowledge of the relevant topics and what data belongs to each topic for the specific source adapter. The DTs within the DT hierarchy serve as the subscribers in this context. Subsequently, the data is removed from the transformed data queue. The “SA request handler” serves as the intermediary between the DT hierarchy and source interface in the scenario of data federation: when a Requires federated data, a request is posed to the SA request handler, which formulates the request and forwards the request to the source interface. Therefore, the source interface is additionally required to send requests to the source system. The response to the federated requests mirrors the data flow described above, except that the SA request handler extracts the transformed data from the queue instead of the topic classifier and sends the data to the DT that raised the initial federated request (Visser et al., 2024).

4.6.3. Services Network

To meet the FRs identified, two services are allocated to the services network, i.e. a notification service and a recall service. The notification service generates notifications to the interested parties based on predetermined events within the system. For example, a DT triggers the notification service when an error has been identified. This provides a layer of abstraction between the components within the system and the interested endpoints receiving the notifications. The recall service identifies DTs based on conditions specified by external suppliers. For example, if faults in physical components are attributed to a specific manufacturing process, the recall service identifies all the DTs that could be affected by that manufacturing process (Visser et al., 2024).

4.6.4. Dashboards and API Service

The dashboards provide each type of user with a dedicated dashboard to fulfill their respective needs. The identified roles within an automotive manufacturing environment are the system configurator, system monitor, and information consumer. The configuration dashboard can be used to configure selected components within the architecture, e.g. updating one of the management services. The monitoring dashboard can be used to monitor the state or performance of the system, e.g. populating a table with all the active DTs in the system. Finally, the information dashboard can be used to visualize information related to results produced by the components within the system, e.g. populating a table with data within a DT. The API service provides a standardized interface for external data endpoints, which could potentially be external to the automotive enterprise. The API service could provide a robust authorization mechanism and broader functionality compared to the gateway service, given its interactions with potentially diverse external endpoints (Visser et al., 2024).

4.6.5. Gateway Service

The gateway service acts as the centralized access point into the DT system. Its main function is to route the requests and responses between the dashboards or API service to and from the management services, DT hierarchy, or services network. Additional functionality can be encapsulated in the gateway service such as authentication, authorization, and logging of ingress and egress activities. A load balancer can be used for the gateway service to improve the component's scalability (276 Visser et al., 2024).

4.6.6. Digital Twin Hierarchy

The DT hierarchy serves as a reflection of reality (Human, 2023, as cited in Visser et al., 2024), i.e. a reflection of the physical system, even though the architecture does not directly interact with the physical system. The first step to develop the DT hierarchy is to conduct a physical decomposition of the system, followed by mapping the physical system to the corresponding DTs within the hierarchy (Visser et al., 2024).

4.6.7. Data Warehousing

The data warehousing service is responsible for managing all warehouse related data, which could be loading data in the warehouse or extracting data from the warehouse upon requests from the custom services. Additionally, the data warehousing service can include transformation steps specifically related to the DT. It is envisaged that this service performs all the ETL related tasks for warehoused data within the DT. The data federation service is responsible for managing all federation activities within the DT. Note that the federation service would be capable of handling data virtualization, i.e. integrating data from a single external source. Because federation happens on query time, e.g. when a request for data that is federated is received, the data federation service can interact with the data source directory to determine where the federated request must be sent to. The response from the source adapter(s). Within a warehousing approach, data is loaded from external sources into an internal database, usually through an extract, transform, and load process (Doan, 2012, as cited in Visser et al., 2024). In data virtualization, the data remains within the source systems and is only accessed at query time by the consumers, through a unified view for data consumers. Data virtualization is achieved through three components: a mediated schema that the consumer interacts with, the source descriptions that maps the mediated schema to the relevant source system's schema, and a wrapper (4, as cited in Visser et al., 2024). The wrapper poses queries to a source, receives responses, and performs lightweight transformations on

the responses. Data virtualization is often associated with the term data federation, with some considering them synonyms but, according to (Lans, 2012, as cited in Visser et al., 2024), data federation always implies multiple data sources, whereas virtualization could imply a single source.

4.6.8. KPIs and Data Analysis

After completing the data transformations for the Initial KPIs, the data warehousing service must determine whether the data belongs to an existing KPI or whether a new KPI is required. If the data belongs to an existing KPI, the data is appended to that KPI's table. Otherwise, a new table is created in the data warehouse for the new KPI and a reference for the KPI is added to the data warehouse. When data is added to the data warehouse, the supply chain's data analysis service is triggered. This service can use a variety of methods to validate the data using the data schema (fact and dimension) and ML models (Visser et al., 2024).

5. Implementation Strategy and Challenges

5.1. Integration into Automotive Industry Supply Chain

Many insights can be drawn from lessons from implementations in other sectors. Through interviews with experts within digital twins in the automotive industry and cross-referencing findings with existing literature, several solutions can be considered when implementing a digital twin (BLOMKVIST, 2020).

5.1.1. Pre-Implementation

Primarily, it is crucial to gauge the needs of the organization so that implementing something as complex and resource intensive as a digital twin can be justified (Edström, personal communication, April 16th, 2020, as cited in BLOMKVIST, 2020). Furthermore, business cases need to be developed that showcase the value of adding specific data or further functionality to the digital twin, as well as the cost of both implementation, operation and maintenance. Having these business cases in place will also be beneficial when it comes to gauging the value added by implementing further functionality to a digital twin down the line (Edström, personal communication, April 16th, 2020, as cited in BLOMKVIST, 2020). When implementing a digital twin, the starting point should be in areas in which there is a lot of knowledge and experience, working with an agile approach, starting small, growing and nurturing skills within the organization. Meanwhile, given the scale of a digital twin implementation, the project must necessarily be backed (if not initiated) from a managerial standpoint before committing to the project (Norrblom, personal communication, April 1st, 2020, as cited in BLOMKVIST, 2020). One of the initial challenges is to create a visual model of the supply chain network, translating the material flows to a visual medium, clearly indicating which actors oversee each corresponding process. The next step is considering the issue from a system engineering standpoint, formulating the idea for the digital twin concept and assessing how it will impact the processes and actors. As one goes deeper into the intricacies of a supply chain, more complexities and dependencies arise. Thus, compiling and presenting a supply chain as a contained visual model is a complex primary stage of a digital twin implementation (Norrblom, personal communication, April 1st, 2020, as cited in BLOMKVIST, 2020). While the model never will be entirely perfect in regards to accuracy, the simplifications in the models that are made should still be done as faithfully to the modelled process as possible. Thus, incorporating personnel with specific detailed knowledge within the processes that are being modelled is crucial, as their insight regarding what to simulate and what to leave out of the future digital twin will be invaluable. An accuracy rating of 80-90% is already incredibly valuable, especially considering that incremental increases in terms of accuracy become increasingly costly when attempting to strive towards perfection. Thus, a final challenge in terms of pre-implementation is deciding which level of accuracy is deemed sufficient for the finished product (Norrblom, personal communication, April 1st, 2020, as cited in BLOMKVIST, 2020).

5.1.2. Knowledge Assessment

If a digital twin is decided upon as the solution of choice the organization itself must then be evaluated in accordance to two distinct metrics, the

technical maturity and expertise within the organization and how far along the processes of digitalization have proceeded within the organization. At this step it is important to have a clear picture regarding the technical systems that are in place today, how (and if) they communicate with one another and how information is gathered from external actors. Furthermore, assessing whether the technical knowledge that is needed to understand, implement and maintain a digital twin exists within the organization is of crucial importance if a digital twin is to be implemented, and cultivating this knowledge must be a priority for any company that is in the business of digital twins (Norrblom, personal communication, April 1st, 2020, as cited in BLOMKVIST, 2020).

5.1.3. Establishing Scope and Objectives

After technical expertise has been established to exist within the company and a clear picture of the current state of technical and digital systems is in place, the next step is to ensure that the scope and objectives of the digital twin is clear. For an implementation to succeed, it is important that knowledge exists within the organization regarding the reasons behind the implementation, the benefits that a digital twin in place will entail and what the road towards implementation will entail for the organization (Kreutzer et al, 2017, as cited in BLOMKVIST, 2020). Furthermore, the choice of method by which to start implementing the digital twin is vital. In large organizations with a complex supply chain involving multiple actors, it may be difficult to decide where to start implementing the twin, as a top-down approach, developing and implementing an all-encompassing digital twin from scratch, may seem an insurmountable task. In situations such as these, approaching the issue from a bottom-up approach may be preferred, creating digital copies and representations of a single process at a time that can then subsequently be interlinked to form a proper digital twin (Edström, personal communication, April 16th, 2020, as cited in BLOMKVIST, 2020). With this approach, it is crucial that there exists a clear consensus regarding how gathered data is structured in terms of information modelling, so that the individual twins easily can be interlinked without having to translate information between the different systems (Edström, personal communication, April 16th, 2020, as cited in BLOMKVIST, 2020).

5.1.4. Developing the digital infrastructure

After all this is in place, the work on developing the digital infrastructure necessary for the digital twin itself can be initiated. Cloud information storage, data security systems and systems able to assess and analyze large quantities of data are all cornerstones for a digital twin, and ensuring these are in place is of vital importance for ensuring functionality of the twin (Edström, personal communication, April 16th, 2020, as cited in BLOMKVIST, 2020). At this stage, regular communication is important to ensure that the work conducted within the various areas of the organization adheres to the previously established goals and project scopes. Quickly establishing a standardized interface is helpful in this regard, along with other tools that serve as an aid for everyone to provide and share information in a standardized manner (Norrblom, personal communication, April 1st, 2020, as cited in BLOMKVIST, 2020).

5.2. Implementation Challenges

Current challenges of using digital twins in this area are mainly reflected in the three aspects of enterprise demand and application willingness, information security, and data integration technology.

From the perspectives of enterprise demand and application willingness, there is currently no related research on the use of digital twins in remanufacturing supply chains, and very few applications and research on the applicability of digital twins to traditional manufacturing supply chains. The potential demand and the willingness of enterprises to go down this route are currently unclear to researchers and potential developers of such platforms. At the same time, very few close-loop remanufacturing supply chains are formed, and inadequate communication between enterprises also increases the difficulties of constructing digital-twin platforms. Therefore, understanding the demand for information-sharing by both upstream and downstream enterprises in the supply chain is pivotal to promoting the rational use of EoL (End-of-Life) products. Further research should use interviews or surveys to analyze the demand for and willingness of

enterprises to use digital twin-related applications in order to improve understanding of the demand for digital twins in remanufacturing supply chains and thus promote the transfer of digital twins from theory to practice (Chen Z and Huang, 2021).

Secondly, the main challenge of using digital twins in remanufacturing supply chains is data integration. Information shared by enterprises at different points along the supply chain includes not only information entered and sent or received manually, such as contracts and orders, but also data generated directly by sensors or Management Information System (MIS) systems, such as logistics, locations and storage volumes (Uhlemann et al., 2017, as cited in Chen Z and Huang, 2021). To realize the dynamic alliance of enterprises in remanufacturing supply chains, it is necessary to build a database to share some of the enterprise's data, which is also the basis for the implementation of digital-twin platforms. Based on this data-integration platform, enterprises can analyze the data and optimize management or production processes based on their demand. However, currently digital twins can only carry out simple information transfers, most of which are between managers and machines. Information-sharing between managers in a complex departmental network has not been achieved (Tao et al., 2018; Toro et al., 2019, as cited in Chen Z and Huang, 2021), nor have cross-enterprise data-integration platforms. Therefore, this article suggests that the establishment of a cross-enterprise data-integration platform based on database technology is the technical prerequisite for building a digital twin for remanufacturing supply chains (Chen Z and Huang, 2021).

The third challenge is security of information. Current applications of digital twins do not need to consider the implications of data-sharing across enterprises for information security, but the digital-twin circular supply chain cannot avoid this challenge, as it may lead to corporate information being leaked (Prabhu et al., 2019, as cited in Chen Z and Huang, 2021). Therefore, finding feasible solutions to ensure the security of data in the data-sharing process is an urgent problem to be solved in current developments of digital twins for remanufacturing supply chains. Mandolla et al. (2019) (as cited in Chen Z and Huang, 2021) propose a digital-twin application framework for the additive manufacture of aircraft components. Due to the widespread distribution of enterprises manufacturing aircraft parts, this research application provides support for the information security of various departments based on the blockchain. Similarly, this article believes that a blockchain-based information-sharing method for remanufacturing supply chains is a potential solution ensuring information security for enterprises on the digital-twin platform of the supply chain (Chen Z and Huang, 2021).

Despite the field of research still being relatively young, it is apparent that there are numerous challenges associated with the successful implementation of a digital twin. Building upon the issues raised by Modoni et al (2018), Kritzinger et al (2018) (as cited in BLOMKVIST, 2020) and Uhlemann et al (2017) (as cited in BLOMKVIST, 2020) the following list of challenges that are likely to be encountered when attempting to implement a digital twin have been identified.

5.2.1. Architecture Challenges

To increase the performance of the whole supply chain, the management processes needs to be autonomous, decentralized and handled collaboratively which allows each constituent to achieve its own desired plan and constraints autonomously and at the same time to pursue the optimum as a whole (Taghipour et al., 2022, as cited in Lugaresi et al., 2023). As a result, it is fundamental to design and agree upon a functional DT architecture that can aid the development of both general components such as data exchange interfaces and application specific services. The structural development should also take into account the operational phases of DTs (i.e., life cycle management) (Romero et al., 2020, as cited in Lugaresi et al., 2023).

5.2.2. Interaction Challenges

While DTs require both on and off-line interaction between the physical and digital world, a small part of existing studies specifically considers it. The development of specific techniques for the physical-digital alignment is essential to overcome this challenge (Tan et al., 2022, as cited in Lugaresi et al., 2023).

5.2.3. Application Challenges

In general, DTs have been implemented sparsely for the improvement of different supply chain functions (e.g., procurement, logistics, distribution, retail). As a result, existing DTs usually regard only one or few echelons of the supply chain, missing holistic opportunities of whole DT approach (Valero et al., 2022, as cited in [Lugaresi et al., 2023](#)). The multi-structural composition of SC networks and the implications for the DT still need to be addressed (e.g., organizational, financial, informational changes). Most existing implementations focus on an asset-centric perspective (e.g., hyper-connected objects, virtual representation). A perspective to enhance sensing and adjusting capabilities of the entire supply chain environment is scarcely studied. While supply chain DTs can offer planning and controlling capabilities at both the tactical and operational levels, the advantages on the strategic decision-making level to provide business intelligence and enhance the business ecosystem need to be further exploited. Last but not least, value chains typically include organizations of different sizes. However, there are many challenges in devising and integrating DTs for small and medium enterprises ([Lugaresi et al., 2023](#)).

5.2.4. Costs

The implementation of a digital twin carries with it a steep initial investment. Investing in any new technology platform can be a costly affair and considering the amount of processes involved in the creation and maintenance of a digital twin the costs can quickly skyrocket ([Kritzing et al., 2018](#), as cited in [BLOMKVIST, 2020](#)). While many of these costs are likely to fall in the future thanks to technological advances the decision to implement a digital twin must be carefully weighed to other solutions that could provide similar benefits for a fraction of the cost. If there are only a handful of critical parameters that are of interest to the organization the best solution might not be an all-encompassing digital twin but rather a contained system of sensors and databases ([Modoni et al., 2018](#), as cited in [BLOMKVIST, 2020](#)). It is difficult to give a general answer regarding how to keep costs down for an implementation of a digital twin. When it comes to operations, the amount of data that is to be processed and how to store it are both hugely impactful factors in terms of cost. In a development phase, asserting which data is necessary to collect and how much data to process is a vital step to reduce development costs. If the amount of data generated by the digital twin results in a constant economic strain, it must be assessed whether the upkeep and maintenance of the twin is worth the cost as compared to the analytical value that it provides ([Edström, personal communication, April 16th, 2020](#), as cited in [BLOMKVIST, 2020](#)).

5.2.5. Accurate Representation

Given the current state of technology, it is difficult to establish a digital twin that can act as a perfect copy of its physical asset, especially as the objects and processes being twinned become increasingly complex ([Modoni et al., 2018](#)). This inevitably results in the developers behind the digital twin having to make assumptions and simplify processes in the underlying model, as financial and technical constraints have to be balanced with the strategic and operational demands on the Digital Twin ([Heutger & Kueckelhaus, 2019](#), as cited in [BLOMKVIST, 2020](#)).

5.2.6. Data Quality

In a digital twin application that relies on data provided by hundreds (or thousands) of IoT sensors it can be difficult to ensure reliable data quality ([Modoni et al., 2018](#), as cited in [BLOMKVIST, 2020](#)). Difficult operational conditions and communication over distant networks means that companies will have to create ways to identify and cull unreliable data and handle inconsistencies in received information ([Uhlemann et al., 2017](#), as cited in [BLOMKVIST, 2020](#)). Using an incorrect dataset can dilute the value of the simulation. While the technology and business opportunities may exist, getting accurate information from the entire ecosystem (both internal and external sources) and ensuring the quality of data poses a major challenge for organizations ([Norrblom, personal communication, April 1st, 2020](#), as cited in [BLOMKVIST, 2020](#)). In addition to the technological challenges, ensuring a solid base regarding information modelling and structuring is key for ensuring data quality. By using APIs, one can achieve a controlled data feed to the database, making it far more likely that the data being saved is of value to the organization ([Edström, personal communication, April 16th, 2020](#), as cited in [BLOMKVIST, 2020](#)). Also, the effectiveness of AI-

enhanced digital twins relies heavily on the accuracy of the input data. Inaccurate data can lead to erroneous predictions, decisions, and optimizations. Ensuring data accuracy involves regular calibration of sensors, validation of data inputs, and robust data cleaning processes. Incomplete data can result in gaps in the digital twin, limiting its utility and accuracy. Ensuring that all relevant data points are captured and integrated into the digital twin is crucial for comprehensive analysis and decision making. Digital twins often require data from a variety of sources, including IoT sensors, ERP systems, Customer Relationship Management (CRM) systems, and external databases. Integrating these diverse data sources into a unified digital twin is complex and requires sophisticated data integration frameworks ([Espinosa-Jaramillo et al., 2024](#)).

5.2.7. Interoperability

As many companies might lack the necessary competencies to develop digital twins inhouse, companies seeking digital twin-solutions then run the risk of entirely relying on simulation- and AI technologies supplied by outside actors ([Uhlemann et al., 2017](#), as cited in [BLOMKVIST, 2020](#)). As companies become increasingly reliant on their digital twin it may prove difficult to get the same functionality from other providers, forcing companies into long-term dependencies on outside actors ([Heutger & Kueckelhaus, 2019](#), as cited in [BLOMKVIST, 2020](#)). Also, Different systems and devices may use various data formats and protocols, making integration challenging. Developing and adopting standardized data formats and communication protocols is essential for seamless interoperability between different components of the digital twin ecosystem. In the automotive industry, standardizing data formats for vehicle telemetry data enables seamless integration of data from different vehicle manufacturers into a unified digital twin platform. Implementing interoperability frameworks that facilitate data exchange and integration between diverse systems and platforms is crucial. These frameworks ensure that data flows smoothly across different parts of the supply chain. The adoption of OPC UA (Open Platform Communications Unified Architecture) in industrial automation facilitates interoperability between machinery from different manufacturers, enabling a cohesive digital twin representation of the entire production line. Integrating digital twins with legacy systems that were not designed for modern data exchange can be particularly challenging. Ensuring compatibility and seamless data flow requires significant effort and investment in system upgrades or middleware solutions ([Espinosa-Jaramillo et al., 2024](#)).

5.2.8. Education

Technological transitions within a company invariably leads to personnel having to adopt and adapt to new ways of working ([Van de Ven and Poole, 1995](#), as cited in [BLOMKVIST, 2020](#)). This inevitably leads to challenges in terms of change management and transferring of knowledge. It is not enough that owners and users of digital twins have the necessary tools and knowledge that they need to operate and maintain the digital twin, they must also be motivated to change both their mindset and ways of working, which in turn necessitates a thorough understanding of the reasons and motivations for the change ([Loup and Koller, 2005](#), as cited in [BLOMKVIST, 2020](#)). It thus becomes apparent that in order to reap all the technical benefits of the digital twin the shift in technology will have to be coupled with a shift in company culture and working procedures ([Uhlemann et al., 2017](#), as cited in [BLOMKVIST, 2020](#)). Explaining the digital twin as a concept may be difficult without referring to concepts and processes that are normally recognizable. To explain a digital twin, [Norrblom](#) gives the example of the automotive sector. Testing and evaluating the functions of a vehicle is vital, and historically the only way to do so is to develop a physical prototype and crash it. Needless to say, it is a costly endeavor to develop a prototype whose only goal is to be crashed, only to then find oneself with a heap of problems that need to be addressed before the vehicle is service-worthy. The process then repeats itself all the way to a research and development- phase, culminating in a new crash prototype. Moving this entire development chain into the digital world, where all the different crash- and service parameters can be taken into account, means that one can develop a serviceable vehicle at a fraction of the cost ([Norrblom, personal communication, April 1st, 2020](#), as cited in [BLOMKVIST, 2020](#)). By using a digital twin, problems can thus be found at an early stage, saving both resources and potential loss of good-will for the organization, should a faulty

product go to market. Badwill is tremendously harmful for organizations, and losing goodwill is a much quicker process than rebuilding it (Norrbom, personal communication, April 1st, 2020, as cited in [BLOMKVIST, 2020](#)). When it comes to launching and spreading knowledge about digital twins internally, one of the more important factors is to get a working product that can be showcased within the organization. As soon as you have a tangible concept, relating to the new methods of working and seeing the benefits that they may bring becomes far easier for veterans within the organization. Having a viable product to show within the organization means that the process of adding more data and use cases to the digital twin is likely to be sped up significantly (Edström, personal communication, April 16th, 2020, as cited in [BLOMKVIST, 2020](#)).

5.2.9. IP Protection

To maximize the benefits of the digital twin, data has to be shared between multiple different instances. If the data being handled is directly correlated to the core competencies of the company it likely contains highly sensitive data, which raises issues of data ownership, protocols for ensuring identity and managing access of users (Modoni et al, 2017, as cited in [BLOMKVIST, 2020](#)).

5.2.10. Digital Security

The wealth of information contained within a digital twin makes them valuable targets for criminal intrusion. Apart from the ever-present risk of theft of data, cases in which the digital twin also has the ability to directly control its physical assets, a hijacked digital twin could have hazardous real-world consequences (Uhlemann et al, 2017, as cited in [BLOMKVIST, 2020](#)). While ensuring proper routines of data security of systems of this complexity may be a challenging prospect for many companies, the risks involved means that ensuring the digital security surrounding the digital twin will always be of the utmost importance (Heutger & Kueckelhaus, 2019, as cited in [BLOMKVIST, 2020](#)). Also, digital twins involve extensive data exchange and connectivity, making them vulnerable to cyber-attacks. Ensuring robust cybersecurity measures, such as encryption, firewalls, and intrusion detection systems, is crucial to protect sensitive data and maintain system integrity. Managing access to the digital twin system is critical to prevent unauthorized access and data breaches. Implementing strict access control policies, role-based access, and multi-factor authentication helps safeguard the system. In a large enterprise, ensuring that only authorized personnel can access specific data within the digital twin prevents internal data misuse and enhances overall security. Digital twin systems must comply with data privacy regulations, such as GDPR (General Data Protection Regulation) in Europe and CCPA (California Consumer Privacy Act) in the US. Ensuring compliance involves implementing data anonymization, consent management, and data protection protocols. A healthcare supply chain management system using digital twins must anonymize patient data and ensure compliance with HIPAA (Health Insurance Portability and Accountability Act) regulations to protect patient privacy ([Espinosa-Jaramillo et al., 2024](#)).

5.2.11. Scalability

The sheer volume of data generated by IoT devices and other sources in large-scale operations can be overwhelming. Processing and analyzing this data to update the digital twin requires substantial computational resources and efficient data handling techniques. Running complex AI and machine learning algorithms on massive datasets necessitates powerful computing infrastructure. Ensuring the digital twin can scale to handle these demands is crucial for maintaining performance. Leveraging cloud computing can provide the necessary scalability, but it introduces latency issues. Edge computing can mitigate this by processing data closer to the source, but it requires substantial investment in edge devices and infrastructure. Transmitting large volumes of data between edge devices, on-premise systems, and cloud infrastructure requires high network bandwidth. Ensuring reliable and fast data transmission is essential for maintaining the performance of the digital twin ([Espinosa-Jaramillo et al., 2024](#)).

5.2.12. Indicators and Analysis

Defining evaluation criteria and KPIs, providing standards, and coherent analysis of technology acceptance by users are among the most important challenges in implementing digital twins in the automotive industry, which

this research has attempted to explain .

5.3. Possible Solutions

To overcome these challenges, Building Information Modelling (BIM) has played a crucial role by providing a digital platform that ensures an enhanced flow of information and visual representations of the process (Bakhshi et al., 2022; Pour Rahimian et al., 2019, as cited in [Bakhshi et al., 2024](#)). However, BIM, if not integrated with other advanced data gathering and management systems, has limitations in capturing real-time updates of assets and resources, as its fundamental task is to represent static models (Brandín and Abrishami, 2021, as cited in [Bakhshi et al., 2024](#)). However, although digital representation technologies such as BIM can store, visualize, and share real-time data collected by IoT sensors and perform specific analyses based on this data, a more comprehensive system is required (Hakimi et al., 2023, as cited in [Bakhshi et al., 2024](#)). Such a system should perform more sophisticated analysis, simulations, and predictions based on more extensive data ([Bakhshi et al., 2024](#)). Also, the steps that an organization should take in order to overcome the challenges that come with the implementation of a digital twin are:

- **Prioritize:** Identify the areas where to start implementing the digital twin. Start small and initially apply it on a limited scale.
- **Data Sources:** Assess the availability of data and how to obtain the data which is necessary but not yet accessible.
- **Framework:** Plan out the data and information structure of the digital twin, so as to ensure future modularity and scalability.
- **Integrate:** Ensure that the existing digital infrastructure can be integrated into the digital twin to make proper use of the existing resources.
- **Collect Data:** Assess which sensors are necessary for the digital twin to function and whether the capability to incorporate, use and maintain this technology exists within the organization.
- **Test:** Test the partial digital twin to assess whether it fulfils the project scope and identify potential issues with the twin.
- **Revise:** Make the necessary changes and adjustments to the partial digital twin. Document lessons learned during the implementation that could be used for the future.
- **Repeat:** Create additional partial digital twin by following steps 1-7 in another area of logistics.
- **Expand:** Expand the Digital Twin by connecting the partial digital twins into an entity.
- **Follow up:** Follow-up, maintain and update the digital twin as necessary ([BLOMKVIST, 2020](#)).

6. Discussion

6.1. Impact on Supply Chain Resilience

The presented framework is a combination of simulation, optimization, and data analytics that constitutes a data-driven vision of managing the disruption risks in SC. The presented model can represent the network state for any given moment in time and allow for complete end-to-end SC visibility to improve resilience and test contingency plans ([Ivanov et al., 2019](#)) . By integrating data from various sources throughout the supply chain with the help of sensors, IoT devices, and historical data, the digital twin can create a virtual model of the supply chain, including suppliers, manufacturers, distributors, and customers. This virtual model can be used to identify vulnerabilities and potential disruptions event in the supply chain, allowing businesses to proactively adjust contingency plans ([Khan, 2024](#)) .

The direct impact of the presented framework is reflected in changes to KPIs. Revenue, sales, service level, fill rate, and costs are typically calculated. Unlike analytical models that usually focus on a particular metric (e.g., costs/profit), simulation enables the simultaneous measurement of all metrics in the same model. Their values can be checked at any chosen moment of the period modelled. This way, disruption duration can be modelled, performance impact measured, and mitigation policies evaluated for efficiency ([Ivanov et al., 2019](#)) .

6.2. Implications for Quality Assurance

Effective quality assurance is vital for automotive companies to remain competitive in an industry marked by rapid technological advancements and fluctuating market demands. Traditional quality assurance methods, which often rely on historical data and static models, are increasingly inadequate for addressing the dynamic nature of modern automotive supply chains. The need for real-time insights, adaptive strategies and especially predictive capabilities has never been more pressing. Consequently, the integration of advanced technologies to optimize supply chain operations has become a strategic imperative for automotive firms aiming to achieve operational excellence and meet evolving market demands (Putha, 2022). Thus, there is a timely motivation to investigate predictive quality assurance in both manufacturing and service supply chains in this uncertain situation (Belhadi et al., 2021). On the other hand, considering the problems faced by traditional quality assurance technology, such as low traceability, and unpredictability, a digital twin approach towards traceability and dynamic assurance for supply chain quality is proposed in this paper. Thus, we designed a conceptual framework to clarify our research avenue.

6.3. Broader Implications

According to Childerhouse et al. (2003) (as cited in Belhadi et al., 2021), the auto industry has traditionally been playing an influential role, and providing lessons to some of the other sectors because of their accumulated experience in managing supply chain disruptions.

The presented model for the digital twin in this research can interface with other emerging technologies in other sectors such as digital twins of aerospace, defense, and general manufacturing to enhance operational performance in many industrial sectors, including the agri-food supply chain. The proposed methods could also increase visibility in the supply chain of other industries, reducing waste and meeting additional sustainability goals. In its proposed solution, using best-in-class modelling of the automotive industry environment and associated risks, it develops virtual replicas of other real-world environments and scenarios. This study claims that KPIs for supply chain resilience are needed to control and manage the company's target plans when a disruption occurs at any time. This study aims to design KPIs to help firms assess the indicators and sub-indicators related to disruption as long-term measures in the supply chain of the automotive industry.

7. Conclusion

7.1. Summary of Findings

Assessing and monitoring quality in the supply chain can help build resilience in volatile, uncertain, complex, and ambiguous (VUCA) business environments. Thus, the quality assurance department has to be an integral part of the supply chain, and the main goal of the supply chain should be to make a quality product in terms of the whole supply chain rather than just the manufacturing process (Sohail et al., 2022). It will be possible only by designing and implementing predictive quality assurance at the supply chain level with the help of digitalization and Industry 4.0 and using smart methods, such as the digital twin, to establish resilience before disruptions enter a production or service system. On the other hand, one of the topics that is highly popular among researchers is the digital supply chain twin for managing the disruption risks and resilience, most of the papers have contributed either theoretical or generic structure. Therefore, this study has drawn attention to the use of machine learning techniques in supply chain resilience, which are much more efficient considering the importance of data in the design and management of the proposed supply chain. Besides, the classification and division of the automotive supply chain into subdivisions will be further used in the paper to analyze subsystems or performance areas with similar functions. Performance areas should be considered by a specific range that can be a part of a business system and not the whole system. The main benefit of dividing a system into several performance areas is the quick identification of bottlenecks and the compilation of more accurate and achievable goals to solve those problems. This research believes that to understand every problem of business, first of all, business processes should be analyzed. According to processes, bottlenecks are known and after that, it's necessary to define KPIs relevant to every bottleneck and department.

Finally, with the help of data, we can solve problems, but it starts with description, diagnosis, prediction and prescription respectively. In the reviews conducted, no article was found that used dividing the supply chain into performance areas with similar functions and determining KPIs appropriate to each area in digital twin applications for predictive quality assurance in a complete supply chain. To fill this gap, the proposed research will design a model for predictive quality assurance for supply chain resilience, construct the conceptual framework for integrating the digital twin into the supply chain, and develop a generic digital supply chain twin model that leverages machine learning to improve supply chain resilience.

7.2. Practical Implication

This study aims to use a set of KPIs that can be used in various divisions to manage the automotive industry for quality assurance and does not require complex data collection methods to measure SCR. This approach allows automotive companies to implement the SCR measurement method and KPIs defined in this study, and thus ensure SCR. Also, by comparing with the IATF standard, it is possible to compare the current and desired status of an automotive company and use the necessary predictions to identify system disruptions. Also, according to the reference model used in this research, the information of each subsystem, such as corporate data, customer data, contracts and databases that will be susceptible to use for analysis is highlighted in green in Fig. 2. Considering the problems faced by traditional quality assurance technology, such as low traceability, and unpredictability, a digital twin approach towards traceability and dynamic assurance for supply chain quality is proposed in this paper. Thus, we designed a conceptual framework, as shown in Fig. 15, to clarify our research avenue.

7.3. Limitations and Future Work

The implementation of digital twin technology within the supply chain is still an emerging area. Based on the literature reviews, there were indeed some limitations and challenges associated with the implementation of digital twin technology in the supply chain which need to be considered (20 Khan, 2024). For example, the ripple effect refers to the impact that a sudden disruption at one node in a supply chain has on the rest of the supply chain. If large industries with extensive supply chains, such as the automotive industry, want to predict and respond appropriately and quickly, they should define KPIs to assess this situation, which can be considered as the next step in future research. On the other hand, while this paper presents the development of a predictive quality assurance (PQA) framework utilizing digital twins to enhance supply chain resilience in the automotive industry, the validation and application of this model within an industrial setting remain as future work. A follow-up study will detail the empirical validation of the proposed framework through a case study involving an automotive supply chain.

8. References

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